

A Cheatsheet for Lie Algebra

Contents

1 Basic Concepts	2
1.1 Preliminaries	2
1.2 Engel's Theorem	7
1.3 Theorems of Lie and Cartan	10
2 Semisimple Algebras	14
2.1 Killing Form	14
2.2 Complete Reducibility of Representations	16
2.3 Representations of $\mathfrak{sl}(2, F)$	21
2.4 Root Space Decomposition	23
3 Root System	28
3.1 Definition	28
3.2 Simple Roots	31
3.3 Classification	37
4 Isomorphism and Conjugacy Theorems	45
4.1 Isomorphism Theorem	45
4.2 Conjugacy Theorem	49
5 Existence Theorem	56
5.1 Universal Enveloping Algebras	56

These notes follow Humphreys, *Introduction to Lie Algebras and Representation Theory* [1]. Throughout, F denotes an algebraically closed field with characteristic 0.

1 Basic Concepts

1.1 Preliminaries

Definition 1.1. A vector space L over a field F with an operation $L \times L \rightarrow L$ $(x, y) \mapsto [xy]$ called the bracket or commutator of x, y is called a Lie algebra over F if the following axioms are satisfied $\left\{ \begin{array}{l} \text{the bracket operator is bilinear} \\ \forall x \in L, [xx] = 0 \text{ where the last} \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = 0 \end{array} \right.$ condition is called the Jacobi identity.

$$\begin{aligned} [x+y, x+y] &= [x, x] + [x, y] + [y, x] + [y, y] && \text{by the linearity} \\ \rightarrow &= [x, y] + [y, x] && [x, x] = 0, \forall x \in L \Leftrightarrow [x, y] = -[y, x] \\ &= 0 && [x, x] = 0, \forall x \in L \end{aligned}$$

and if $\text{char } F \neq 2$, the converse also holds.

Definition 1.2. Two Lie algebras L, L' over F are isomorphic if there exists a vector space isomorphism $\phi: L \rightarrow L'$ satisfying $\phi([xy]) = [\phi(x)\phi(y)], \forall x, y \in L$.

Definition 1.3. A subalgebra of a Lie algebra L is a subspace K of L s.t. $[xy] \in K, \forall x, y \in K$.

$\rightarrow K$ is a Lie algebra with the inherited operations.

e.g. **The General Linear Algebra** $\mathfrak{gl}(V)$

If V is a finite dimensional vector space over F , the set of linear transformations $V \rightarrow V$ is denoted by $\text{End}V$, which is of dimension n^2 where $\dim V = n$ and is a ring relative to the usual product operation. Define a new operation $[x, y] = xy - yx$, called the bracket of x, y . With this operation, $\text{End}V$ becomes a Lie algebra over F .

$\rightarrow \left\{ \begin{array}{l} \forall x \in L, [x, x] = xx - xx = 0 \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = x(yz - zy) + y(zx - xz) + z(xy - yx) = 0 \end{array} \right.$ This Lie algebra is called the general linear algebra $\mathfrak{gl}(V)$. It is identified with the set of all $n \times n$ matrices over F , denoted $\mathfrak{gl}(n, F)$. Then $[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{li}e_{kj}$ where e_{ij} is a standard basis for matrices.

Subalgebras of the General Linear Algebra

- **the Special Linear Algebra** is the set of endomorphisms of V having trace zero and is denoted $\mathfrak{sl}(V) = \mathfrak{sl}(l+1, F)$ where $\dim V = l+1$.

Throughout, $T(\cdot)$ denotes the trace. Since $\left\{ \begin{array}{l} T(xy) = T(yx) \\ T(x+y) = T(x) + T(y) \end{array} \right.$, it is a

subalgebra of $\mathfrak{gl}(V)$.

Take all e_{ij} , $i \neq j$ along with $h_i = e_{ii} - e_{i+1,i+1}$. Then this is the basis for $\mathfrak{sl}(V)$ and therefore it has dimension $l + (l+1)^2 - (l+1)$.

- **the symplectic algebra** is the set of endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$, denoted $\mathfrak{sp}(V) = \mathfrak{sp}(2l, F)$, where $\dim V = 2l$ and f has a skew-symmetric form $s = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Then for $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$ where $m, n, p, q \in$

$\mathfrak{gl}(l, F)$, $sx = -x's$ given it being symplectic. In other words, $\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} =$

$$-\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \Leftrightarrow \begin{cases} n' = n \\ p' = p \\ m' = -q \end{cases}. \text{ Then the basis consists of}$$

$$\begin{cases} m' = -q \\ n = n' \end{cases} \begin{cases} e_{ii} - e_{l+i, l+i}, 1 \leq i \leq l \\ e_{ij} - e_{l+j, l+i} \\ e_{i, l+i} \\ e_{i, l+j} + e_{j, l+i} \end{cases} \begin{matrix} l + l^2 - l \\ \\ l + \frac{1}{2}(l(l-1)) \end{matrix}. \text{ Since } n \text{ and } p \text{ share}$$

the basis, the total dimension becomes $2l^2 + l$.

- **the Orthogonal Algebra** is the set of endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$, denoted $\mathfrak{o}(V) = \mathfrak{o}(2l, F)$ where $\dim V = 2l$ and f has a form $s = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$. Then for $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$ where $m, n, p, q \in \mathfrak{gl}(l, F)$, $sx = -x's$

given it being orthogonal. In other words, $\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} = -\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$.

Then the basis consists of

$$\begin{cases} m' = -q \\ n = -n' \end{cases} \begin{cases} \text{similar with the symplectic case} \\ \text{diagonal} = 0 \\ e_{i, l+j} - e_{j, l+i} \end{cases} \begin{matrix} l^2 \\ \\ \frac{1}{2}(l(l-1)) \end{matrix}. \text{ Since } n \text{ and } p$$

share the basis, the total dimension becomes $2l^2 - l$.

- $\begin{cases} \text{the set of upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i > j & \mathfrak{t}(n, F) \\ \text{the set of strictly upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i \geq j & \mathfrak{n}(n, F) \\ \text{the set of all diagonal matrices} & \mathfrak{d}(n, F) \end{cases} \rightarrow$

$$\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$$

pf. The basis for $\mathfrak{t}(n, F)$ consists of all e_{ij} s.t. $i \leq j$ which has dimension $\frac{1}{2}n(n+1)$. The basis for $\mathfrak{d}(n, F)$ consists of all e_{ii} which has dimension n , while the basis for $\mathfrak{n}(n, F)$ consists of all e_{ij} s.t. $i < j$ which has dimension $\frac{1}{2}n(n-1)$. Then the sum of bases for $\mathfrak{n}(n, F)$, $\mathfrak{d}(n, F)$ gives the basis for $\mathfrak{t}(n, F)$ and the stmt follows. Moreover the sum is direct: $\mathfrak{n}(n, F)$ is nonabelian while $\mathfrak{d}(n, F)$ is abelian, and their bases are disjoint (e_{ij} with $i < j$ versus e_{ii}), so $\mathfrak{n}(n, F) \cap \mathfrak{d}(n, F) = 0$ and $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$.

Definition 1.4. For any field F , an F -algebra is a vector space $\mathcal{U}$ over F endowed with a bilinear operation $\mathcal{U} \times \mathcal{U} \rightarrow \mathcal{U}$, usually denoted by juxtaposition. By a derivation of $\mathcal{U}$, we mean a linear map $\delta : \mathcal{U} \rightarrow \mathcal{U}$ satisfying the product rule $\delta(ab) = a\delta(b) + \delta(a)b$.

Since a Lie algebra L is an F -algebra in the above sense, $\text{Der}L$ is defined. If $x \in L$, $y \mapsto [xy]$ is an endomorphism of L , denoted adx . Then $\text{adx} \in \text{Der}L$ by the Jacobi identity. Derivations of this form are called **inner**, all others **outer**.

Definition 1.5. The map $\begin{matrix} L & \rightarrow & \text{Der}L \\ x & \mapsto & \text{adx} \end{matrix}$ is called the adjoint representation of L .

Definition 1.6. A Lie algebra L over F is called abelian if $\forall x, y \in L$, $[xy] = 0$.

Definition 1.7. A subspace I of a Lie algebra L is called an ideal of L if $x \in L$, $y \in I$ imply $[xy] \in I$.

$\rightarrow [yx] = -[xy] \in I$ by skew-symmetry.

e.g. $\begin{cases} Z(L) = \{z \in L \mid [xz] = 0, \forall x \in L\}, \text{ the centre of } L \\ [LL] \text{ the derived algebra of } L \end{cases}$

Then L is abelian iff $[LL] = 0$. At the other extreme, when $L = \mathfrak{sl}(n, F)$, $[LL] = L$.

If I, J are two ideals of a Lie algebra L , then $\begin{cases} I + J = \{x + y \mid x \in I, y \in J\} \\ [IJ] = \{\sum [x_i y_i] \mid x_i \in I, y_i \in J\} \end{cases}$ are ideals.

Definition 1.8. A Lie algebra L is called simple if it has no ideals except itself and 0, and $[LL] \neq 0$.

e.g. $L = \mathfrak{sl}(2, F)$, $\text{char} F \neq 2$. Then it has a standard basis $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then the multiplication table is determined by the equations $\begin{cases} [xy] = h \\ [hx] = 2x \\ [hy] = -2y \end{cases}$. If $I \neq 0$ is an ideal of L , let $ax + by + ch$ be an arbitrary

nonzero element of I . Then applying $\begin{cases} \text{adx twice, we get } -2bx \in I \\ \text{ady twice, we get } -2ay \in I \end{cases}$. If a or b is nonzero, that means either x or y is in I , and therefore $L = I$ by the above multiplication table. If $a = b = 0$, $ch \in I$ and this also implies $L = I$ by the above multiplication table. This shows L is simple.

If L is not simple, it is possible to factor out a nonzero proper ideal I and obtain a **quotient algebra** L/I : as vector space L/I is just the quotient space, while its Lie multiplication is defined by $[x + I, y + I] = [xy] + I$.

Definition 1.9. For any Lie algebras L, L' over F , a linear transformation $\phi : L \rightarrow L'$ is called a *homomorphism* if $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$. In addition, if $\ker \phi = 0$, ϕ is called a *monomorphism*, or if $\text{Im} \phi = L'$, ϕ is called an *epimorphism*. It is an *isomorphism* if it is both *mono-* and *epi-*.

$\ker \phi$ is an ideal of L .

pf. $\forall x \in \ker \phi$, $\forall y \in L$, $\phi([xy]) = [\phi(x)\phi(y)] = 0$ so $[xy] \in \ker \phi$ and therefore the stmt holds.

$\text{Im} \phi$ is a subalgebra of L' .

Proposition 1.1. 1. If $\phi : L \rightarrow L'$ is a homomorphism of Lie algebras, then $L/\ker \phi \cong \text{Im} \phi$. If I is any ideal of L included in $\ker \phi$, there exists a unique

homomorphism $\psi : L/I \rightarrow L'$ making the following diagram commute:

$$\begin{array}{ccc} L & \xrightarrow{\phi} & L' \\ & \searrow \pi & \uparrow \psi \\ & & L/I \end{array}$$

2. If I, J are ideals of L s.t. $I \subset J$, then J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J .

3. If I, J are ideals of L , there is a natural isomorphism between $(I + J)/J$ and $I/(I \cap J)$.

Definition 1.10. A representation of a Lie algebra L is a homomorphism $\phi : L \rightarrow \mathfrak{gl}(V)$.

For L finite-dimensional, we have an adjoint representation $\text{ad} : L \rightarrow \mathfrak{gl}(L)$. Then its kernel consists of all $x \in L$ for which $\text{adx} = 0$, i.e. $[xy] = 0$. So $\ker \text{ad} = Z(L)$. Then, if L is simple, $Z(L) = 0$ so that $\text{ad}L : L \rightarrow \mathfrak{gl}(L)$ is a monomorphism. This means that any simple Lie algebra is isomorphic to a linear Lie algebra.

Definition 1.11. An automorphism of L is an isomorphism of L onto itself. The group of automorphisms of L is denoted $\text{Aut}L$.

If $g \in \text{GL}(V)$ is any invertible endomorphism of V and if moreover $gLg^{-1} = L$, then $x \mapsto gxg^{-1}$ is an automorphism of L .

Assume $\text{char} F = 0$. Suppose $x \in L$ is an element for which adx is nilpotent, i.e. $(\text{adx})^k = 0$, $\exists k \in \mathbb{Z}_{++}$. Then $\exp(\text{adx}) = 1 + \text{adx} + \frac{(\text{adx})^2}{2!} + \dots + \frac{(\text{adx})^{k-1}}{(k-1)!}$.

CLAIM : $\exp(\text{adx}) \in \text{Aut}L$.

$$\begin{aligned}
[\exp \delta(x), \exp \delta(y)] &= \left[\sum_{i=0}^{k-1} \frac{\delta^i x}{i!}, \sum_{j=0}^{k-1} \frac{\delta^j y}{j!} \right] \\
&= \sum_{n=0}^{2k-2} \left(\sum_{i=0}^n \left[\frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \right) \\
\text{pf.} \quad &= \sum_{n=0}^{2k-2} \frac{\delta^n [xy]}{n!} \quad \frac{\delta^n}{n!} [xy] = \sum_{i=0}^n \left[\frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \quad \text{where } \text{adx} \\
&= \sum_{n=0}^{k-1} \frac{\delta^n [xy]}{n!} \quad \delta^k = 0 \\
&= \exp \delta([xy])
\end{aligned}$$

is replaced by an arbitrary nilpotent derivation δ of L . Also given $\exp \delta = 1 + \eta$, its inverse is $1 - \eta + \eta^2 - \dots \pm \eta^{k-1}$ exists and therefore $\exp \delta$ is invertible. This gives the claim.

Leibniz rule

$$\frac{\delta^n}{n!}(xy) = \sum_{i=0}^n \frac{\delta^i x}{i!} \frac{\delta^{n-i} y}{(n-i)!}$$

pf. when $n = 0$, it trivially holds.

when $n > 0$, $\delta(\frac{\delta^{n-1}}{(n-1)!}(xy)) = \delta(\sum_{i=0}^{n-1} \frac{\delta^i x}{i!} \frac{\delta^{n-1-i} y}{(n-1-i)!}) = \sum_{i=0}^{n-1} \frac{\delta^{i+1} x}{i!} \frac{\delta^{n-1-i} y}{(n-1-i)!} + \sum_{i=0}^{n-1} \frac{\delta^i x}{i!} \frac{\delta^{n-i} y}{(n-1-i)!} =$
 $n \sum_{i=0}^n \frac{\delta^i x}{i!} \frac{\delta^{n-i} y}{(n-i)!}$ by the binomial identity. So the Leibniz rule follows by induction.

Definition 1.12. An automorphism of the form $\exp(\text{adx})$, adx nilpotent is called inner and the subgroup of $\text{Aut} L$ generated by these is denoted $\text{Int} L$, the set of inner automorphisms.

→ it is a normal subgroup ($\phi \in \text{Aut} L$, $x \in L$, $\phi(\text{adx})\phi^{-1} = \text{ad}\phi(x) \rightarrow \phi \exp(\text{adx})\phi^{-1} = \exp(\text{ad}\phi(x))$).

CLAIM : If $L \subset \mathfrak{gl}(V)$ is an arbitrary linear Lie algebra, and $x \in L$ is nilpotent, then $(\exp x)y(\exp x)^{-1} = \exp \text{ad}(x)(y)$, $\forall y \in L$.

pf. $\text{adx} = \lambda_x + \rho_{-x}$ where $\begin{cases} \lambda_x & \text{left multiplication by } x \\ \rho_{-x} & \text{right multiplication by } -x \end{cases} \in \text{End}(\text{End} V)$. Then $\exp \text{adx} = \exp(\lambda_x + \rho_{-x}) = \exp \lambda_x \exp \rho_{-x} = \lambda_{\exp x} \cdot \rho_{\exp -x}$ and the stmt follows.

Definition 1.13. Define a sequence of ideals of L called the derived series by $L^{(0)} = L$, $L^{(1)} = [LL]$, $L^{(2)} = [L^{(1)}L^{(1)}]$, $\dots$, $L^{(i)} = [L^{(i-1)}L^{(i-1)}]$. Then L is said to be solvable if $L^{(n)} = 0$, $\exists n \in \mathbb{Z}_{++}$.

e.g. $\mathfrak{t}(n, F)$ is solvable.

pf. The basis for it consists of the matrix units e_{ij} , $i \leq j$ and the dimension is $\frac{n(n+1)}{2}$. Since $[e_{ii}e_{il}] = e_{il}$, $\forall i < l$, $\mathfrak{n}(n, F) \subset [LL]$ where it is the subalgebra of upper triangular nilpotent matrices. Since $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$ and $\mathfrak{d}(n, F)$ is abelian, $\mathfrak{n}(n, F)$ is equal to the derived algebra of L . Define the level of e_{ij} by $j - i$. In the formula of commutators, assume $i < j, k < l$. If $i = l$, $[e_{ij}e_{kl}] = -e_{kj}$ with the level $j - k = j - i + l - k \geq 2$, two brackets of level $\geq 2^i$ must give level $\geq 2^{i-1}$.

So now consider the case where $i \neq l$. Then $[e_{ij}e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$. Each e_{il} is a commutator of two matrices whose levels add up to that of e_{il} . Thus $L^{(2)}$ is spanned by e_{ij} of level ≥ 2 , $L^{(i)}$ by those of level $\geq 2^{i-1}$. Since $L^{(i)} = 0$ whenever $2^{i-1} > n - 1$ and therefore the stmt follows.

Proposition 1.2. *Let L be a Lie algebra.*

1. *If L is solvable, then so are all subalgebras and homomorphic images of L .*
2. *If I is a solvable ideal of L s.t. L/I is solvable, then L itself is solvable.*
3. *If I, J are solvable ideals of L , then so is $I + J$.*

pf.

- If K is a subalgebra of L , $K^{(i)} \subset L^{(i)}$ and therefore the subalgebras are also solvable. If $\phi : L \rightarrow M$ is an epimorphism, an easy induction on i shows $\phi(L^{(i)}) = M^{(i)}$ and therefore its homomorphic images are also solvable.
- Say $(L/I)^{(n)} = 0$. For the canonical homomorphism $\pi : L \rightarrow L/I$, 1 implies $\pi(L^{(n)}) = 0$, i.e. $L^{(n)} \subset I = \ker \pi$. By hypothesis, $\exists m \in \mathbb{Z}_{++}$ s.t. $I^{(m)} = 0 \rightarrow (L^{(n)})^{(m)} = L^{(n+m)} = 0$ so 2 follows.
- There exists an isomorphism between $(I + J)/J$ and $I/(I \cap J)$. As a homomorphic image of I , the RHS is solvable so $(I + J)/J$ is solvable. Then so is $I + J$ by 2.

$\rightarrow$ let L be an arbitrary Lie algebra and let S be a maximal solvable ideal. If I is any other solvable ideal of L , by the previous proposition 3, $S + I = S$ by maximality, hence $I \subset S$. Thus there exists a unique maximal solvable ideal of L , called the **radical** of L , denoted $\text{Rad}L$. If $\text{Rad}L = 0$, L is called **semisimple**.

1.2 Engel's Theorem

Definition 1.14. *Define a sequence of ideals of L called the descending central series by $L^0 = L, L^1 = [LL], \dots, L^i = [LL^{i-1}]$. L is called **nilpotent** if $\exists n \in \mathbb{Z}_{++}$ s.t. $L^n = 0$.*

$\rightarrow L^{(i)} \subset L^i, \forall i$ so nilpotent algebras are solvable but not vice versa.

pf. $L = \mathfrak{t}(n, F)$ is solvable. But $L^{(1)} = L^1 = \mathfrak{n}(n, F) = [LL^1] = L^2 = \dots \forall i \geq 1$. Thus it is not nilpotent.

Proposition 1.3. *Let L be a Lie algebra.*

1. If L is nilpotent, then so are all subalgebras and homomorphic images of L .
2. If $L/Z(L)$ is nilpotent, then so is L .
3. If L is nilpotent, and nonzero, then $Z(L) \neq 0$.

pf.

- by the previous proposition, part 1
- $L^n \subset Z(L)$ implies $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$
- for the descending central series, $\exists n \in \mathbb{Z}_{++}$ s.t. $L^{n-1} = [LL^{n-2}] \neq 0, L^n = [LL^{n-1}] = 0$. This means $0 \neq L^{n-1} \subset Z(L)$ and therefore $Z(L) \neq 0$.

$\rightarrow \exists n \in \mathbb{Z}_{++}$ s.t. $\text{adx}_1 \dots \text{adx}_n(y) = 0, \forall x_i, y \in L$ implies L to be nilpotent.

Definition 1.15. If L is any Lie algebra and $x \in L$, we call x ad-nilpotent if adx is a nilpotent endomorphism.

Lemma 1.1. Let $x \in \mathfrak{gl}(V)$ be a nilpotent endomorphism. Then adx is also nilpotent.

pf. Associate two endomorphisms $\begin{cases} \lambda_x & \text{left translation} \\ \rho_x & \text{right translation} \end{cases}$ to x . Then they are nilpotent since x is. Moreover $\lambda_x \rho_x(y) = xyx = \rho_x \lambda_x(y)$, i.e. they commute. In any ring the sum or difference of two commuting nilpotents is again nilpotent since x, y nilpotent with m, n s.t. $x^m = 0 = y^n, (x \pm y)^{m+n} = 0$. So $\text{adx} = \lambda_x - \rho_x$ is nilpotent.

Theorem 1.1. Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If L consists of nilpotent endomorphisms and $V \neq 0$, then there exists nonzero $v \in V$ for which $Lv = 0$.

pf.

- If $\dim L = 0$, the theorem is trivial.
- If $\dim L = 1$, a single nilpotent endomorphism has a nonzero vector in its kernel.
- If $\dim L > 1$, $\exists K \subsetneq L$ proper subalgebra of L . By the previous lemma, K acts via ad as a Lie algebra of nilpotent linear transformation on the vector space L , hence also on the vector space L/K . Because $\dim K < \dim L$, by the inductive hypothesis, $\exists x \in L$ with $x + K \neq K$ such that $x + K$ is killed by the image of K in $\mathfrak{gl}(L/K)$. Then $\forall y \in K, [yx] \in K$ while $x \notin K$, i.e. K is properly

included in $N_L(K)$. Take K to be a maximal proper subalgebra of L . By the previous argument, $N_L(K) = L$, i.e. K is an ideal of L . If $\dim L/K$ were greater than 1, then the inverse image in L of one dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction. Thus K has codimension 1. Thus $L = K + Fz$, $z \in L \setminus K$. By induction, $W = \{v \in V \mid Kv = 0\}$ is nonzero. Since K is an ideal, W is stable under L , i.e. $\forall x \in L, \forall y \in K, \forall w \in W, yxw = xyw - [xy]w = 0$. Choose $z \in L \setminus K$ as above, so the nilpotent endomorphism z has an eigenvector, i.e. $\exists v \in W$ for which $zv = 0$. Then $Lv = 0$ as desired.

Theorem 1.2 (Engel). *If all elements of L are ad-nilpotent, then L is nilpotent.*

pf. For any Lie algebra L all of whose elements are ad-nilpotent, the algebra $\text{ad}L \subset \mathfrak{gl}(L)$ satisfies the hypothesis of the previous theorem. Thus $\exists 0 \neq x \in L$ s.t. $[Lx] = 0$, i.e. $Z(L) \neq 0$. Now $L/Z(L)$ consists of ad-nilpotent elements and has smaller dimension than L . By inductive hypothesis, $L/Z(L)$ is nilpotent. The previous proposition, part 2, then implies that L itself is nilpotent.

Definition 1.16. *If V is a finite dimensional vector space, a flag in V is a chain of subspaces $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V$, $\dim V_i = i$. If $x \in \text{End}V$ we say x stabilises this flag provided $xV_i \subset V_i$, $\forall i$.*

Corollary 1.1. *Under the hypotheses of the theorem, there exists a flag (V_i) in V stable under L with $xV_i \subset V_{i-1}$, $\forall i$.*

pf. By Theorem 1.1, there exists $0 \neq v \in V$ s.t. $Lv = 0$. Set $V_1 = Fv$ and let $W = V/V_1$. Then the induced action of L on W is also by nilpotent endomorphisms. By induction on $\dim V$, W has a flag stabilised by L , whose inverse image in V does the trick.

Lemma 1.2. *Let L be nilpotent, K an ideal of L . Then if $K \neq 0$, $K \cap Z(L) \neq 0$.*

pf. L acts on K via the adjoint representation, so by Theorem 1.1, $\exists x \in K$ s.t. $[Lx] = 0$ so $x \in K \cap Z(L)$.

CLAIM : Let L be nilpotent, K a proper subalgebra of L . Then $N_L(K)$ includes K properly.

pf. By Lemma 1.1, K acts via ad as a Lie algebra of nilpotent linear transformations on the vector space L , hence also on the vector space L/K . By Theorem 1.1, $\exists x' \in L/K$ s.t. $Kx' = 0$. This implies $\forall y \in K, [yx] \in K$, $x \in L \setminus K$. Thus $x \in N_L(K) \setminus K$ and therefore it is properly contained.

CLAIM : Let L be nilpotent. Then L has an ideal of codimension 1.

pf. Let K be a maximal proper subalgebra of L . By the previous claim, $N_L(K) = L$. Then $[Ly] \subset K$, $\forall y \in K$ and therefore K is an ideal of L . If $\dim L/K$ were greater than 1, the inverse image in L of a one dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction.

1.3 Theorems of Lie and Cartan

Assume throughout this subsection and the sequel that $\text{char } F = 0$.

Theorem 1.3. *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If $V \neq 0$, then V contains a common eigenvector for all the endomorphisms in L .*

pf.

- $\dim L = 0$, it holds trivially
- $\dim L = 1$, its proper subalgebra $[LL] = 0$ and L is abelian of dimension 1 and an eigenvector for a basis vector of L gives the theorem.
- $\dim L > 1$, since L is solvable, L properly includes $[LL]$. $L/[LL]$ being abelian, any subspace is automatically an ideal. Take a subspace of codimension 1, then its inverse image K is an ideal of codimension 1 in L . By the inductive hypothesis, $\exists v \in V$, $\exists \lambda : K \rightarrow F$ s.t. $xv = \lambda(x)v$, $\forall x \in K$. Fix this λ and set $W = \{w \in V | xw = \lambda(x)w, \forall x \in K\} \neq 0$.

– **CLAIM : L leaves W invariant.**

pf. $\forall w \in W$, $x \in L$, $\forall y \in K$, $yxw = xyw - [xy]w = \lambda(y)xw - \lambda([xy])w = \lambda(y)(xw)$ iff $\lambda([xy]) = 0$. So we only need to prove $\lambda([xy]) = 0$. Fix $w \in W$, $x \in L$. Let $n \in \mathbb{Z}_{++}$ be the smallest integer for which $w, xw, \dots, x^n w$ are linearly dependent. Let W_i be the subspace of V spanned by $w, xw, \dots, x^{i-1}w$, so $\dim W_n = n$, $W_n = W_{n+i}$, $\forall i \geq 0$ and x maps W_n into W_n . Relative to the basis $w, xw, \dots, x^{n-1}w$ of W_n ,

$$\begin{aligned} yx^i w &= yxx^{i-1}w = xyx^{i-1}w - [xy]x^{i-1}w \\ &= x\lambda(y)x^{i-1}w + xw' - [xy]x^{i-1}w \quad \text{by the inductive hypothesis,} \\ &\equiv \lambda(y)x^i w \pmod{W_i} \quad x \text{ maps } W_{i-1} \text{ into } W_i \end{aligned}$$

where $w' \in W_{i-1}$. This means $y \in K$ is represented by an upper triangular matrix whose diagonal entries are equal to $\lambda(y)$. Then $T|_{W_n}(y) = n\lambda(y)$. In particular, this is true for elements of K of the special form $[xy]$. But x, y both stabilise W_n , so $[xy]$ acts on W_n as the commutator of two endomorphisms of W_n and therefore its trace is 0. So $n\lambda([xy]) = 0$. Given $\text{char } F = 0$, $\lambda([xy]) = 0$.

Then write $L = K + Fz$. Since F is algebraically closed, $\exists v_0 \in W$ an eigenvector of z . Then v_0 is obviously a common eigenvector for L .

Corollary 1.2 (Lie). *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, $\dim V = n < \infty$. Then L stabilises some flag in V .*

pf. By the previous theorem with induction on $\dim V$.

Corollary 1.3. *Let L be solvable. Then there exists a chain of ideals of L , $0 = L_0 \subsetneq \dots \subsetneq L_n = L$ s.t. $\dim L_i = i$.*

Corollary 1.4. *Let L be solvable. Then $x \in [LL]$ implies $\text{ad}_L x$ is nilpotent. In particular, $[LL]$ is nilpotent.*

pf. Find a flag of ideals in the previous corollary. Relative to a basis $(x_1, \dots, x_n)$ of L for which $(x_1, \dots, x_i)$ spans L_i , the matrices of ad_L lie in $\mathfrak{t}(n, F)$. Thus the matrices of $[\text{ad}_L, \text{ad}_L] = \text{ad}_L[LL] \subset \mathfrak{n}(n, F)$ the derived algebra of $\mathfrak{t}(n, F)$. Thus $\text{ad}_L x$ is nilpotent for $x \in [LL]$. $\text{ad}_{[LL]} x$ is nilpotent, so $[LL]$ is nilpotent by Engel's theorem.

Definition 1.17. $x \in \text{End} V$ is semisimple iff the roots of its minimal polynomial over F are all distinct.

$\Leftrightarrow$ given F being algebraically closed, x is semisimple iff x is diagonalisable.

Proposition 1.4. *Let V be a finite dimensional vector space over F , $x \in \text{End} V$.*

1. *There exists unique $x_s, x_n \in \text{End} V$ satisfying the conditions; $x = x_s + x_n$, $\begin{cases} x_s & \text{semisimple} \\ x_n & \text{nilpotent} \end{cases}$ which commute.*
2. *There exist polynomials $p(T), q(T)$ in one indeterminate, without constant term, s.t. $x_s = p(x)$, $x_n = q(x)$. In particular, x_s, x_n commute with any endomorphism commuting with x .*
3. *If $A \subset B \subset V$ are subspaces, and x maps B into A , then x_s, x_n also map B into A .*

pf. Let $a_1, \dots, a_l$ be the distinct eigenvalues of x , so the characteristic polynomial is $\prod (T - a_i)^{m_i}$ where m_i denote the multiplicity. If $V_i = \ker(x - a_i \cdot 1)^{m_i}$, V is the direct sum of the subspaces $V_1, \dots, V_l$ each stable under x . On V_i , x clearly has characteristic polynomial $(T - a_i)^{m_i}$. By CRT, we get $p(T) \equiv \begin{cases} a_i & \text{mod } (T - a_i)^{m_i} \\ 0 & \text{mod } T \end{cases}$. Set $q(T) = T - p(T)$. Then $p(T), q(T)$ has zero constant term since $p(T) \equiv 0 \pmod T$.

- 2,3

Set $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$. Since they are polynomials in x , they commute with each other as well as all endomorphisms which commute with x .

They stabilise all subspaces of V stabilised by x , in particular the spaces V_i . The congruence $p(T) \equiv a_i \pmod{(T - a_i)^{m_i}}$ shows the restriction of $x_s - a_i \cdot 1$ to V_i is zero for all i , hence x_s acts diagonally on V_i with single eigenvalue a_i . By definition, $x_n = x - x_s$ which makes it clear that x_n is nilpotent (on V_i since $x_s = a_i I$, $x_n|_{V_i} = (x - a_i I)|_{V_i}$ which is nilpotent by definition of the generalised eigenspace V_i . Since $V = \oplus_i V_i$, x_n is nilpotent on V). Because $p(T), q(T)$ have no constant term, 3 follows.

- 1

Existence follows by the previous argument. If there exists $x = s + n$ another decomposition, $x_s - s = n - x_n$. Then by 2, all endomorphisms in the equation commute. Sums of commuting semisimple/nilpotent endomorphisms are semisimple/nilpotent, whereas only 0 can be both semisimple and nilpotent. Thus $x_s = s$, $x_n = n$.

→ the decomposition $x = x_s + x_n$ is called the **Jordan-Chevalley decomposition** of x .

Consider the adjoint representation of the Lie algebra $\mathfrak{gl}(V)$, V finite dimensional. If $x \in \mathfrak{gl}(V)$ is nilpotent, then so is adx by the lemma 1.1. Similarly, **if x is semisimple, then so is adx** .

pf. Choose a basis $(v_1, \dots, v_n)$ of V relative to which x has matrix $\text{diag}(a_1, \dots, a_n)$. Let $\{e_{ij}\}$ be the standard basis of $\mathfrak{gl}(V)$ relative to $(v_1, \dots, v_n)$. Then $e_{ij}(v_k) = \delta_{jk} v_i$. Then $\text{adx}(e_{ij}) = (a_i - a_j)e_{ij}$ so adx has diagonal matrix, relative to the chosen basis of $\mathfrak{gl}(V)$.

Lemma 1.3. *Let $x \in \text{End} V$, $x = x_s + x_n$ its Jordan decomposition. Then $\text{adx} = \text{adx}_s + \text{adx}_n$ is the Jordan decomposition of adx .*

pf. Since $\text{adx}_s, \text{adx}_n$ are semisimple and nilpotent respectively, they commute, since $[\text{adx}_s, \text{adx}_n] = \text{ad}[x_s, x_n] = 0$. Thus by the previous proposition 1, the lemma follows.

Lemma 1.4. *Let $\mathcal{U}$ be a finite dimensional F -algebra. Then $\text{Der} \mathcal{U}$ contains the semisimple and nilpotent parts in $\text{End} \mathcal{U}$ of all its elements.*

pf. If $\delta \in \text{Der} \mathcal{U}$, let $\sigma, \nu \in \text{End} \mathcal{U}$ be its semisimple and nilpotent parts respectively. If $a \in F$, set $\mathcal{U}_a = \{x \in \mathcal{U} | (\delta - a1)^k x = 0, \exists k\}$. Then $\mathcal{U}$ is the direct sum

of those $\mathcal{U}_a$ for which a is an eigenvalue of δ and σ acts on $\mathcal{U}_a$ as scalar multiplication by a . For arbitrary $a, b \in F$, $\mathcal{U}_a \mathcal{U}_b \subset \mathcal{U}_{a+b}$ since $(\delta - (a+b)1)^n(xy) = \sum_{i=0}^n \binom{n}{i} ((\delta - a)1)^{n-i}x \cdot ((\delta - b)1)^i y$, $\forall x, y \in \mathcal{U}$. If $x \in \mathcal{U}_a$, $y \in \mathcal{U}_b$, $\sigma(xy) = (a+b)xy$ because $xy \in \mathcal{U}_{a+b}$. On the other hand, $\sigma(x)y + x\sigma(y) = (a+b)xy$. By directness of the sum $\mathcal{U} = \bigoplus \mathcal{U}_a$, $\sigma \in \text{Der } \mathcal{U}$. Then $\nu = \delta - \sigma \in \text{Der } \mathcal{U}$.

Lemma 1.5. *Let $A \subset B$ be two subspaces of $\mathfrak{gl}(V)$, $\dim V < \infty$. Set $M = \{x \in \mathfrak{gl}(V) \mid [xB] \subset A\}$. Suppose $x \in M$ satisfies $T(xy) = 0$, $\forall y \in M$. Then x is nilpotent.*

pf. Let $x = s + n$ be the Jordan decomposition of x . Fix a basis $v_1, \dots, v_m$ of V relative to which s has matrix $\text{diag}(a_1, \dots, a_m)$. Let E be the vector subspace of F over the prime field $\mathbb{Q}$ spanned by the eigenvalues $a_1, \dots, a_m$. Consider the dual space E^* , whose elements are the linear functions $f : E \rightarrow \mathbb{Q}$. Given such $f \in E^*$, let y be any element of $\mathfrak{gl}(V)$ whose matrix relative to the given basis is $\text{diag}(f(a_1), \dots, f(a_m))$. If $\{e_{ij}\}$ is the corresponding basis of $\mathfrak{gl}(V)$,
$$\begin{cases} \text{ads}(e_{ij}) = (a_i - a_j)e_{ij} \\ \text{ady}(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}.$$
 Let $r(T) \in F[T]$ be a polynomial without constant term satisfying $r(a_i - a_j) = f(a_i) - f(a_j)$ for all pairs i, j . The existence of such $r(T)$ follows from Lagrange interpolation. Then $\text{ady} = r(\text{ads})$. Since ads is the semisimple part of adx by the lemma 1.3, it can be written as a polynomial in adx without constant term by the proposition 1.4. Therefore $\text{ady} = r(\text{ads})$ is also a polynomial in adx without constant term. By the hypothesis of the lemma, adx maps B into A , so $\text{ady}(B) \subset A$ i.e. $y \in M$. By the hypothesis of the lemma, $T(xy) = 0 \rightarrow \sum a_i f(a_i) = 0$. The LHS is a $\mathbb{Q}$ -linear combination of elements of E . Applying f to the above equation gives $\sum (f(a_i))^2 = 0$. But the numbers $f(a_i)$ are rational, so this forces all of them to be 0, and therefore f must be 0 because the a_i span E . Then E^* is zero and its dual E is also 0 and the lemma follows.

Lemma 1.6. *$T([xy]z) = T(x[yz])$ for all endomorphisms x, y, z of a finite-dimensional vector space.*

pf.
$$\begin{cases} [xy]z = xyz - yxz \rightarrow T([xy]z) = T(xyz) - T(yxz) \\ x[yz] = xyz - xzy \rightarrow T(x[yz]) = T(xyz) - T(xzy) \end{cases} \text{ are equivalent given } T(ab) = T(ba).$$

Theorem 1.4 (Cartan's Criterion). *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. Suppose that $T(xy) = 0$, $\forall x \in [LL]$, $y \in L$. Then L is solvable.*

pf. Let $\begin{cases} A = [LL] \\ B = L \end{cases}$, so $M = \{x \in \mathfrak{gl}(V) \mid [xL] \subset [LL]\}$. Then $L \subset M$ trivially. By the hypothesis $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. If $[xy]$ is a typical generator of $[LL]$, and if $z \in M$, the equation above shows $T([xy]z) = T(x[yz]) = T([yz]x)$. By the definition of M , $[yz] \in [LL]$ so $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in M$. Then by the previous lemma, every $x \in [LL]$ is nilpotent. Then by Engel's theorem, $[LL]$ is nilpotent and therefore L is solvable.

2 Semisimple Algebras

2.1 Killing Form

Definition 2.1. Let L be any Lie algebra. If $x, y \in L$, define the Killing form $\kappa(x, y) = T(\text{adxady})$.

→ it is symmetric bilinear and associative in the sense that $\kappa([xy], z) = \kappa(x, [yz])$.

Lemma 2.1. Let I be an ideal of L . If κ is the Killing form of L and κ_I be the Killing form of I , then $\kappa_I = \kappa|_{I \times I}$.

pf. If W is a subspace of a finite dimensional vector space V , and ϕ an endomorphism of V mapping V into W , then $T\phi = T\phi|_W$, since we can extend a basis of W to a basis of V and calculate ϕ . Then if $x, y \in I$, adxady is an endomorphism of L , mapping L into I , so its trace $\kappa(x, y)$ coincides with the trace $\kappa_I(x, y)$ of $\text{adxady}|_I = \text{ad}_I x \text{ad}_I y$.

Definition 2.2. A symmetric bilinear form $\beta(x, y)$ is called nondegenerate if its radical S is 0, where $S = \{x \in L | \beta(x, y) = 0, \forall y \in L\}$.

→ S is an ideal of L , given the associativity of the Killing form.

Theorem 2.1. Let L be a Lie algebra. Then L is semisimple iff its Killing form is nondegenerate.

pf.

- Assume L is semisimple. Then $\text{Rad}L = 0$. Let S be the radical of κ . By definition, $T(\text{adxady}) = 0, \forall x \in S, \forall y \in L$. By the Cartan criterion, $\text{ad}_L S$ is solvable, hence S is solvable. Since S is an ideal of L , $S \subset \text{Rad}L = 0$ and κ is nondegenerate.
- Let $S = 0$ and I be an abelian ideal of L . Suppose $x \in I, y \in L$. Then adxady maps $L \rightarrow L \rightarrow I$ and $(\text{adxady})^2$ maps L into $[II] = 0$. This means adxady is nilpotent, hence $0 = T(\text{adxady}) = \kappa(x, y)$ so $I \subset S = 0$.

→ $S \subset \text{Rad}L$ but not vice versa.

pf. Let L be a Lie algebra with basis x, y, z s.t.
$$\begin{cases} [xy] = z \\ [xz] = -y \\ [yz] = 0 \end{cases} \text{ . Then } \text{adx} =$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \text{ady} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \text{ad}(z) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and therefore its } \kappa = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

So $\text{Rad}L = L \not\subset \text{span}(y, z) = S \equiv \text{radical of } \kappa$.

Definition 2.3. A Lie algebra L is said to be the direct sum of ideals $I_1, \dots, I_t$ provided $L = I_1 + \dots + I_t$, denoted $L = I_1 \oplus \dots \oplus I_t$.

$$\rightarrow [I_i I_j] \subset I_i \cap I_j = 0, \forall i \neq j.$$

Theorem 2.2. Let L be semisimple. Then there exist ideals $L_1, \dots, L_t$ of L which are simple s.t. $L = L_1 \oplus \dots \oplus L_t$. Every simple ideal of L coincides with one of the L_i . Moreover, the Killing form of L_i is the restriction of κ to $L_i \times L_i$.

pf.

- **CLAIM : Let I be an arbitrary ideal of L . Then $L = I \oplus I^\perp$.**

pf. $I^\perp = \{x \in L : \kappa(x, y) = 0, \forall y \in I\}$ is also an ideal by the associativity of κ . By Cartan's criterion applied to $I \cap I^\perp$, it is solvable, hence 0 as an ideal of the semisimple L . Thus since $\dim I + \dim I^\perp = \dim L$, $L = I \oplus I^\perp$.

- Existence

If L has no proper nonzero ideal, then L itself is simple and there is nothing to prove. Otherwise, let L_1 be a minimal nonzero ideal. Then by the first claim, $L = L_1 \oplus L_1^\perp$. In particular, any ideal of L_1 is also an ideal of L and therefore L_1 is semisimple, and by minimality it is simple. For the same reason $L_1^\perp$ is semisimple. By inductive hypothesis, it splits into a direct sum of simple ideals, which are also ideals of L .

- Uniqueness

If I is any simple ideal of L , then $[IL]$ is also an ideal of L , nonzero because $Z(L) = 0$. Then $[IL] = I$ by simplicity. On the other hand, $[IL] = [IL_1] + \dots + [IL_t]$ so all but one summand must be 0. Say $[IL_i] = I$. Then $I \subset L_i$, which forces $I = L_i$ by the simplicity of L_i .

Corollary 2.1. If L is semisimple, $L = [LL]$ and all ideals and homomorphic images of L are semisimple. Moreover, each ideal of L is a sum of certain simple ideals of L .

For any Lie algebra L , $\text{ad}L$ is an ideal in $\text{Der}L$.

pf. $\forall y, [\delta, \text{adx}](y) = \delta \text{adx}(y) - \text{adx}\delta(y) = \delta([xy]) - [x, \delta y] = [\delta x, y] + [x, \delta y] - [x, \delta y] = [\delta x, y] = \text{ad}(\delta x)(y) \Leftrightarrow [\delta, \text{adx}] = \text{ad}(\delta x), \forall x \in L, \forall \delta \in \text{Der}L$. So the stmt follows.

Theorem 2.3. If L is semisimple, then $\text{ad}L = \text{Der}L$.

pf. Since L is semisimple, $Z(L) = 0$. Thus $L \rightarrow \text{ad}L$ is an isomorphism of Lie algebras. In particular, $M = \text{ad}L$ has nondegenerate Killing form by the theorem 2.1. If $D = \text{Der}L$, $[DM] \subset M$. Then by the lemma 2.1, κ_M is the restriction to $M \times M$ of the Killing form κ_D of D . In particular, if $I = M^\perp$ is the subspace of D orthogonal to M under κ_D , then the nondegeneracy of κ_M forces $I \cap M = 0$. Both I, M are ideals of D so $[IM] = 0$. If $\delta \in I$, $\text{ad}\delta x = 0$, $\forall x \in L$ by the previous stmt, so $\delta x = 0$, $\forall x \in L$ because ad is one to one and $\delta = 0$. So $I = 0$, $\text{Der}L = M = \text{ad}L$.

→ For any semisimple Lie algebra L , if $\mathcal{U}$ is any F -algebra of finite dimension, $\text{Der}\mathcal{U}$ contains the semisimple and nilpotent parts in $\text{End}\mathcal{U}$ of all its elements. In particular, since $\text{Der}L$ coincides with $\text{ad}L$ while $L \rightarrow \text{ad}L$ is one to one, each $x \in L$ determines unique elements $s, n \in L$ s.t. $\text{ad}x = \text{ad}s + \text{ad}n$ is the Jordan decomposition of $\text{ad}x$. This means $x = s + n$, $[sn] = 0$, $\begin{cases} s & \text{ad-semisimple} \\ n & \text{ad-nilpotent} \end{cases}$ and $\begin{cases} s = x_s & \text{the semisimple part} \\ n = x_n & \text{the nilpotent part} \end{cases}$ of x .

2.2 Complete Reducibility of Representations

Definition 2.4. Let L be a Lie algebra. A vector space V , endowed with an operation $\begin{matrix} L \times V & \rightarrow & V \\ (x, v) & \mapsto & xv \end{matrix}$ is called an L -module if the following conditions are satisfied: $\begin{cases} (ax + by)v = a(xv) + b(yv), & x(av + bw) = a(xv) + b(xw) \\ [xy]v = xyv - yxv \end{cases}, \forall x, y \in L. \forall v, w \in V, a, b \in F.$

e.g. If $\phi : L \rightarrow \text{gl}(V)$ is a representation of L , then V may be viewed as an L -module via the action $xv = \phi(x)(v)$. Conversely, given an L -module V , this equation defines a representation $\phi : L \rightarrow \text{gl}(V)$.

Definition 2.5. A homomorphism of L -modules is a linear map $\phi : V \rightarrow W$ s.t. $\phi(xv) = x\phi(v)$. When ϕ is an isomorphism of vector spaces, we call it an isomorphism of L -modules.

Definition 2.6. An L -module V is called irreducible if it has precisely two L -submodules: itself and 0 .

Definition 2.7. V is called completely reducible if V is a direct sum of irreducible L -submodules or equivalently if each L -submodule W of V has a complement W' .

→ if it is a direct sum of irreducible L -submodules, any submodule W of it is also a direct sum of irreducible L -submodules. Then the summand that does not belong to its submodule also makes an L -submodule $W^\perp$ s.t. $W \cap W^\perp = 0$, $W + W^\perp =$

V so $V = W \oplus W^\perp$. Conversely, if any submodule of V has its complement, pick a minimal submodule of V . If there is no proper submodule, V is irreducible and there is nothing to prove. Otherwise, pick a minimal proper submodule V_1 of V . Then by minimality, it is irreducible. Then it has complement $V_1^\perp$ with $\dim V_1^\perp < \dim V$. Then by the inductive hypothesis it is a direct sum of irreducible L -submodules and therefore $V = V_1 \oplus V_1^\perp$ is.

Given a representation $\phi : L \rightarrow \mathfrak{gl}(V)$, the associative algebra generated by $\phi(L)$ in $\text{End}V$ leaves invariant precisely the same subspaces as L . Therefore, all the usual results for modules over associative rings hold for L as well.

SCHUR'S LEMMA : Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be irreducible. Then the only endomorphisms of V commuting with all $\phi(x)$ are the scalars.

L itself is an L -module, and then an L -submodule is just an ideal, so a simple algebra L is irreducible as L -module, while a semisimple algebra is completely reducible.

MAKING NEW MODULES out of OLD ONES

- dual module

Let V be an L -module. Then the dual vector space V^* becomes an L -module with $(xf)(v) = -f(xv)$, $\forall f \in V^*$. Then $([xy]f)(v) = -f([xy]v) = -f(xyv - yxv) = -f(xyv) + f(yxv) = (xf)(yv) - (yf)(xv) = -(yx f)(v) + (xy f)(v) = ((xy - yx)f)(v)$.

- tensor-product

If V, W are L -modules, let $V \otimes W$ be the tensor product over F of the underlying vector spaces. For Lie algebras, define $x(v \otimes w) = xv \otimes w + v \otimes xw$. Then $[xy](v \otimes w) = [xy]v \otimes w + v \otimes [xy]w = (xyv - yxv) \otimes w + v \otimes (xyw - yxw) = (xyv \otimes w + v \otimes xyw) - (yxv \otimes w + v \otimes yxw) = (xy - yx)(v \otimes w)$.

→ given a vector space V over F , $\exists$ an isomorphism
$$\begin{array}{ccc} V^* \otimes V & \rightarrow & \text{End}V \\ f \otimes v & \mapsto & (w \mapsto f(w)v) \end{array}$$
 enables us to consider $\text{End}V$ as an L -module via the isomorphism. The action of L on $\text{End}V$ is $(x \cdot \psi)(v) = x\psi(v) - \psi(xv)$, $\forall \psi \in \text{End}V$, $\forall v \in V$.

Let L be semisimple and $\phi : L \rightarrow \mathfrak{gl}(V)$ be a faithful representation of L . Define a symmetric bilinear form $\beta(x, y) = T(\phi(x)\phi(y))$ on L . The form β is associative, so its radical S is an ideal of L . Moreover $\phi(S) \cong S$ is solvable, so $S = 0$ and β is non-degenerate. Let L be semisimple, β any nondegenerate symmetric associative bilinear form on L . If $(x_1, \dots, x_n)$ is a basis of L , there is a uniquely determined dual basis $(y_1, \dots, y_n)$ relative to β satisfying $\beta(x_i, y_j) = \delta_{ij}$. If $x \in L$,
$$\begin{cases} [xx_i] = \sum_j a_{ij} x_j \\ [xy_i] = \sum_j b_{ij} y_j \end{cases}.$$

$$\begin{aligned}
a_{ik} &= \sum_j a_{ij} \beta(x_j, y_k) = \beta([xx_i], y_k) && \text{by the definition of } \beta \\
\text{Then} \quad &= \beta(-[x_i x], y_k) && \text{by the skew-symmetry of bracket} \\
&= \beta(x_i, -[xy_k]) && \text{by the associativity of } \beta \\
&= -\sum_j b_{kj} \beta(x_i, y_j) && = -b_{ki}
\end{aligned}$$

$L \rightarrow \mathfrak{gl}(V)$ is any representation of L , write $c_\phi(\beta) = \sum_i \phi(x_i) \phi(y_i) \in \text{End} V$ where

$$\begin{aligned}
[\phi(x) c_\phi(\beta)] &= \sum_i [\phi(x) \phi(x_i)] \phi(y_i) + \sum_i \phi(x_i) [\phi(x) \phi(y_i)] \\
x_i, y_i \text{ run over dual bases relative to } \beta. \text{ Then} &= \sum_{ij} a_{ij} \phi(x_j) \phi(y_i) + \sum_{ij} b_{ij} \phi(x_i) \phi(y_j) \\
&= 0
\end{aligned}$$

Thus $c_\phi(\beta)$ is an endomorphism of V commuting with $\phi(L)$.

$\rightarrow c_\phi(\beta)$ is called the Casimir element of ϕ and its trace is $\sum_i T(\phi(x_i) \phi(y_i)) = \sum_i \beta(x_i, y_i) = \dim L$. If ϕ is also irreducible, by Schur's lemma, it should be scalar.

e.g. $L = \mathfrak{sl}(2, F)$, $V = F^2$, ϕ is the identity map $L \rightarrow \mathfrak{gl}(V)$. Let (x, h, y) be the standard basis of L . Then its dual basis relative to the trace form is $(y, h/2, x)$ so

$$c_\phi = xy + \frac{h^2}{2} + yx = \begin{bmatrix} 3/2 & 0 \\ 0 & 3/2 \end{bmatrix}.$$

When ϕ is not faithful, we can factor out $\ker \phi$ and get L' the sum of the remaining simple ideals after factoring out $\ker \phi$. Then $\phi|_{L'}$ is faithful and we can get the Casimir element for L' . Then $\phi(L) = \phi(L')$ commute with the Casimir element.

Lemma 2.2. *Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra L . Then $\phi(L) \subset \mathfrak{sl}(V)$. In particular, L acts trivially on any one dimensional L -module.*

pf. $L = [LL]$ and $\mathfrak{sl}(V)$ is the derived algebra of $\mathfrak{gl}(V)$, the lemma follows.

Theorem 2.4 (Weyl). *Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a finite dimensional representation of a semisimple Lie algebra. Then ϕ is completely reducible.*

pf.

- V has an L -submodule W of codimension 1

Since by the previous lemma, L acts trivially on V/W , wlog we may denote this module F and get a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow F \rightarrow 0$.

– **CLAIM : wlog we can assume W is an irreducible L -module.**

Let W' be a proper nonzero submodule of W . Then we get another s.e.s. $0 \rightarrow W/W' \rightarrow V/W' \rightarrow F \rightarrow 0$. By the inductive hypothesis, it splits, so there is a one dimensional L -submodule of V/W' complementary to W/W' . Let it be $\tilde{W}/W'$. Then there exists another s.e.s $0 \rightarrow$

$W' \rightarrow \tilde{W} \rightarrow F \rightarrow 0$ with $\dim W' < \dim W$, so by the inductive hypothesis, there exists a submodule X complementary to W' , i.e. $\tilde{W} = W' \oplus X$. But $V/W' = W/W' \oplus \tilde{W}/W'$ so $V = W \oplus X$, given $W \cap X = 0$ and dimension matching.

Let $c = c_\phi$ be the Casimir element of ϕ . Since c commutes with $\phi(L)$, c is actually an L -module endomorphism of V . In particular, $c(W) \subset W$ and $\ker c$ is an L -submodule of V . Because L acts trivially on V/W , c must do likewise as a linear combination of products of elements $\phi(x)$. So c has trace 0 on V/W . On the other hand, c acts as a scalar on the irreducible L -submodule W by Schur's lemma. This cannot be 0, because that would force $T|_V(c) = 0$, a contradiction. Thus $\ker c$ is a one-dimensional L -submodule of V which intersects W trivially, which is the desired complement of W .

- General case

Let W be a nonzero submodule of V . Then there exists a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow V/W \rightarrow 0$. Let $\text{hom}(V, W)$ be the space of linear maps $V \rightarrow W$, viewed as L -module. Let $\mathcal{V}$ be the subspace of $\text{hom}(V, W)$ consisting of those maps whose restriction to W is a scalar multiplication. Then $f|_W = a1_W, \forall x \in L, \forall w \in W, (xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0 \rightarrow (xf)|_W = 0$ so $\mathcal{V}$ is an L -submodule. Let $\mathcal{W}$ be the subspace of $\mathcal{V}$ consisting of those f whose restriction to W is zero. Then $\mathcal{W}$ is also an L -submodule and L maps $\mathcal{V}$ to $\mathcal{W}$. Moreover $\mathcal{V}/\mathcal{W}$ has dimension one because $f \in \mathcal{V}$ is determined by the scalar $f|_W \bmod \mathcal{W}$. This gives the first situation. Then $\mathcal{V}$ has a one-dimensional submodule complementary to $\mathcal{W}$. Let $f : V \rightarrow W$ span it, so after multiplying by a nonzero scalar, we may assume $f|_W = 1_W$. Since f maps V into W and acts as 1_W on W , we can say $V = W \oplus \ker f$.

Theorem 2.5. *Let $L \subset \mathfrak{gl}(V)$ be a semisimple linear Lie algebra with V finite dimensional. Then L contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and usual Jordan decompositions in L coincide.*

pf. Let $x \in L$ arbitrary, with Jordan decomposition $x = x_s + x_n$ in $\mathfrak{gl}(V)$. Since $\text{adx}(L) \subset L$, $\begin{cases} \text{adx}_s(L) \subset L \\ \text{adx}_n(L) \subset L \end{cases}$ where $\text{ad} = \text{ad}_{\mathfrak{gl}(V)}$, i.e. $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$, which is a Lie subalgebra of $\mathfrak{gl}(V)$ including L as an ideal. If W is any L -submodule of V , define $L_W = \{y \in \mathfrak{gl}(V) | y(W) \subset W, T(y|_W) = 0\}$. For example, $L_V = \mathfrak{sl}(V)$. Since $L = [LL]$, L lies in all such L_W . Set $L' =$ intersection of N with all spaces L_W . Then L' is a subalgebra of N including L as an ideal. Then if $x \in L$, x_s, x_n also lie in L_W and therefore in L' .

- **CLAIM :** $L = L'$.

pf. L' being finite dimensional L -module, by Weyl's theorem $L' = L + M$ for some L -submodule M , where the sum is direct. But $[LL'] \subset L$ given $L' \subset L$ so the action of L on M is trivial. Let W be any irreducible L -submodule of V . If $y \in M$, $[Ly] = 0$ so by Schur's lemma, y acts on W as a scalar. On the other hand, $T(y|_W) = 0$ because $y \in L_W$. Therefore y acts on W as zero. V can be written as a direct sum of irreducible L -submodules by Weyl's theorem so $y = 0$. Thus $M = 0$, $L = L'$.

Corollary 2.2. *Let L be a semisimple Lie algebra, $\phi : L \rightarrow \mathfrak{gl}(V)$ a finite dimensional representation of L . If $x = s + n$ is the abstract Jordan decomposition of $x \in L$, $\phi(x) = \phi(s) + \phi(n)$ is the usual Jordan decomposition of $\phi(x)$.*

pf. The algebra $\phi(L)$ is spanned by the eigenvectors of $\text{ad}_{\phi(L)}\phi(s)$ since L has this property relative to ads , and therefore $\text{ad}_{\phi(L)}\phi(s)$ is semisimple. Similarly, $\text{ad}_{\phi(L)}\phi(n)$ is nilpotent and it commutes with $\text{ad}_{\phi(L)}\phi(s)$. Thus $\phi(x) = \phi(s) + \phi(n)$ is the abstract Jordan decomposition of $\phi(x)$ in the semisimple Lie algebra $\phi(L)$. Then by the previous theorem, the corollary follows.

Definition 2.8. *A Lie algebra L for which $\text{Rad}L = Z(L)$ is called reductive.*

e.g. $\left\{ \begin{array}{l} L \text{ abelian} \\ L = \mathfrak{gl}(n, F) \end{array} \right\}$, then it is reductive.

- **If L is reductive, L is completely reducible $\text{ad}L$ -module.**

pf. By the hypothesis $L/Z(L)$ is semisimple. Since $\text{ad}L$ acts trivially on $Z(L)$, the adjoint representation of L on L factors through $L/Z(L)$. Thus L is an $L/Z(L)$ -module. $\phi : L/Z(L) \rightarrow \mathfrak{gl}(V)$ is a representation of a semisimple Lie algebra. Then ϕ is completely reducible by Weyl's theorem. Thus L is completely reducible.

- **If L is a completely reducible $\text{ad}L$ -module, then L is reductive.**

pf. There is an adjoint representation $\phi : L \rightarrow \mathfrak{gl}(L)$, let $R = \text{Rad}L$ be its radical. Then since it is an ideal of L , it is an $\text{ad}L$ -submodule of L . Then, $L = R \oplus S$ since L is completely reducible. Because $\text{ad}L$ -submodules of the adjoint module are ideals, S is also an ideal. Thus $\left\{ \begin{array}{l} [RS] \subset R \\ [RS] \subseteq S \end{array} \right\}$. Thus $[RS] \subseteq R \cap S = 0 \rightarrow [RS] = 0$. Since R is solvable, $[RR] \subsetneq R$ unless $R = 0$ and $[RR]$ is also an $\text{ad}L$ -submodule. By complete reducibility, choose an $\text{ad}L$ -submodule $T \subseteq R$ s.t. $R = [RR] \oplus T$. Given T an ideal, $\left\{ \begin{array}{l} [RT] \subseteq T \\ [RT] \subseteq [RR] \end{array} \right\} \rightarrow$

$[RT] = 0$, $R = [RR] \oplus T$. Thus $[RR] = [R[RR]]$. If $[RR] \neq 0$, $R^{(1)} = R^{(2)} = \dots$, a contradiction. Thus $[RR] = 0$ and $R \subseteq Z(L)$. Since $Z(L) \subseteq R$ the stmt follows.

- **If L is reductive, then all finite dimensional representations of L in which $Z(L)$ is represented by semisimple endomorphisms are completely reducible.**

pf. By the first stmt, it is a completely reducible $\text{ad}L$ -module and $L = Z(L) \oplus [LL]$ where $[LL]$ is semisimple. Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be finite dimensional with every element of $Z(L)$ represented semisimply. By Weyl's theorem, it is completely reducible as $[LL]$ -module. $[Z(L), L] = 0$ implies $[\phi(z)\phi(x)] = 0$, $\forall z \in Z(L)$, $\forall x \in [LL]$. Since $\phi(z)$ are commuting semisimple endomorphisms over an algebraically closed field they are simultaneously diagonalisable. Thus $V = \bigoplus_{\lambda} V_{\lambda}$ where each V_{λ} is a common eigenspace for $Z(L)$. Since $\phi(z)$ commute with $\phi([LL])$, $xV_{\lambda} \subseteq V_{\lambda}$, $\forall x \in [LL]$. So V_{λ} is an $[LL]$ -module and by Weyl's theorem $V_{\lambda} = \bigoplus_j V_{\lambda,j}$ where $V_{\lambda,j}$ are irreducible $[LL]$ -modules. But $Z(L)$ acts on all $V_{\lambda,j}$ by the scalar $\lambda(z)$, $V_{\lambda,j}$ is actually an L -submodule. If it is a proper L -submodule, then it is an $[LL]$ -submodule, contradicting irreducibility. Hence $V = \bigoplus_{\lambda,j} V_{\lambda,j}$ is a decomposition into irreducible L -modules.

2.3 Representations of $\mathfrak{sl}(2, F)$

Let L denote $\mathfrak{sl}(2, F)$ whose standard basis consists of $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $[hx] = 2x$, $[hy] = -2y$, $[xy] = h$.

Let V be an arbitrary L -module. Since h is semisimple, h acts diagonally on V by the previous corollary. Then V is decomposed as direct sum of eigenspaces $V_{\lambda} = \{v \in V | hv = \lambda v\}$, $\lambda \in F$. Whenever $V_{\lambda} \neq 0$, λ is called a weight of h in V and we call V_{λ} a weight space.

Lemma 2.3. If $v \in V_{\lambda}$, $\begin{cases} xv \in V_{\lambda+2} \\ yv \in V_{\lambda-2} \end{cases}$.

pf. $\begin{cases} h(xv) = [hx]v + xhv = 2xv + \lambda xv = (\lambda + 2)xv \\ h(yv) = [hy]v + yhv = -2yv + \lambda yv = (\lambda - 2)yv \end{cases}$
 $\rightarrow x, y$ are represented by nilpotent endomorphisms of V .

Since $\dim V < \infty$ and $V = \bigoplus_{\lambda \in F} V_{\lambda}$ is direct, there must exist $V_{\lambda} \neq 0$ s.t. $V_{\lambda+2} = 0$. For such λ , any nonzero vector in V_{λ} will be called a maximal vector of weight

λ .

Assume V is an irreducible L -module. Choose a maximal vector, say $v_0 \in V$ and set $v_{-1} = 0$, $v_i = \frac{1}{i!}y^i v_0$, $\forall i \geq 0$.

Lemma 2.4. $\begin{cases} (i) & hv_i = (\lambda - 2i)v_i \\ (ii) & yv_i = (i+1)v_{i+1} \\ (iii) & xv_i = (\lambda - i + 1)v_{i-1} \end{cases}$

pf.

- for $i = 0$, $hv_0 = \lambda v_0$
- $i > 0$, $hv_i = \frac{1}{i}hyv_{i-1} = \frac{1}{i}([hy]v_{i-1} + yhv_{i-1}) = \frac{1}{i}(-2yv_{i-1} + (\lambda - 2(i-1))yv_{i-1}) = \frac{1}{i}((\lambda - 2i)yv_{i-1}) = (\lambda - 2i)v_i$
- $(i+1)v_{i+1} = (i+1)\frac{1}{(i+1)!}y^{i+1}v_0 = yv_i$
- $i = 0 \rightarrow xv_0 = (\lambda + 1)v_{-1} = (\lambda + 1)0 = 0$
 $ixv_i = xyv_{i-1} = [xy]v_{i-1} + yxv_{i-1}$ definition of bracket
 $= hv_{i-1} + yxv_{i-1}$
 $i > 0 \rightarrow = (\lambda - 2(i-1))v_{i-1} + (\lambda - i + 2)yv_{i-2}$ i and inductive hypothesis
 $= (\lambda - 2i + 2)v_{i-1} + (i-1)(\lambda - i + 2)v_{i-1}$ by (ii)
 $= i(\lambda - i + 1)v_{i-1}$

$\rightarrow$ (i) implies the nonzero v_i are all linearly independent. But $\dim V < \infty$. Let m be the smallest integer for which $\begin{cases} v_m \neq 0 \\ v_{m+1} = 0 \end{cases}$. Then the subspace of V with basis $(v_0, \dots, v_m)$ is an L -submodule different from 0. Because V is irreducible, this subspace must be all of V and moreover relative to the ordered basis $(v_0, \dots, v_m)$, the matrices of the endomorphisms representing x, y, h can be written down explicitly. Also for $i = m+1$, $xv_i = 0$ in the LHS of (iii), whereas the RHS is $(\lambda - m)v_m$. Since $v_m \neq 0$, $\lambda = m$, i.e. the weight of a maximal vector is a nonnegative integer. We call it the highest weight of V . Moreover each weight μ occurs with multiplicity one by (i). In particular, since V determines λ uniquely, the maximal vector v_0 is the only possible one in V .

Theorem 2.6. *Let V be an irreducible module for $L = \mathfrak{sl}(2, F)$.*

1. *Relative to h , V is the direct sum of weight spaces V_μ , $\mu = m, m-2, \dots, -(m-2), -m$ where $m+1 = \dim V$ and $\dim V_\mu = 1$ for each such μ .*
2. *V has up to nonzero scalar multiples a unique maximal vector, whose weight is m .*
3. *The action of L on V is given explicitly by the above formulas, if the basis is chosen in the prescribed fashion. In particular, there exists at most one irreducible L -module up to isomorphism of each possible dimension $m+1$, $m \geq 0$.*

pf. By the preceding paragraph.

Corollary 2.3. *Let V be any finite dimensional L -module, $L = \mathfrak{sl}(2, F)$. Then the eigenvalues of h on V are all integers and each occurs along with its negative. Moreover, in any decomposition of V into direct sum of irreducible submodules, the number of summands is precisely $\dim V_0 + \dim V_1$.*

pf. If $V = 0$ then nothing to prove. If $V \neq 0$, by Weyl's theorem, we can write V as direct sum of irreducible submodules. Then by the previous theorem, the first part of the corollary follows. Each irreducible L -module has a unique occurrence of either the weight 0 or else the weight 1, but not both so the last part of the corollary follows.

Existence of the maximal vector

pf. Look at the subalgebra B spanned by h, x . It is a solvable subalgebra of $\mathfrak{gl}(V)$ with $\dim V < \infty$. Since $V \neq 0$, V contains a common eigenvector v for all the endomorphisms in B by Lie's theorem ($[BB] = Fx$, $[FxFx] = 0$).

• **CLAIM : v is a maximal vector.**

pf. Since v is a common eigenvector for B , $\exists \lambda, \mu \in F$ s.t. $\begin{cases} hv = \lambda v \\ xv = \mu v \end{cases}$. Then $2xv = [hx]v = hxv - xhv = h(\mu v) - x(\lambda v) = \mu\lambda v - \lambda\mu v = 0$. Since $\text{char } F = 0$, $xv = 0$ so v is a maximal vector.

2.4 Root Space Decomposition

Let L denote a semisimple Lie algebra. If L consisted entirely of nilpotent elements, then L would be nilpotent by Engel's theorem. This not being the case, we can find $x \in L$ whose semisimple part x_s in the abstract Jordan decomposition is nonzero. Then L possesses nonzero subalgebras consisting of semisimple elements. Call such a subalgebra **toral**.

Lemma 2.5. *A toral subalgebra of L is abelian.*

pf. Let T be toral. Since $\text{ad } x$, $x \in T$ is diagonalisable since $\text{ad } x$ is semisimple and F is algebraically closed, then if $\text{ad}_T x$ has no nonzero eigenvalues, then $\text{ad}_T x = 0$, $\forall x \in T$. Suppose the contrary. Then $\exists 0 \neq y \in T$ s.t. $[xy] = ay$, $a \neq 0$. Then $\text{ad}_T y(x) = -ay$ shows that $\text{ad}_T y(x)$ is an eigenvector of $\text{ad}_T y$ of eigenvalue 0 ($\text{ad}_T y(\text{ad}_T y(x)) = \text{ad}_T y(-ay) = 0$). On the other hand, we can write x as a linear combination of eigenvectors of $\text{ad}_T y$ since y is semisimple. Thus after applying $\text{ad}_T y$ to x , all that is left is a combination of eigenvectors which belong to nonzero eigenvalues, if any, a contradiction. Thus the lemma follows.

e.g. $L = \mathfrak{sl}(2, F)$. Let H be a maximal toral subalgebra. Then $x, y \notin H$ since they are nilpotent. But $Fh \subset H$. This forces $Fh = H$.

Fix a maximal toral subalgebra H of L , i.e. a toral subalgebra not properly included in any other. Since H is abelian by the previous lemma, $\text{ad}_L H$ is a commuting family of semisimple endomorphisms of L . Then $\text{ad}_L H$ is simultaneously diagonalisable, i.e. L is the direct sum of the subspaces $L_\alpha = \{x \in L \mid [hx] = \alpha(h)x, \forall h \in H\}$, where α ranges over H^* . Here $L_0 = C_L(H)$, the centraliser of H . Then $H \subseteq L_0$ by the lemma. The set of all nonzero $\alpha \in H^*$ for which $L_\alpha \neq 0$ is denoted by Φ .

Definition 2.9. *The elements of Φ are called the roots of L relative to H and a root space decomposition is*

$$L = C_L(H) \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$$

Proposition 2.1. $\forall \alpha, \beta \in H^*, [L_\alpha L_\beta] \subset L_{\alpha+\beta}$. If $x \in L_\alpha, \alpha \neq 0$, then adx is nilpotent. If $\alpha, \beta \in H^*$ and $\alpha + \beta \neq 0$, then L_α is orthogonal to L_β relative to the Killing form κ of L .

pf.

- $\forall \begin{cases} x \in L_\alpha \\ y \in L_\beta \end{cases}, h \in H$ implies

$$\begin{aligned} \text{adh}([xy]) &= [h[xy]] = -[x[yh]] - [y[hx]] && \text{by the Jacobi identity} \\ &= [[hx]y] + [x[hy]] && \text{by the skew-symmetry of bracket} \\ &= \alpha(h)[xy] + \beta(h)[xy] \\ &= (\alpha + \beta)(h)[xy] \end{aligned}$$
- find $h \in H$ for which $(\alpha + \beta)(h) \neq 0$. Then if $x \in L_\alpha, y \in L_\beta, \kappa([hx], y) = -\kappa([xh], y) = -\kappa(x, [hy]) \Leftrightarrow \alpha(h)\kappa(x, y) = -\beta(h)\kappa(x, y) \Leftrightarrow (\alpha + \beta)(h)\kappa(x, y) = 0$, which forces $\kappa(x, y) = 0$.

Corollary 2.4. *The restriction of the Killing form to $L_0 = C_L(H)$ is nondegenerate.*

pf. κ is nondegenerate. But L_0 is orthogonal to $L_\alpha, \forall \alpha \in \Phi$ by the previous proposition. If $z \in L_0$ is orthogonal to L_0 , then $\kappa(z, L) = 0 \rightarrow z = 0$.

Proposition 2.2. *Let H be a maximal toral subalgebra of L . Then $H = C_L(H)$.*

pf. Denote $C = C_L(H)$.

- **C contains the semisimple and nilpotent parts of its elements.**

pf. $x \in C_L(H)$ then adx maps the subspace of H of L into the subspace 0. Then $\begin{cases} (\text{adx})_s \\ (\text{adx})_n \end{cases}$ have the same property by the proposition 1.4. Also $\begin{cases} (\text{adx})_s = \text{adx}_s \\ (\text{adx})_n = \text{adx}_n \end{cases}$, so the claim follows.

- **All semisimple elements of C lie in H .**

pf. If x is semisimple and centralises H , $H + Fx$ is toral since the sum of commuting semisimple elements are again semisimple. By the maximality of H , $H + Fx = H \rightarrow x \in H$.

- **The restriction of κ to H is nondegenerate.**

pf. Let $\kappa(h, H) = 0$ for some $h \in H$. If $x \in C$ is nilpotent, then $\begin{cases} [xH] = 0 \\ \text{adx nilpotent} \end{cases}$ implies $T(\text{adx} \text{ad} y) = 0$, $\forall y \in H$ or $\kappa(x, H) = 0$ by the previous lemma. Then the previous statements imply $\kappa(h, C) = 0$ whence $h = 0$.

- **C is nilpotent.**

pf. If $x \in C$ is semisimple, $x \in H$ by the second statement, and $\text{ad}_C x = 0$ is certainly nilpotent. On the other hand, if $x \in C$ is nilpotent, then $\text{ad}_C x$ is a fortiori nilpotent. Now let $x \in C$ be arbitrary, $x = x_s + x_n$. Since $x_s, x_n \in C$ by the first statement, $\text{ad}_C x$ is the sum of commuting nilpotents and is therefore itself nilpotent. Then by Engel's theorem, C is nilpotent.

Since κ is associative and $[HC] = 0$, $\kappa(H, [CC]) = 0$. Then since the restriction of κ to H is nondegenerate, $H \cap [CC] = 0$. If C is not abelian, $[CC] \neq 0$. Then since C is nilpotent, $Z(C) \cap [CC] \neq 0$ by the lemma 1.2. Let $z \neq 0$ lie in this intersection. Then z cannot be semisimple. Its nilpotent part n is therefore nonzero and lies in C , hence also lies in $Z(C)$. Then the lemma implies $\kappa(n, C) = 0$, a contradiction to the previous corollary. So C is abelian. If $C \neq H$, C contains a nonzero nilpotent element x by the first and second statements. Then $\kappa(x, y) = T(\text{adx} \text{ad} y) = 0$, $\forall y \in C$, a contradiction to the previous corollary. So $C = H$ and the proposition follows.

Corollary 2.5. *The restriction of κ to H is nondegenerate.*

$$\rightarrow H \sim H^*$$

$$\forall \phi \in H^*, \exists ! t_\phi \in H \text{ s.t. } \phi(h) = \kappa(t_\phi, h), \forall h \in H.$$

Proposition 2.3. 1. Φ spans H^*

2. If $\alpha \in \Phi$, $-\alpha \in \Phi$
3. Let $\alpha \in \Phi, x \in L_\alpha, y \in L_{-\alpha}$. Then $[xy] = \kappa(x, y)t_\alpha$
4. If $\alpha \in \Phi$, $[L_\alpha L_{-\alpha}]$ is one dimensional with basis t_α
5. $\alpha(t_\alpha) = \kappa(t_\alpha, t_\alpha) \neq 0, \forall \alpha \in \Phi$
6. If $\alpha \in \Phi$, and x_α is any nonzero element of L_α , then there exists $y_\alpha \in L_{-\alpha}$ s.t. $x_\alpha, y_\alpha, h_\alpha = [x_\alpha y_\alpha]$ spans a three dimensional simple subalgebra of L isomorphic to $\mathfrak{sl}(2, F)$ via $x_\alpha \mapsto \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, y_\alpha \mapsto \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, h_\alpha \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
7. $h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$

pf.

- If Φ fails to span H^* , by the duality, $\exists h \in H$ s.t. $\alpha(h) = 0, \forall \alpha \in \Phi$. Then $[hL_\alpha] = 0, \forall \alpha \in \Phi$. Since $[hH] = 0, [hL] = 0$, i.e. $h \in Z(L) = 0$ a contradiction.
- Let $\alpha \in \Phi$. If $-\alpha \notin \Phi$, $\kappa(L_\alpha, L_\beta) = 0, \forall \beta \in H^*$ by the proposition 2.1. Therefore $\kappa(L_\alpha, L) = 0$, a contradiction to the nondegeneracy of κ
- Let $\alpha \in \Phi, x \in L_\alpha, y \in L_{-\alpha}$. Let $h \in H$ be arbitrary. Then

$$\begin{aligned} \kappa(h, [xy]) &= \kappa([hx], y) && \text{by the associativity of } \kappa \\ &= \alpha(h)\kappa(x, y) && = \kappa(t_\alpha, h)\kappa(x, y) \\ &= \kappa(\kappa(x, y)t_\alpha, h) && \text{by the bilinearity of } \kappa \\ &= \kappa(h, \kappa(x, y)t_\alpha) && \text{by the symmetry of } \kappa \end{aligned} \Leftrightarrow H \perp [xy] - \kappa(x, y)t_\alpha \text{ and}$$
 therefore $[xy] = \kappa(x, y)t_\alpha$ by the previous corollary.
- By 3, t_α spans $[L_\alpha L_{-\alpha}]$, provided $[L_\alpha L_{-\alpha}] \neq 0$. Let $0 \neq x \in L_\alpha$. If $\kappa(x, L_{-\alpha}) = 0, \kappa(x, L) = 0$ a contradiction to the nondegeneracy of κ . Thus $\exists 0 \neq y \in L_{-\alpha}$ s.t. $\kappa(x, y) \neq 0$. Then $[xy] \neq 0$.
- Suppose the contrary. Then $\alpha(t_\alpha) = 0$ so that $[t_\alpha x] = 0 = [t_\alpha y], \forall x \in L_\alpha, \forall y \in L_{-\alpha}$. By 4, $\exists x \in L_\alpha, y \in L_{-\alpha}$ s.t. $\kappa(x, y) \neq 0$. By modifying scalar, we may assume $\kappa(x, y) = 1$. Then $[xy] = t_\alpha$ by 3. Then the subspace of L spanned by x, y, t_α is a three dimensional solvable algebra, $S \cong \text{ad}_L S \subset \mathfrak{gl}(L)$. In particular $\text{ad}_L S$ is nilpotent $\forall s \in [SS]$. So $\text{ad}_L t_\alpha$ is both semisimple and nilpotent, i.e. $\text{ad}_L t_\alpha = 0$. This implies $t_\alpha \in Z(L) = 0$, a contradiction. Thus $\alpha(t_\alpha) \neq 0$.
- Given $x_\alpha \in L_\alpha, \exists y \in L_{-\alpha}$ s.t. $\kappa(x_\alpha, y_\alpha) = \frac{2}{\kappa(t_\alpha, t_\alpha)}$ by 5 and the fact that $\kappa(x_\alpha, L_{-\alpha}) \neq 0$. Set $h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$. Then $[x_\alpha y_\alpha] = h_\alpha$ by 3. Also $[h_\alpha x_\alpha] = \frac{2}{\alpha(t_\alpha)}[t_\alpha x_\alpha] = \frac{2\alpha(t_\alpha)}{\alpha(t_\alpha)}x_\alpha = 2x_\alpha$ and similarly $[h_\alpha y_\alpha] = -2y_\alpha$. So $x_\alpha, y_\alpha, h_\alpha$ span a three dimensional subalgebra of L with the same multiplication table as $\mathfrak{sl}(2, F)$.

- $\kappa(t_\alpha, h) = \alpha(h)$ implies $t_\alpha = -t_{-\alpha}$.

For each pair of roots $\pm\alpha$, let $S_\alpha \cong \mathfrak{sl}(2, F)$ be a subalgebra of L constructed as in the previous proposition. Then by Weyl's theorem and theorem 2.6., we have a complete description of all finite dimensional S_α -modules.

Fix $\alpha \in \Phi$. Consider the subspace M of L spanned by H along with all root spaces of the form $L_{c\alpha}$, $c \in F^*$. This is an S_α submodule of L by the proposition 2.1.

- Proposition 2.4.** 1. $\alpha \in \Phi$ implies $\dim L_\alpha = 1$. In particular, $S_\alpha = L_\alpha + L_{-\alpha} + H$ and for given nonzero $x_\alpha \in L_\alpha$, there exists a unique $y_\alpha \in L_{-\alpha}$ s.t. $[x_\alpha y_\alpha] = h_\alpha$.
2. if $\alpha \in \Phi$, the only scalar multiples of α which are roots are $\pm\alpha$
3. If $\alpha, \beta \in \Phi$, $\beta(h_\alpha) \in \mathbb{Z}$ and $\beta - \beta(h_\alpha)\alpha \in \Phi$. Here $\beta(h_\alpha)$ are called Cartan integers.
4. If $\alpha, \beta, \alpha + \beta \in \Phi$, $[L_\alpha L_\beta] = L_{\alpha+\beta}$
5. Let $\alpha, \beta \in \Phi$ s.t. $\beta \neq \pm\alpha$. Let r, q be the largest integers for which $\beta - r\alpha, \beta + q\alpha$ are roots respectively. Then all $\beta + i\alpha \in \Phi$ and $\beta(h_\alpha) = r - q$.

pf.

- 1,2

The weights of h_α on M are integers 0 and $2c = c\alpha(h_\alpha)$. In particular, all c occurring here must be integral multiples of $1/2$. Given $\ker \alpha \subset H$, S_α acts trivially on $\ker \alpha$, a subspace of codimension 1 in H complementary to Fh_α , while on the other hand S_α is itself an irreducible S_α -submodule of M . Thus $\ker \alpha, S_\alpha$ exhaust the occurrences of the weight 0 for h_α . Therefore the only even weights occurring in M are $0, \pm 2$. This proves 2α is not a root, i.e. twice a root is never a root. But then $\frac{\alpha}{2}$ cannot be a root either, so 1 cannot occur as a weight of h_α in M . Since $M = H + S_\alpha$ by the corollary 2.3., $\dim L_\alpha = 1$ and the only multiples of a root α which are roots are $\pm\alpha$.

- 3,4

Let $\beta \neq \pm\alpha$. Set $K = \sum_{i \in \mathbb{Z}} L_{\beta+i\alpha}$. Then each root space is one dimensional and no $\beta + i\alpha$ can equal 0. So K is an S_α -submodule of L , with one dimensional weight spaces for the distinct integral weights $\beta(h_\alpha) + 2i$ where $\beta + i\alpha \in \Phi$, $i \in \mathbb{Z}$. Not both 0 and 1 can occur as weights of this form so by the corollary 2.3., K is irreducible. The highest weight must be $\beta(h_\alpha) + 2q$ and the lowest must

be $\beta(h_\alpha) - 2r$ if q, r is the largest integer for which $\beta + q\alpha$ or $\beta - r\alpha$ respectively is a root. Moreover, the weights on K form an arithmetic progression with difference 2 by the theorem 2.6., and therefore roots $\beta + i\alpha$ form a string $\beta - r\alpha, \dots, \beta, \dots, \beta + q\alpha$. Then $(\beta - r\alpha)(h_\alpha) = -(\beta + q\alpha)(h_\alpha) \Leftrightarrow \beta(h_\alpha) = r - q$. If $\alpha, \beta, \alpha + \beta \in \Phi$, then $\text{ad}L_\alpha$ maps L_β onto $L_{\alpha+\beta}$, i.e. $[L_\alpha L_\beta] = L_{\alpha+\beta}$.

Here bracket of two nilpotents $[x_\alpha y_\alpha] = h_\alpha$ gives the semisimple element. We can view nilpotent elements as shifting weights. In particular, here x_α shifts weight by α , while y_α shifts weight by $-\alpha$. Since by the proposition 2.1, $[L_\alpha L_\beta] \subset L_{\alpha+\beta}$, $[x_\alpha y_\alpha] \in L_{\alpha-\alpha} = L_0$, the abelian subalgebra where the weight is fixed.

Let L be a semisimple Lie algebra over the algebraically closed field F of characteristic 0, H be a maximal toral subalgebra, $\Phi \subset H^*$ be the set of roots of L relative to H and $L = H \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$ be the root space decomposition. Since the restriction to H of the Killing form is nondegenerate, we may transfer the form to H^* , letting $(\gamma, \delta) = \kappa(t_\gamma, t_\delta)$, $\forall \delta, \gamma \in H^*$. Since Φ spans H^* , choose a basis $\alpha_1, \dots, \alpha_l$ of H^* consisting of roots. If $\beta \in \Phi$, $\beta = \sum_{i=1}^l c_i \alpha_i$ for some $c_i \in F$.

CLAIM : $c_i \in \mathbb{Q}$

pf. $\forall j$, $(\beta, \alpha_j) = \sum_{i=1}^l c_i (\alpha_i, \alpha_j)$, so multiplying both sides by $\frac{2}{(\alpha_j, \alpha_j)}$ yields $\frac{2(\beta, \alpha_j)}{(\alpha_j, \alpha_j)} = \sum_i \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)} c_i$, a system of l equations in l unknowns c_i with integral coefficients. Since $(\alpha_1, \dots, \alpha_l)$ is a basis of H^* , and the form is nondegenerate, the matrix $((\alpha_i, \alpha_j))$ is nonsingular. So the same is true for the coefficient matrix of this system of equations. So the equations possess a unique solution over $\mathbb{Q}$.

$\rightarrow \mathbb{Q}$ -subspace $E_{\mathbb{Q}}$ of H^* spanned by all the roots has $\mathbb{Q}$ -dimension $l = \dim_F H^*$. Also $\forall \lambda, \mu \in H^*$, $(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \sum \alpha(t_\lambda) \alpha(t_\mu) = \sum (\alpha, \lambda) (\alpha, \mu)$ summing over $\alpha \in \Phi$. In particular, for $\beta \in \Phi$, $(\beta, \beta) = \sum (\alpha, \beta)^2$. So dividing both sides by $(\beta, \beta)^2$ gives $\frac{1}{(\beta, \beta)} = \sum \frac{(\alpha, \beta)^2}{(\beta, \beta)^2} = \frac{1}{4} \sum \left(\frac{2(\alpha, \beta)}{(\beta, \beta)} \right)^2 \in \mathbb{Q}$ given $\frac{2(\alpha, \beta)}{(\beta, \beta)} \in \mathbb{Z}$ by the previous proposition 3. Therefore $(\beta, \beta) \in \mathbb{Q}$, $(\alpha, \beta) \in \mathbb{Q}$. Thus all inner products of vectors in $E_{\mathbb{Q}}$ are rational, so we get a nondegenerate form on $E_{\mathbb{Q}}$. Also $(\lambda, \lambda) = \sum (\alpha, \lambda)^2$ gives that the form on $E_{\mathbb{Q}}$ is positive definite.

3 Root System

3.1 Definition

Consider a fixed Euclidean space E , i.e. a finite dimensional vector space over $\mathbb{R}$ endowed with a positive definite symmetric bilinear form (α, β) .

Definition 3.1. A reflection in E is an invertible linear transformation leaving

pointwise fixed a hyperplane and sending any vector orthogonal to that hyperplane into its negative.

→ a reflection is orthogonal in the sense that it preserves the inner product on E .

Each nonzero $\alpha \in E$ determines a reflection σ_α s.t. $\sigma_\alpha(\beta) = \beta - \frac{2\langle\beta,\alpha\rangle}{\langle\alpha,\alpha\rangle}\alpha \equiv \beta - \langle\beta,\alpha\rangle\alpha$ with reflecting hyperplane $P_\alpha = \{\beta \in E | \langle\beta,\alpha\rangle = 0\}$.

Lemma 3.1. *Let Φ be a finite set which spans E . Suppose all reflections σ_α , $\forall \alpha \in \Phi$ leave Φ invariant. If $\sigma \in GL(E)$ leaves Φ invariant, fixes pointwise a hyperplane P of E and sends some nonzero $\alpha \in \Phi$ to its negative, then $\sigma = \sigma_\alpha$.*

pf. Let $\tau = \sigma\sigma_\alpha = \sigma\sigma_\alpha^{-1}$. Then $\begin{cases} \tau(\Phi) = \Phi \\ \tau(\alpha) = \alpha \end{cases}$ and τ acts as the identity on $\begin{cases} \text{the subspace } \mathbb{R}\alpha \\ \text{the quotient } E/\mathbb{R}\alpha \end{cases}$. So all eigenvalues of τ are 1 and the minimal polynomial of τ divides $(T-1)^l$ where $l = \dim E$. On the other hand, since Φ is finite, not all vectors $\beta, \tau(\beta), \dots, \tau^k(\beta)$ can be distinct, so some power of τ fixes β . Choose k large enough so that τ^k fixes all $\beta \in \Phi$. Because Φ spans E , $\tau^k = 1$ so the minimal polynomial of τ divides $T^k - 1$. In sum, τ has minimal polynomial $T - 1 = \gcd(T^k - 1, (T - 1)^l)$, i.e. $\tau = 1$ and the lemma follows.

Definition 3.2. *A subset Φ of the Euclidean space E is called a root system in E if Φ is finite, spans E and $0 \notin \Phi$ if the following axioms are satisfied $\begin{cases} \text{if } \alpha \in \Phi, \text{ the only multiples of } \alpha \text{ in } \Phi \text{ are } \pm\alpha \\ \text{if } \alpha \in \Phi, \text{ the reflection } \sigma_\alpha \text{ leaves } \Phi \text{ invariant} \\ \text{if } \alpha, \beta \in \Phi, \langle\beta, \alpha\rangle \in \mathbb{Z} \end{cases}$.*

Let Φ be a root system in E . Denote by $\mathcal{W}$ the subgroup of $GL(E)$ generated by the reflections σ_α , $\forall \alpha \in \Phi$. Then by the third axiom in the definition, $\mathcal{W}$ permutes the set Φ . This allows us to identify $\mathcal{W}$ with a subgroup of the symmetric group on Φ given the fact that only the identity fixes all the roots by the first axiom; in particular, $\mathcal{W}$ is finite. $\mathcal{W}$ is called the **Weyl group** of Φ .

Lemma 3.2. *Let Φ be a root system in E , with Weyl group $\mathcal{W}$. If $\sigma \in GL(E)$ leaves Φ invariant, then $\sigma\sigma_\alpha\sigma^{-1} = \sigma_{\sigma(\alpha)}$, $\forall \alpha \in \Phi$ and $\langle\beta, \alpha\rangle = \langle\sigma(\beta), \sigma(\alpha)\rangle$, $\forall \alpha, \beta \in \Phi$.*

pf. $\begin{aligned} \sigma\sigma_\alpha\sigma^{-1}(\sigma(\beta)) &= \sigma\sigma_\alpha(\beta) \in \Phi && \text{since } \sigma_\alpha(\beta) \in \Phi \\ &= \sigma(\beta - \langle\beta, \alpha\rangle\alpha) && \text{by the definition} \\ &= \sigma(\beta) - \langle\beta, \alpha\rangle\sigma(\alpha) \end{aligned}$. Since $\sigma(\beta)$ runs over Φ

as β runs over Φ , this implies $\sigma\sigma_\alpha\sigma^{-1}$ leaves Φ invariant while fixing pointwise the hyperplane $\sigma(P_\alpha)$ and sending $\sigma(\alpha)$ to $-\sigma(\alpha)$. Then by the previous lemma, $\sigma\sigma_\alpha\sigma^{-1} = \sigma_{\sigma(\alpha)}$. Then $\sigma_{\sigma(\alpha)}(\sigma(\beta)) = \sigma(\beta) - \langle\sigma(\beta), \sigma(\alpha)\rangle\sigma(\alpha) = \sigma(\beta) - \langle\beta, \alpha\rangle\sigma(\alpha)$ gives the last identity.

Definition 3.3. Call (Φ, E) , (Φ', E') isomorphic if there exists a vector space isomorphism $\phi : E \rightarrow E'$ sending Φ onto Φ' s.t. $\langle \phi(\beta), \phi(\alpha) \rangle = \langle \beta, \alpha \rangle$, $\forall \alpha, \beta \in \Phi$.

$\rightarrow \sigma_{\phi(\alpha)}(\phi(\beta)) = \phi(\sigma_\alpha(\beta))$ so there exists a natural isomorphism $\sigma \mapsto \phi \circ \sigma \circ \phi^{-1}$ of Weyl groups.

CLAIM : $\mathcal{W}$ is a normal subgroup of $\text{Aut}\Phi$, where $\text{Aut}\Phi$ is the group of automorphisms of Φ , i.e. isomorphisms of Φ onto itself.

pf. Since $\mathcal{W}$ is a subgroup of the symmetric group on Φ , and by the previous lemma, $\langle \sigma(\beta), \sigma(\alpha) \rangle = \langle \beta, \alpha \rangle$, $\mathcal{W} \subset \text{Aut}\Phi$. Also by the isomorphism defined in the previous stmt, it is a normal subgroup.

Definition 3.4. Define $\alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle}$. Call $\Phi^\vee = \{\alpha^\vee | \alpha \in \Phi\}$ the dual of Φ .

$\rightarrow$ it is a root system in E whose Weyl group is isomorphic to $\mathcal{W}$.

pf. By the duality, the first and second axiom of the root system follows trivially.

CLAIM : $\langle \alpha^\vee, \beta^\vee \rangle = \frac{2\langle \alpha^\vee, \beta^\vee \rangle}{\langle \beta^\vee, \beta^\vee \rangle} = \frac{2(\frac{2\alpha}{\langle \alpha, \alpha \rangle}, \frac{2\beta}{\langle \beta, \beta \rangle})}{(\frac{2\beta}{\langle \beta, \beta \rangle}, \frac{2\beta}{\langle \beta, \beta \rangle})} = \frac{2\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} = \langle \beta, \alpha \rangle$.

Thus, if $\alpha^\vee, \beta^\vee \in \Phi^\vee$, $\langle \beta^\vee, \alpha^\vee \rangle = \langle \alpha, \beta \rangle \in \mathbb{Z}$ since $\alpha, \beta \in \Phi$. Also, if $\alpha^\vee \in \Phi^\vee$, $\sigma_{\alpha^\vee}(\beta^\vee) = \beta^\vee - \langle \beta^\vee, \alpha^\vee \rangle \alpha^\vee = \frac{2\beta}{\langle \beta, \beta \rangle} - \langle \alpha, \beta \rangle \frac{2\alpha}{\langle \alpha, \alpha \rangle}$ Let $\gamma = \beta - \langle \beta, \alpha \rangle \alpha \in \Phi$. Then it equals $\beta -$

$\langle \alpha^\vee, \beta^\vee \rangle \alpha$ and $\gamma^\vee = \frac{2\gamma}{\langle \gamma, \gamma \rangle} = \frac{2\beta - 2\langle \beta, \alpha \rangle \alpha}{\langle \beta, \beta \rangle + \langle \beta, \alpha \rangle^2 \frac{2\alpha}{\langle \alpha, \alpha \rangle} - 2\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle} = \frac{2\beta - \frac{4\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha}{\langle \beta, \beta \rangle + 2\langle \beta, \alpha \rangle \langle \beta, \alpha \rangle - 2\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle} = \frac{2\beta}{\langle \beta, \beta \rangle} - \frac{4\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle \langle \beta, \beta \rangle} \alpha = \sigma_{\alpha^\vee}(\beta^\vee)$. Then it leaves $\Phi^\vee$ invariant. Thus it is a root system.

For all $\beta \in E$, $\sigma_{\alpha^\vee}(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha^\vee = \beta - \frac{2\langle \beta, \alpha^\vee \rangle}{\langle \alpha^\vee, \alpha^\vee \rangle} \frac{2\alpha}{\langle \alpha, \alpha \rangle} = \beta - \langle \beta, \alpha \rangle \alpha = \sigma_\alpha(\beta)$ and the Weyl group is isomorphic to $\mathcal{W}$.

Definition 3.5. Call $l = \dim E$ the rank of the root system Φ .

Given θ , the angle between vectors $\alpha, \beta \in E$, $\begin{cases} \langle \beta, \alpha \rangle = 2 \frac{\|\beta\|}{\|\alpha\|} \cos \theta \\ \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4 \cos^2 \theta \end{cases}$. This

narrows down the possibilities for the values for $\langle \alpha, \beta \rangle, \langle \beta, \alpha \rangle, \theta, (\frac{\|\beta\|}{\|\alpha\|})^2$. For non-proportional α, β with $\|\beta\| \geq \|\alpha\|$:

$\langle \alpha, \beta \rangle$	$\langle \beta, \alpha \rangle$	θ	$\ \beta\ ^2 / \ \alpha\ ^2$
0	0	$\pi/2$	undetermined
1	1	$\pi/3$	1
-1	-1	$2\pi/3$	1
1	2	$\pi/4$	2
-1	-2	$3\pi/4$	2
1	3	$\pi/6$	3
-1	-3	$5\pi/6$	3

Lemma 3.3. *Let α, β be nonproportional roots. If $(\alpha, \beta) > 0$, i.e. if the angle between α, β is strictly acute, then $\alpha - \beta$ is a root. If $(\alpha, \beta) < 0$, $\alpha + \beta$ is a root.*

pf. For $(\alpha, \beta) > 0$, it is possible iff $\langle \alpha, \beta \rangle$ or $\langle \beta, \alpha \rangle$ equals 1 by the above table.

$\begin{cases} \langle \alpha, \beta \rangle = 1 \rightarrow \sigma_\beta(\alpha) = \alpha - \beta \in \Phi \\ \langle \beta, \alpha \rangle = 1 \rightarrow \beta - \alpha \in \Phi \rightarrow \sigma_{\beta-\alpha}(\beta - \alpha) = \alpha - \beta \in \Phi \end{cases}$. So the first part follows. The second follows by the first with replacing β with $-\beta$.

3.2 Simple Roots

Let Φ denote a root system of rank l in a Euclidean space E with Weyl group $\mathcal{W}$.

Definition 3.6. *A subset Δ of Φ is called a base if*

$\begin{cases} \Delta \text{ is a basis of } E \\ \text{each root } \beta = \sum_{\alpha \in \Delta} k_\alpha \alpha \text{ with integral coefficients } k_\alpha \text{ all nonnegative or nonpositive} \end{cases}$.
The roots in Δ are called simple.

$\rightarrow$ define the height of a root by $\text{ht}\beta = \sum_{\alpha \in \Delta} k_\alpha$. If $\begin{cases} \forall k_\alpha \geq 0 \\ \forall k_\alpha \leq 0 \end{cases}$, β is called
 $\begin{cases} \text{positive} \\ \text{negative} \end{cases}$ and write $\begin{cases} \beta > 0 \\ \beta < 0 \end{cases}$. The collection of $\begin{cases} \text{positive} \\ \text{negative} \end{cases}$ roots is denoted
 $\begin{cases} \Phi^+ \\ \Phi^- \end{cases}$. If α, β are positive roots, and $\alpha + \beta$ is a root, it is also positive. Δ defines a partial order on E by $\beta < \alpha$ iff $\alpha - \beta$ is a sum of positive roots or $\beta = \alpha$.

Lemma 3.4. *If Δ is a base of Φ , then $(\alpha, \beta) \leq 0$, $\forall \alpha \neq \beta \in \Delta$ and $\alpha - \beta$ is not a root.*

pf. Assume the contrary. Then $(\alpha, \beta) > 0$. Since $\alpha \neq \beta$, by the hypothesis and $\alpha \neq -\beta$ trivially, by the previous lemma, $\alpha - \beta$ is a root, a contradiction to the second condition in the definition of a base.

Definition 3.7. $\forall \gamma \in E$, define $\Phi^+(\gamma) = \{\alpha \in \Phi \mid (\gamma, \alpha) > 0\}$ = the set of roots lying on the positive side of the hyperplane orthogonal to γ . Call $\gamma \in E$ regular if $\gamma \in E \setminus \cup_{\alpha \in \Phi} P_\alpha$, singular otherwise.

$\rightarrow$ if γ is regular, $\Phi = \Phi^+(\gamma) \cup -\Phi^+(\gamma)$.

Definition 3.8. Call $\alpha \in \Phi^+(\gamma)$ decomposable if $\alpha = \beta_1 + \beta_2$, $\beta_i \in \Phi^+(\gamma)$, indecomposable otherwise.

CLAIM : The intersection of positive open half-spaces associated with any basis $\gamma_1, \dots, \gamma_l$ of E is nonvoid.

pf. If δ_i is the projection of γ_i on the orthogonal complement of the subspace spanned by all basis vectors except γ_i , consider $\gamma = \sum r_i \delta_i$ with all $r_i > 0$. Here $\delta_i \perp \gamma_j$, $i \neq j$ so only one term survives which is positive. Then such γ belongs to the intersection of these half-spaces, which is therefore nonvoid.

Theorem 3.1. *Let $\gamma \in E$ be regular. Then the set $\Delta(\gamma)$ of all indecomposable roots in $\Phi^+(\gamma)$ is a base of Φ .*

pf.

- each root in $\Phi^+(\gamma)$ is a nonnegative $\mathbb{Z}$ -linear combination of $\Delta(\gamma)$.

pf. Assume the contrary. Then $\exists \alpha \in \Phi^+(\gamma)$ s.t. it cannot be written in such a way. Choose α so that (γ, α) is as small as possible. Then α cannot be in $\Delta(\gamma)$, so $\alpha = \beta_1 + \beta_2$, $\beta_i \in \Phi^+(\gamma)$ and therefore $(\gamma, \alpha) = (\gamma, \beta_1) + (\gamma, \beta_2)$. Then each of (γ, β_i) is positive, so β_1, β_2 must each be a nonnegative $\mathbb{Z}$ -linear combination of $\Delta(\gamma)$ by the minimality of (γ, α) , which implies so is α , a contradiction.

- if $\alpha, \beta \in \Delta(\gamma)$, $(\alpha, \beta) \leq 0$ unless $\alpha = \beta$.

pf. Assume the contrary. Then by the acute-angle lemma, $\alpha - \beta$ is a root, since β clearly cannot be $-\alpha$. So $\alpha - \beta$ or $\beta - \alpha$ is in $\Phi^+(\gamma)$. Then either $\alpha = \beta + (\alpha - \beta)$ is decomposable or $\beta = \alpha + (\beta - \alpha)$ is decomposable, a contradiction to $\alpha, \beta \in \Delta(\gamma)$.

- $\Delta(\gamma)$ is a linearly independent set.

pf. Suppose $\sum r_\alpha \alpha = 0$ for some $r_\alpha \in \mathbb{R}$. Separating the indices α for which $r_\alpha > 0$ from those for which $r_\alpha < 0$, we can rewrite $\sum s_\alpha \alpha = \sum t_\beta \beta$, $s_\alpha, t_\beta > 0$. Call $\epsilon = \sum s_\alpha \alpha$. Then $(\epsilon, \epsilon) = \sum_{\alpha, \beta} s_\alpha t_\beta (\alpha, \beta) \leq 0$ by the second stmt. This forces $\epsilon = 0$. Then $0 = (\gamma, \epsilon) = \sum s_\alpha (\gamma, \alpha)$ forcing $s_\alpha = 0$, $\forall \alpha$. Similarly, $t_\beta = 0$.

Since $\Phi = \Phi^+(\gamma) \cup -\Phi^+(\gamma)$, by the first stmt, it satisfies the second condition of the base and also implies it spans E . By the third stmt, it is linearly independent and therefore it is a basis for E and $\Delta(\gamma)$ is a base of Φ .

→ **each base Δ of Φ has the form $\Delta(\gamma)$ for some regular $\gamma \in E$.**

pf. Given Δ , select $\gamma \in E$ so that $(\gamma, \alpha) > 0$, $\forall \alpha \in \Delta$. This is possible since the intersection of positive open half-spaces associated with any basis of E is nonvoid.

Then γ is regular and $\begin{cases} \Phi^+ \subset \Phi^+(\gamma) \\ \Phi^- \subset -\Phi^+(\gamma) \end{cases}$ by the second condition in the definition of the base so equality must hold in each instance. Since $\Phi^+ = \Phi^+(\gamma)$, Δ consists of indecomposable elements, i.e. $\Delta \subset \Delta(\gamma)$. But $\text{Card} \Delta = \text{Card} \Delta(\gamma) = l$, so $\Delta = \Delta(\gamma)$.

Definition 3.9. *The connected components of $E \setminus \bigcup_{\alpha \in \Phi} P_\alpha$ are called the open Weyl chambers of E .*

→ each regular $\gamma \in E$ belongs to precisely one Weyl chamber, denoted $\mathcal{C}(\gamma)$.

Weyl chambers are in natural one to one correspondence with bases. ($\mathcal{C}(\gamma) = \mathcal{C}(\gamma')$ is equivalent to γ, γ' lie on the same side of each hyperplane $P_\alpha, \forall \alpha \in \Phi \Leftrightarrow \Phi^+(\gamma) = \Phi^+(\gamma') \Leftrightarrow \Delta(\gamma) = \Delta(\gamma')$). If $\Delta = \Delta(\gamma)$, $\mathcal{C}(\Delta) = \mathcal{C}(\gamma)$ is called the fundamental Weyl chamber relative to Δ . The Weyl group sends one Weyl chamber onto another, i.e. $\sigma(\mathcal{C}(\gamma)) = \mathcal{C}(\sigma\gamma)$, $\forall \sigma \in \mathcal{W}$, γ regular. On the other hand, $\mathcal{W}$ permutes bases, i.e. σ sends Δ to $\sigma(\Delta)$, which is again a base. So we have $\sigma(\Delta(\gamma)) = \Delta(\sigma\gamma)$ because $(\sigma\gamma, \sigma\alpha) = (\gamma, \alpha)$.

CLAIM : Let $\Phi^\vee$ be the dual system of Φ , $\Delta^\vee = \{\alpha^\vee | \alpha \in \Delta\}$. Then $\Delta^\vee$ is a base of $\Phi^\vee$.

pf. Let $\gamma \in E$ be regular, belong to the Weyl chamber $\mathcal{C}(\gamma)$. Then $\gamma^\vee$ belongs to the Weyl chamber $\mathcal{C}(\gamma^\vee)$, i.e. there is a one to one correspondence between Weyl chamber of Φ to ones of $\Phi^\vee$. Assume that $\exists \beta \in \Delta = \Delta(\gamma)$ s.t. $\beta^\vee$ not simple. Then by the lemma 3.5., $\exists \delta$ in the base of $\Phi^\vee$ s.t. $\beta^\vee - \delta$ is a positive root of $\Phi^\vee$. By duality, $\exists \mu \in \Phi$ s.t. $\frac{2\mu}{(\mu, \mu)} = \delta$. This implies $\frac{2\beta}{(\beta, \beta)} - \frac{2\mu}{(\mu, \mu)}$ is a root of $\Phi^\vee$ hence has all nonnegative coefficients for simple roots $\nu \in \Delta$ by the chamber correspondence. This is only possible when β has simple roots which coincide with μ which is impossible given β simple. Hence the claim follows.

Let Δ be a fixed base of Φ .

Lemma 3.5. *If α is positive but not simple, then $\alpha - \beta$ is a positive root for some $\beta \in \Delta$.*

pf. If $(\alpha, \beta) \leq 0$, $\forall \beta \in \Delta$, $\Delta \cup \{\alpha\}$ is a linearly independent set (set $\Delta' = \Delta \cup \{\alpha\}$ and $\sum_{\gamma \in \Delta'} k_\gamma \gamma = 0$. Then we can separate γ whose coefficient is positive from ones with negative coefficients. Then inner product gives all those coefficients should be zero, which implies linear independence), a contradiction to the definition of Δ . So $\exists \beta \in \Delta$ s.t. $(\alpha, \beta) > 0$ and then by the acute-angle lemma $\alpha - \beta \in \Phi$. Write $\alpha = \sum_{\gamma \in \Delta} k_\gamma \gamma$ with all $k_\gamma \geq 0$ and at least one $\gamma \neq \beta$ with $k_\gamma > 0$. Subtracting β from α yields a $\mathbb{Z}$ -linear combination of simple roots with at least one positive coefficient. This forces all coefficients to be nonnegative by the uniqueness of the expression.

Corollary 3.1. *Each $\beta \in \Phi^+$ can be written in the form $\alpha_1 + \dots + \alpha_k$ with $\alpha_i \in \Delta$ (not necessarily distinct) in such a way that each partial sum $\alpha_1 + \dots + \alpha_i$ is a root.*

pf. When $\text{ht}\beta = 1$, β itself is in Δ and there is nothing to prove, When $\text{ht}\beta > 1$, since it is positive root and not simple, by the previous lemma, $\exists \beta' \in \Delta$ s.t. $\beta - \beta'$ is a positive root. Then $\text{ht}(\beta - \beta') < \text{ht}\beta$ and therefore by the inductive hypothesis, it can be written as a sum of simple roots s.t. its partial sum is also a root. Then $\beta = \beta' + \text{such sum}$ gives the desired form.

Lemma 3.6. *Let α be simple. Then σ_α permutes the positive roots other than α .*

pf. Let $\beta \in \Phi^+ \setminus \{\alpha\}$, $\beta = \sum_{\gamma \in \Delta} k_\gamma \gamma$, $k_\gamma \in \mathbb{Z}_{\geq 0}$. Then β is not proportional to α . Therefore $k_\gamma \neq 0$, for some $\gamma \neq \alpha$. But the coefficient of γ in $\sigma_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$ is still k_γ , i.e. $\sigma_\alpha(\beta)$ has at least one positive coefficient relative to Δ , forcing it to be positive. Moreover, $\sigma_\alpha(\beta) \neq \alpha$ since α is the image of $-\alpha$.

Corollary 3.2. *Set $\delta = \frac{1}{2} \sum_{\beta > 0} \beta$. Then $\sigma_\alpha(\delta) = \delta - \alpha$, $\forall \alpha \in \Delta$.*

pf. By the previous lemma, σ_α permutes all positive roots other than α so the sum of the positive roots other than α is left invariant. Since α is sent to $-\alpha$, the corollary follows.

Lemma 3.7. *Let $\alpha_1, \dots, \alpha_t \in \Delta$. Write $\sigma_i = \sigma_{\alpha_i}$. If $\sigma_1 \dots \sigma_{t-1}(\alpha_t)$ is negative, then for some index $1 \leq s < t$, $\sigma_1 \dots \sigma_t = \sigma_1 \dots \sigma_{s-1} \sigma_{s+1} \dots \sigma_{t-1}$.*

pf. Write $\beta_i = \begin{cases} \sigma_{i+1} \dots \sigma_{t-1}(\alpha_t), & 0 \leq i \leq t-2 \\ \alpha_t, & i = t-1 \end{cases}$. Since $\beta_0 < 0$, $0 < \beta_{t-1}$, we can find the smallest index s for which $0 < \beta_s$. Then $\sigma_s(\beta_s) = \beta_{s-1} < 0$ and by the previous lemma $\beta_s = \alpha_s$. Then since $\sigma \in \mathcal{W}$ implies $\sigma_{\sigma(\alpha)} = \sigma \sigma_\alpha \sigma^{-1}$, $\sigma_s = (\sigma_{s+1} \dots \sigma_{t-1}) \sigma_t (\sigma_{t-1} \dots \sigma_{s+1})$ which yields the lemma.

Corollary 3.3. *If $\sigma = \sigma_1 \dots \sigma_t$ is an expression for $\sigma \in \mathcal{W}$ in terms of reflections corresponding to simple roots, with t as small as possible, then $\sigma(\alpha_t) < 0$.*

pf. The minimality of t , together with the previous lemma, gives the corollary.

Theorem 3.2. 1. *If $\gamma \in E$ regular, there exists $\sigma \in \mathcal{W}$ s.t. $(\sigma(\gamma), \alpha) > 0$, $\forall \alpha \in \Delta$*

2. *If Δ' is another base of Φ , then $\sigma(\Delta') = \Delta$ for some $\sigma \in \mathcal{W}$*

3. *If α is any root, there exists $\sigma \in \mathcal{W}$ s.t. $\sigma(\alpha) \in \Delta$*

4. *$\mathcal{W}$ is generated by the σ_α , $\forall \alpha \in \Delta$*

5. *If $\sigma(\Delta) = \Delta$, $\sigma \in \mathcal{W}$, then $\sigma = 1$.*

pf. Define $\mathcal{W}'$ a subgroup of $\mathcal{W}$ generated by all σ_α , $\alpha \in \Delta$.

- Write $\delta = \frac{1}{2} \sum_{\alpha < 0} \alpha$ and choose $\sigma \in \mathcal{W}'$ for which $(\sigma(\gamma), \delta)$ is as big as possible. If α is simple, $\sigma_\alpha \sigma \in \mathcal{W}'$ so the choice of σ implies $(\sigma(\gamma), \delta) \geq (\sigma_\alpha \sigma(\gamma), \delta) = (\sigma(\gamma), \sigma_\alpha(\delta)) = (\sigma(\gamma), \delta - \alpha)$ by the corollary 3.2. The right side equals $(\sigma(\gamma), \delta) - (\sigma(\gamma), \alpha)$ which implies $(\sigma(\gamma), \alpha) \geq 0$, $\forall \alpha \in \Delta$. But $(\sigma(\gamma), \alpha) \neq 0$ since otherwise γ would be orthogonal to $\sigma^{-1}\alpha$. So all the inequalities are strict. Therefore, $\sigma(\gamma)$ lies in the fundamental Weyl chamber $\mathcal{C}(\Delta)$ and σ sends $\mathcal{C}(\gamma)$ to $\mathcal{C}(\Delta)$ as desired.

- Since $\mathcal{W}'$ permutes the Weyl chambers transitively by 1, it also permutes the bases of Φ transitively.
- Since the only roots proportional to α are $\pm\alpha$, the hyperplanes P_β , $\beta \neq \pm\alpha$ are distinct from P_α so there exists $\gamma \in P_\alpha$, $\gamma \notin P_\beta$. Choose γ' close enough to γ so that $(\gamma', \alpha) = \epsilon > 0$ while $|(\gamma', \beta)| > \epsilon$, $\forall \beta \neq \pm\alpha$. Then α belongs to the base $\Delta(\gamma')$ since ϵ is the smallest value $(\gamma', \cdot)$ takes on positive roots, so α cannot be decomposable. So each root belongs to at least one base, and therefore by 2, 3 follows.
- By 3, find $\sigma \in \mathcal{W}'$ s.t. $\beta = \sigma(\alpha) \in \Delta$. Then $\sigma_\beta = \sigma_{\sigma(\alpha)} = \sigma\sigma_\alpha\sigma^{-1}$ so $\sigma_\alpha = \sigma^{-1}\sigma_\beta\sigma \in \mathcal{W}'$, i.e. $\mathcal{W} = \mathcal{W}'$.
- Let $\sigma(\Delta) = \Delta$, but $\sigma \neq 1$. If σ is written minimally as a product of one or more simple reflections, then by the corollary 3.3., a contradiction. So $\sigma = 1$.

When $\sigma \in \mathcal{W}$ is written as $\sigma_{\alpha_1} \dots \sigma_{\alpha_t}$, $\alpha_i \in \Delta$, t minimal, we call the expression **reduced** and write $l(\sigma) = t$, which is the **length** of σ relative to Δ . By definition, $l(1) = 0$.

Define $n(\sigma) =$ number of positive roots α for which $\sigma(\alpha) < 0$.

Lemma 3.8. $\forall \sigma \in \mathcal{W}$, $l(\sigma) = n(\sigma)$.

pf. When $l(\sigma) = 0$, lemma follows trivially since $l(\sigma) = 0$ implies $\sigma = 1$.
When $l(\sigma) > 0$, write σ in reduced form as $\sigma = \sigma_{\alpha_1} \dots \sigma_{\alpha_t}$ and set $\alpha = \alpha_t$. By the corollary 3.3., $\sigma(\alpha) < 0$. Then by the lemma 3.6., $n(\sigma\sigma_\alpha) = n(\sigma) - 1$. On the other hand, $l(\sigma\sigma_\alpha) = l(\sigma) - 1 < l(\sigma)$ so by the inductive hypothesis, $l(\sigma\sigma_\alpha) = n(\sigma\sigma_\alpha)$. Then $l(\sigma) = n(\sigma)$.

The closure of the fundamental Weyl chamber $\overline{\mathcal{C}(\Delta)}$ relative to Δ is a fundamental domain for the action of $\mathcal{W}$ on E , i.e. each vector in E is $\mathcal{W}$ -conjugate to precisely one point of this set.

Lemma 3.9. Let $\lambda, \mu \in \overline{\mathcal{C}(\Delta)}$. If $\sigma\lambda = \mu$ for some $\sigma \in \mathcal{W}$, then σ is a product of simple reflections which fix λ ; in particular, $\lambda = \mu$.

pf. When $l(\sigma) = 0$, $\sigma = 1$ and the lemma trivially holds.
When $l(\sigma) > 0$, by the previous lemma, σ must send some positive root to a negative root; so σ cannot send all simple roots to positive roots. Say $\sigma\alpha < 0$, $\alpha \in \Delta$. Then $0 \geq (\mu, \sigma\alpha) = (\sigma^{-1}\mu, \alpha) = (\lambda, \alpha) \geq 0$, $\lambda, \mu \in \overline{\mathcal{C}(\Delta)}$. This forces $(\lambda, \alpha) = 0$, $\sigma_\alpha\lambda = \lambda$, hence $(\sigma\sigma_\alpha)\lambda = \mu$. Then by the lemma 3.6. and 3.8., $l(\sigma\sigma_\alpha) = l(\sigma) - 1$ so the inductive hypothesis can be applied and get $\sigma\sigma_\alpha$ is a product of simple reflections fixing

λ , so $\sigma = (\sigma\sigma_\alpha)\sigma_\alpha$ is too. Thus $\lambda = \mu$.

Φ is called irreducible if it cannot be partitioned into the union of two proper subsets s.t. each root in one set is orthogonal to each root in the other.

CLAIM : Φ is irreducible iff Δ cannot be partitioned in the way stated above. pf.

- Let $\Phi = \Phi_1 \cup \Phi_2$ with $(\Phi_1, \Phi_2) = 0$. Unless Δ is wholly contained in one factor, this induces a similar partition of Δ . But $\Delta \subset \Phi_1$ implies $(\Delta, \Phi_2) = 0$, i.e. $(E, \Phi_2) = 0$ since Δ spans E , a contradiction. So the partition of Δ exists.
- Let Φ be irreducible, but $\Delta = \Delta_1 \cup \Delta_2$ with $(\Delta_1, \Delta_2) = 0$. Each root is conjugate to a simple root by the previous theorem, so $\Phi = \Phi_1 \cup \Phi_2$, Φ_i the set of roots having a conjugate in Δ_i . Since $(\alpha, \beta) = 0$ implies $\sigma_\alpha \sigma_\beta = \sigma_\beta \sigma_\alpha$, and $\mathcal{W}$ is generated by σ_α , each root in Φ_i is obtained from one in Δ_i by adding or subtracting elements of Δ_i . Therefore Φ_i lies in the subspace E_i of E spanned by Δ_i and $(\Phi_1, \Phi_2) = 0$. This forces either Φ_1 or Φ_2 equals $\emptyset$, i.e. $\Delta_1 = \emptyset$ or $\Delta_2 = \emptyset$.

CLAIM : Let Φ be irreducible. Then $\Phi^\vee$ is also irreducible. Also, if Φ has all roots of equal length, so does $\Phi^\vee$ and $\Phi^\vee \cong \Phi$.

pf. If $\Phi^\vee$ is reducible, $\Phi^\vee = \Phi_1^\vee \cup \Phi_2^\vee$, s.t. $(\Phi_1^\vee, \Phi_2^\vee) = 0$. Then $\Delta^\vee = \Delta_1^\vee \cup \Delta_2^\vee$ with $(\Delta_1^\vee, \Delta_2^\vee) = 0$. Since $\alpha^\vee$ is a positive multiple of α , this in turn implies $(\Delta_1, \Delta_2) = 0$, a contradiction to the irreducibility of Δ and therefore Φ .

By the definition of the length, $(\alpha^\vee, \alpha^\vee) = \frac{4}{(\alpha, \alpha)} = \frac{4}{c}$ where $c = (\alpha, \alpha)$ gives that if Φ has all roots of equal length, so does $\Phi^\vee$.

With all lengths equal $\langle \alpha, \beta \rangle = \langle \beta, \alpha \rangle$, so the Cartan matrix equals its transpose, which is the Cartan matrix of $\Phi^\vee$. Thus $\Phi^\vee \cong \Phi$ by Proposition 3.1 below.

Lemma 3.10. *Let Φ be irreducible. Relative to the partial ordering $<$, there is a unique maximal root β . Moreover, if $\beta = \sum_{\alpha \in \Delta} k_\alpha \alpha$, then all $k_\alpha > 0$.*

pf. Let $\beta = \sum k_\alpha \alpha$ be a maximal root in the ordering. Then $0 < \beta$. If $\begin{cases} \Delta_1 = \{\alpha \in \Delta | k_\alpha > 0\} \\ \Delta_2 = \{\alpha \in \Delta | k_\alpha = 0\} \end{cases}$,

then $\Delta = \Delta_1 \cup \Delta_2$ is a partition. Suppose Δ_2 is nonvoid. Then $(\alpha, \beta) \leq 0$, $\forall \alpha \in \Delta_2$ by the lemma 3.4. Then, since Φ is irreducible, at least one $\alpha \in \Delta_2$ must be nonorthogonal to Δ_1 , forcing $(\alpha, \alpha') < 0$ for some $\alpha' \in \Delta_1$. This implies $(\alpha, \beta) = \sum_{\alpha' \in \Delta_1} k_{\alpha'} (\alpha, \alpha') \leq 0$ and at least one term < 0 and therefore $(\alpha, \beta) < 0$ and $\alpha + \beta$ is a root by the acute-angle lemma, a contradiction to the maximality of β . Therefore, Δ_2 is empty and all $k_\alpha > 0$. Also, by the acute angle lemma, $(\alpha, \beta) \geq 0$, $\forall \alpha \in \Delta$ with $(\alpha, \beta) > 0$ for at least one α , since Δ spans E . Now let β' be another maximal root. Then by the previous argument, β' involves at least one $\alpha \in \Delta$ for which $(\alpha, \beta) > 0$.

So $(\beta', \beta) > 0$ and $\beta - \beta'$ is a root by the acute-angle lemma unless $\beta = \beta'$. But if $\beta - \beta'$ is a root, either $\beta < \beta'$ or $\beta' < \beta$, a contradiction. So β is unique.

Lemma 3.11. *Let Φ be irreducible. Then $\mathcal{W}$ acts irreducibly on E . In particular, the $\mathcal{W}$ -orbit of a root α spans E .*

pf. Let E' be a nonzero subspace of E invariant under $\mathcal{W}$. The orthogonal complement E'' of E' is also $\mathcal{W}$ -invariant, and $E = E' \oplus E''$. Then $\forall \alpha \in \Phi$ either $\alpha \in E'$ or $E' \subset P_\alpha$ since $\sigma_\alpha(E') = E'$. Thus $\alpha \notin E'$ implies $\alpha \in E''$, so each root lies in one subspace or the other. This partitions Φ into orthogonal subsets, forcing one or the other to be empty. Since Φ spans E , $E' = E$.

The span of the $\mathcal{W}$ -orbit of a root is a nonzero $\mathcal{W}$ -invariant subspace of E , so the second stnt follows from the first.

Lemma 3.12. *Let Φ be irreducible. Then at most two root lengths occur in Φ and all roots of a given length are conjugate under $\mathcal{W}$.*

pf. If α, β are arbitrary roots, then not all $\sigma(\alpha)$, $\sigma \in \mathcal{W}$ can be orthogonal to β since $\sigma(\alpha)$ span E by the previous lemma. If $(\alpha, \beta) \neq 0$, the possible ratios of squared root lengths of α, β are $1, 2, 3, 1/2, 1/3$. These imply that at most two root lengths occur in Φ since the presence of three root lengths would yield also a ratio $3/2$.

Let α, β have equal length. After replacing one of these by a $\mathcal{W}$ -conjugate, we may assume them to be nonorthogonal and distinct, otherwise there is nothing to prove. Then $\langle \alpha, \beta \rangle = \langle \beta, \alpha \rangle = \pm 1$. Replacing β by $-\beta = \sigma_\beta(\beta)$ we may assume $\langle \alpha, \beta \rangle = 1$. Thus $(\sigma_\alpha \sigma_\beta \sigma_\alpha)(\beta) = \sigma_\alpha \sigma_\beta(\beta - \alpha) = \sigma_\alpha(-\beta - \alpha + \beta) = \alpha$.

→ when Φ is irreducible with two distinct root lengths, we speak of long and short roots.

Lemma 3.13. *Let Φ be irreducible, with two distinct root lengths. Then the maximal root β in the lemma 3.10. is long.*

pf. Let $\alpha \in \Phi$ be arbitrary. Then it suffices to show $(\beta, \beta) \geq (\alpha, \alpha)$, $\forall \alpha$. So we may replace α by a $\mathcal{W}$ -conjugate lying in the closure of the fundamental Weyl chamber relative to Δ . Since $0 < \beta - \alpha$ by the lemma 3.10., $(\gamma, \beta - \alpha) \geq 0$, $\forall \gamma \in \overline{\mathcal{C}(\Delta)}$. Then when $\gamma = \beta$, $\gamma = \alpha$, $(\beta, \beta) \geq (\beta, \alpha) \geq (\alpha, \alpha)$.

3.3 Classification

Denote Φ a root system of rank l , $\mathcal{W}$ its Weyl group, Δ , a base of Φ .

Definition 3.10. *For a fixed ordering $(\alpha_1, \dots, \alpha_l)$ of the simple roots, the matrix $(\langle \alpha_i, \alpha_j \rangle)$ is called the Cartan matrix of Φ , and its entries are called Cartan integers.*

$$\begin{array}{lcl} A_1 \times A_1 & \rightarrow & \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ \text{e.g.} & & A_2 \rightarrow \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\ & & B_2 \rightarrow \begin{bmatrix} 2 & -2 \\ -1 & 2 \end{bmatrix} \\ & & G_2 \rightarrow \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \end{array} \cdot$$

Proposition 3.1. *Let $\Phi' \subset E'$ be another root system, with base $\Delta' = \{\alpha'_1, \dots, \alpha'_l\}$. If $\langle \alpha_i, \alpha_j \rangle = \langle \alpha'_i, \alpha'_j \rangle$, $\forall 1 \leq i, j \leq l$, then the bijection $\alpha_i \mapsto \alpha'_i$ extends uniquely to an isomorphism $\phi : E \rightarrow E'$ mapping Φ onto Φ' and satisfying $\langle \phi(\alpha), \phi(\beta) \rangle = \langle \alpha, \beta \rangle$, $\forall \alpha, \beta \in \Phi$. Therefore, the Cartan matrix of Φ determines Φ up to isomorphism.*

pf. Since $\begin{cases} \Delta \text{ is a basis of } E \\ \Delta' \text{ is a basis of } E' \end{cases}$, there is a unique vector space isomorphism $\phi : E \rightarrow E'$ sending α_i to α'_i , $\forall 1 \leq i \leq l$. If $\alpha, \beta \in \Delta$, by the hypothesis $\sigma_{\phi(\alpha)}(\phi(\beta)) = \sigma_{\alpha'}(\beta') = \beta' - \langle \beta', \alpha' \rangle \alpha' = \phi(\beta) - \langle \beta, \alpha \rangle \phi(\alpha) = \phi(\beta - \langle \beta, \alpha \rangle \alpha) = \phi(\sigma_\alpha(\beta))$, i.e. $\forall \alpha \in \Delta$,

$$\begin{array}{ccc} E & \xrightarrow{\phi} & E' \\ \downarrow \sigma_\alpha & & \downarrow \sigma_{\phi(\alpha)} \\ E & \xrightarrow{\phi} & E' \end{array} \text{ commutes. The respective Weyl groups } \mathcal{W}, \mathcal{W}' \text{ are generated by}$$

simple reflections by the previous theorem. Write $\beta = \sigma(\alpha)$ with $\sigma \in \mathcal{W}$, $\alpha \in \Delta$. Then $\phi(\beta) = (\phi \circ \sigma \circ \phi^{-1})(\phi(\alpha)) \in \Phi'$ and therefore ϕ maps Φ onto Φ' . Moreover, the formula for a reflection shows ϕ preserves all Cartan integers.

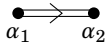
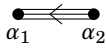
$\rightarrow$ it is possible to recover Φ from the Cartan integers. For example, start with the roots of height 1, i.e. the simple roots. For any pair $\alpha_i \neq \alpha_j$, the integer r for the α_j -string through α_i is 0, i.e. $\alpha_i - \alpha_j$ is not a root by the lemma 3.4. So the integer $q = -\langle \alpha_i, \alpha_j \rangle$. This lets us write down all roots α of height 2, hence all integers $\langle \alpha, \alpha_j \rangle$. For each root α of height 2, the integer r for the α_j -string through α can be determined easily, since α_j can be subtracted at most once and then q is found because $r - q = \langle \alpha, \alpha_j \rangle$. By the corollary 3.1., all positive roots are eventually obtained if we repeat this process.

Define the Coxeter graph of Φ to be a graph having l vertices, the i th joined to the j th by $\langle \alpha_i, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle$ edges.

$$\begin{array}{lcl} A_1 \times A_1 & \bullet & \bullet \\ \text{e.g.} & & A_2 \bullet \text{---} \bullet \\ & & B_2 \bullet \text{====} \bullet \\ & & G_2 \bullet \text{=====} \bullet \end{array} \cdot$$

The Coxeter graph determines the numbers $\langle \alpha_i, \alpha_j \rangle$ in case all roots have equal length, since then $\langle \alpha_i, \alpha_j \rangle = \langle \alpha_j, \alpha_i \rangle$. Whenever a double or triple edge occurs in

the Coxeter graph of Φ , we can add an arrow pointing to the shorter of the two roots. This allows us to recover the Cartan integers; we call the resulting figure the **Dynkin diagram** of Φ .

e.g. B_2 
 G_2 

Φ is irreducible iff Φ cannot be partitioned into two proper orthogonal subsets. Also Φ is irreducible iff its Coxeter graph is connected. In general, there will be a number of connected components of the Coxeter graph.

Proposition 3.2. Φ decomposes as the union of irreducible root systems Φ_i in subspaces E_i of E s.t. $E = E_1 \oplus \cdots \oplus E_t$ orthogonal direct sum.

pf. For the Coxeter graph for Φ , let $\Delta = \Delta_1 \cup \cdots \cup \Delta_t$ be the partition of Δ into mutually orthogonal subsets, namely the one given by the connected components of the Coxeter graph. If E_i is the span of Δ_i , $E = E_1 \oplus \cdots \oplus E_t$ orthogonal direct sum. Since every root is $\mathcal{W}$ -conjugate to a simple root by the theorem 3.2, it is the case.

Moreover, the $\mathbb{Z}$ -linear combinations of Δ_i which are roots obviously form a root system in E_i , whose Weyl group is the restriction to E_i of the subgroup of $\mathcal{W}$, generated by all σ_α , $\alpha \in \Delta_i$. Each E_i is $\mathcal{W}$ -invariant since $\alpha \notin \Delta_i$ implies σ_α acts trivially on E_i , so each root lies in one of the E_i , i.e. $\Phi = \Phi_1 \cup \cdots \cup \Phi_t$.

Theorem 3.3. If Φ is an irreducible root system of rank l , its Dynkin diagram is one of the following :

$$\begin{array}{ll}
A_l, l \geq 1 & \bullet \text{---} \bullet \cdots \bullet \text{---} \bullet \\
& \quad 1 \quad 2 \quad \quad \quad l-1 \quad l \\
B_l, l \geq 2 & \bullet \text{---} \bullet \cdots \bullet \text{---} \bullet \\
& \quad 1 \quad 2 \quad \quad \quad l-1 \quad l \\
C_l, l \geq 3 & \bullet \text{---} \bullet \cdots \bullet \text{---} \bullet \\
& \quad 1 \quad 2 \quad \quad \quad l-1 \quad l \\
D_l, l \geq 4 & \bullet \text{---} \bullet \cdots \bullet \text{---} \bullet \\
& \quad 1 \quad 2 \quad \quad \quad l-2 \quad l-1 \quad l \\
E_6 & \begin{array}{c} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \\ \quad \quad \quad \bullet \\ \quad \quad \quad 2 \end{array} \\
& \quad 1 \quad 3 \quad 4 \quad 5 \quad 6 \\
E_7 & \begin{array}{c} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \\ \quad \quad \quad \bullet \\ \quad \quad \quad 2 \end{array} \\
& \quad 1 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \\
E_8 & \begin{array}{c} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \\ \quad \quad \quad \bullet \\ \quad \quad \quad 2 \end{array} \\
& \quad 1 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \\
F_4 & \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \\
& \quad 1 \quad 2 \quad 3 \quad 4 \\
G_2 & \bullet \text{---} \bullet \\
& \quad 1 \quad 2
\end{array}
\begin{array}{l}
\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -2 \\ & & & -1 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & & -2 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 & -1 \\ & & & -1 & 2 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & & -1 & & \\ -1 & 2 & & -1 & \\ & -1 & -1 & 2 & -1 \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & & -1 & & \\ -1 & 2 & & -1 & \\ & -1 & -1 & 2 & -1 \\ & & -1 & 2 & -1 \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -2 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{pmatrix} \\
\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}
\end{array}$$

In the Cartan matrices, blank entries are 0.

pf. Assume E is a Euclidean space, $\mathcal{U} = \{\epsilon_1, \dots, \epsilon_n\}$ is a set of n linearly independent unit vectors which satisfy $\begin{cases} (\epsilon_i, \epsilon_j) \leq 0 \\ 4(\epsilon_i, \epsilon_j)^2 \in \{0, 1, 2, 3\} \end{cases}, i \neq j$. Such a set of vectors is called admissible. We attach a graph Γ to the set $\mathcal{U}$ with vertices i, j joined by $4(\epsilon_i, \epsilon_j)^2$ edges.

- If some of the ϵ_i are discarded, the remaining ones still form an admissible set, whose graph is obtained from Γ by omitting the corresponding vertices and all incident edges.
- The number of pairs of vertices in Γ connected by one edge is strictly less than n .

pf. Set $\epsilon = \sum_{i=1}^n \epsilon_i$. Since the ϵ_i are linearly independent, $\epsilon \neq 0$. So $0 < (\epsilon, \epsilon) = n + 2 \sum_{i < j} (\epsilon_i, \epsilon_j)$. Let i, j be a pair of distinct indices for which $(\epsilon_i, \epsilon_j) \neq 0$. Then $4(\epsilon_i, \epsilon_j)^2 = 1, 2, 3$ so $2(\epsilon_i, \epsilon_j) \leq -1$. Then the number of such pairs cannot exceed $n - 1$.

- Γ contains no cycles.

pf. A cycle would be the graph Γ' of an admissible subset $\mathcal{U}'$ of $\mathcal{U}$ and then Γ' violates the second stmt with n replaced by $\text{Card } \mathcal{U}'$.

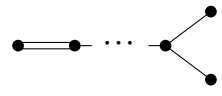
- No more than three edges can originate at a given vertex of Γ .

pf. Say $\epsilon \in \mathcal{U}$ and $\eta_1, \dots, \eta_k$ are the vectors in $\mathcal{U}$ connected to ϵ , i.e. $(\epsilon, \eta_i) < 0$ with $\epsilon, \eta_1, \dots, \eta_k$ all distinct. Then no two η 's can be connected so $(\eta_i, \eta_j) = 0$, $i \neq j$. Because $\mathcal{U}$ is linearly independent, some unit vector η_0 in the span of $\epsilon, \eta_1, \dots, \eta_k$ is orthogonal to $\eta_1, \dots, \eta_k$; clearly $(\epsilon, \eta_0) \neq 0$ for such η_0 . Then $\epsilon = \sum_{i=0}^k (\epsilon, \eta_i) \eta_i$, so $1 = (\epsilon, \epsilon) = \sum (\epsilon, \eta_i)^2$. This forces $\sum (\epsilon, \eta_i)^2 < 1 \Leftrightarrow \sum 4(\epsilon, \eta_i)^2 < 4$. But $4(\epsilon, \eta_i)^2$ is the number of edges joining ϵ to η_i in Γ .

- The only connected graph Γ of an admissible set $\mathcal{U}$ which can contain a triple edge is the Coxeter graph G_2 by the previous stmt.
- Let $\{\epsilon_1, \dots, \epsilon_k\} \subset \mathcal{U}$ have as subgraph a simple chain in Γ . If $\mathcal{U}' = (\mathcal{U} \setminus \{\epsilon_1, \dots, \epsilon_k\}) \cup \{\epsilon\}$, $\epsilon = \sum_{i=1}^k \epsilon_i$, then $\mathcal{U}'$ is admissible.

pf. Linear independence of $\mathcal{U}'$ is obvious. By hypothesis, $2(\epsilon_i, \epsilon_{i+1}) = -1$, $1 \leq i \leq k-1$ so $(\epsilon, \epsilon) = k + 2 \sum_{i < j} (\epsilon_i, \epsilon_j) = k - (k-1) = 1$. So ϵ is a unit vector. Any $\eta \in \mathcal{U} \setminus \{\epsilon_1, \dots, \epsilon_k\}$ can be connected to at most one of ϵ_i by the third stmt, so $(\eta, \epsilon) = 0$ or else $(\eta, \epsilon) = (\eta, \epsilon_i)$. In either case, $4(\eta, \epsilon)^2 \in \{0, 1, 2, 3\}$.

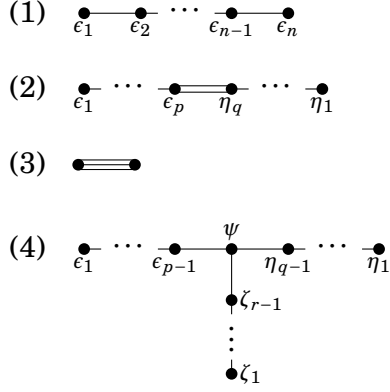
→ this excludes the subgraph of the form



since by the above stmt we replace the simple chain by a

single vertex, yielding graphs which violate the fourth stmt.

- Any connected graph Γ of an admissible set has one of the following forms:



pf. By the fifth stmt, we have the unique occurrence of triple edge. A connected graph containing more than one double edge would contain a subgraph which is not allowed by the previous stmt, so at most one double edge occurs. Moreover, if Γ has a double edge, it cannot also have a node by the previous stmt, so the second graph pictured is the only possibility. Finally, let Γ only have single edges; if Γ has no node, it must be a simple chain. It cannot contain more than one node, so the fourth graph is the only remaining possibility.

- The only connected Γ of type 2 is F_4 or B_n .

pf. Set $\begin{cases} \epsilon = \sum_{i=1}^p i\epsilon_i \\ \eta = \sum_{i=1}^q i\eta_i \end{cases}$. Then $2(\epsilon_i, \epsilon_{i+1}) = -1 = 2(\eta_i, \eta_{i+1})$ and other pairs are orthogonal, so $\begin{cases} (\epsilon, \epsilon) = \sum_{i=1}^p i^2 - \sum_{i=1}^{p-1} i(i+1) = \frac{p(p+1)}{2} \\ (\eta, \eta) = \frac{q(q+1)}{2} \end{cases}$. Since $4(\epsilon_p, \eta_q)^2 = 2$, $(\epsilon, \eta)^2 = p^2 q^2 (\epsilon_p, \eta_q)^2 = \frac{p^2 q^2}{2} < (\epsilon, \epsilon)(\eta, \eta) = \frac{p(p+1)q(q+1)}{4}$ by Cauchy-Schwarz inequality. Thus $(p-1)(q-1) < 2$. The possibilities are $\begin{cases} p = q = 2 & F_4 \\ p = 1, q \text{ arbitrary or vice versa} & B_l \end{cases}$.

- The only connected Γ of type 4 is the Coxeter graph D_n or E_n .

pf. Set $\begin{cases} \epsilon = \sum_{i=1}^p i\epsilon_i \\ \eta = \sum_{i=1}^q i\eta_i \\ \xi = \sum_{i=1}^r i\xi_i \end{cases}$. Then they are mutually orthogonal, linearly independent vectors and $\exists \psi$ not in their span. Then we can obtain $\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3 < 1$ where θ_i are the angles between ψ and ϵ, η, ξ . Then with $p-1$ in place of p , we can have $\begin{cases} (\epsilon, \epsilon) = \frac{p(p-1)}{2} \\ (\eta, \eta) = \frac{q(q-1)}{2} \\ (\xi, \xi) = \frac{r(r-1)}{2} \end{cases}$. Therefore, $\cos^2 \theta_1 =$

$$\frac{(\epsilon, \psi)^2}{(\epsilon, \epsilon)(\psi, \psi)} = \frac{(p-1)^2(\epsilon_{p-1}, \psi)^2}{(\epsilon, \epsilon)} = \frac{1}{4} \frac{2(p-1)^2}{p(p-1)} = \frac{(p-1)}{2p} = \frac{1}{2} \left(1 - \frac{1}{p}\right) \text{ and similarly, } \begin{cases} \cos^2 \theta_2 = \frac{1}{2} \left(1 - \frac{1}{q}\right) \\ \cos^2 \theta_3 = \frac{1}{2} \left(1 - \frac{1}{r}\right) \end{cases}.$$

Thus the previous inequality becomes $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$. By changing labels, we may assume $\frac{1}{p} \leq \frac{1}{q} \leq \frac{1}{r}$. Then $\frac{3}{2} \geq \frac{3}{r} > 1 \rightarrow r = 2$. Then $\frac{1}{p} + \frac{1}{q} > \frac{1}{2}$, $\frac{2}{q} > \frac{1}{2}$ and $2 \leq q < 4$. If $q = 2$, it allows any p , which gives D_l . If $q = 3$, $\frac{1}{p} > \frac{1}{6} \rightarrow p < 6$. So the possible triples (p, q, r) turn out to be D_n, E_n .

E is a subspace of $\mathbb{R}^n$, where $\mathbb{R}^n$ has standard basis $\epsilon_1, \dots, \epsilon_n$, a set of orthonormal unit vectors. Then the $\mathbb{Z}$ -span of this basis is a lattice, denoted I . Then Φ is defined to be the set of all vectors in I having specified lengths. Since the group I is discrete in the topology of $\mathbb{R}^n$ while the set of vectors in $\mathbb{R}^n$ having one or two given lengths is compact, Φ is obviously finite, and will exclude 0 by definition. In any case, Φ spans E , so the first condition of the root system is satisfied. The choice of lengths also makes the second condition satisfied.

Theorem 3.4. *For each Dynkin diagram of type $A - G$ there exists an irreducible root system having the given diagram.*

pf.

- A_l

pf. Let E be the l -dimensional subspace of $\mathbb{R}^{l+1}$ orthogonal to the vector $\epsilon_1 + \dots + \epsilon_{l+1}$. Let $I' = I \cap E$, and take Φ to be the set of all vectors $\alpha \in I'$ for which $(\alpha, \alpha) = 2$. Then $\Phi = \{\epsilon_i - \epsilon_j, i \neq j\}$. The vectors $\alpha_i = \epsilon_i - \epsilon_{i+1}$, $1 \leq i \leq l$ are independent and $\epsilon_i - \epsilon_j = (\epsilon_i - \epsilon_{i+1}) + \dots + (\epsilon_{j-1} - \epsilon_j)$, $i < j$ which shows they form a base of Φ . It is clear that the Cartan matrix is A_l . Finally the reflection w.r.t. α_i permutes the subscripts $i, i+1$ and leaves all other subscripts fixed. Thus σ_{α_i} corresponds to the transposition $(i, i+1)$ in the symmetric group $\mathcal{S}_{l+1}$; these transpositions generate $\mathcal{S}_{l+1}$ so there is a natural isomorphism of $\mathcal{W}$ onto $\mathcal{S}_{l+1}$.

- B_l

pf. Let $E = \mathbb{R}^l$, $\Phi = \{\alpha \in I | (\alpha, \alpha) \in \{1, 2\}\}$. Then Φ consists of the vectors $\pm \epsilon_i$ of squared length 1 and the vectors $\pm(\epsilon_i \pm \epsilon_j)$, $i \neq j$ of squared length 2. The l

vectors $\epsilon_1 - \epsilon_2, \dots, \epsilon_{l-1} - \epsilon_l, \epsilon_l$ are independent; $\begin{cases} \text{a short root} & \epsilon_i = (\epsilon_i - \epsilon_{i+1}) + \dots + (\epsilon_{l-1} - \epsilon_l) + \epsilon_l \\ \text{a long root} & \epsilon_i + \epsilon_j = (\epsilon_i - \epsilon_{i+1}) + \dots + (\epsilon_{j-1} - \epsilon_j) + \epsilon_j \end{cases}$

The Cartan matrix for this ordered base is B_l and $\mathcal{W}$ acts as the group of all permutations and sign changes of the set $\{\epsilon_1, \dots, \epsilon_l\}$ so $\mathcal{W}$ is isomorphic to the semidirect product of $(\mathbb{Z}/2\mathbb{Z})^l$ and $\mathcal{S}_l$.

- C_l

pf. In $E = \mathbb{R}^l$, the set of all $\pm 2\epsilon_i$ and $\pm(\epsilon_i \pm \epsilon_j)$, $i \neq j$ forms a root system of type C_l with base $\{\epsilon_1 - \epsilon_2, \dots, \epsilon_{l-1} - \epsilon_l, 2\epsilon_l\}$. The Weyl group is isomorphic to that of B_l .

- D_l

pf. Let $E = \mathbb{R}^l$, $\Phi = \{\alpha \in I | (\alpha, \alpha) = 2\} = \{\pm(\epsilon_i \pm \epsilon_j), i \neq j\}$. For a base take the l independent vectors $\epsilon_1 - \epsilon_2, \dots, \epsilon_{l-1} - \epsilon_l, \epsilon_{l-1} + \epsilon_l$. The Weyl group is the group of permutations and sign changes involving only even numbers of signs of the set $\{\epsilon_1, \dots, \epsilon_l\}$. So $\mathcal{W}$ is isomorphic to the semidirect product of $(\mathbb{Z}/2\mathbb{Z})^{l-1}$ and $\mathcal{S}_l$.

- E_l

pf. E_6, E_7 can be identified canonically with subsystems of E_8 , so it suffices to construct E_8 . Take $E = \mathbb{R}^8$, $I' = I + \mathbb{Z}((\epsilon_1 + \dots + \epsilon_8)/2)$. $I'' =$ subgroup of I' consisting of all elements $\sum c_i \epsilon_i + \frac{c}{2}(\epsilon_1 + \dots + \epsilon_8)$ for which $\sum c_i$ is an even integer. Define $\Phi = \{\alpha \in I'' | (\alpha, \alpha) = 2\}$. Then Φ consists of the vectors $\pm(\epsilon_i \pm \epsilon_j)$, $i \neq j$ along with $\frac{1}{2} \sum_{i=1}^8 (-1)^{k(i)} \epsilon_i$, $k(i) \in \{0, 1\}$ add up to an even integer. All inner products here are in $\mathbb{Z}$. As a base we take $\{\frac{1}{2}(\epsilon_1 + \epsilon_8 - (\epsilon_2 + \dots + \epsilon_7)), \epsilon_1 + \epsilon_2, \epsilon_2 - \epsilon_1, \epsilon_3 - \epsilon_2, \epsilon_4 - \epsilon_3, \epsilon_5 - \epsilon_4, \epsilon_6 - \epsilon_5, \epsilon_7 - \epsilon_6\}$.

- F_4

pf. Let $E = \mathbb{R}^4$, $I' = I + \mathbb{Z}(\frac{1}{2}(\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4))$, $\Phi = \{\alpha \in I' | (\alpha, \alpha) \in \{1, 2\}\}$. Then Φ consists of all $\pm \epsilon_i$, $\pm(\epsilon_i \pm \epsilon_j)$, $i \neq j$ as well as all $\pm \frac{1}{2}(\epsilon_1 \pm \epsilon_2 \pm \epsilon_3 \pm \epsilon_4)$, where the signs may be chosen independently. By inspection, all numbers $\langle \alpha, \beta \rangle$ are integral. As a base take $\{\epsilon_2 - \epsilon_3, \epsilon_3 - \epsilon_4, \epsilon_4, \frac{1}{2}(\epsilon_1 - \epsilon_2 - \epsilon_3 - \epsilon_4)\}$.

- G_2

Take E to be the subspace of $\mathbb{R}^3$ orthogonal to $\epsilon_1 + \epsilon_2 + \epsilon_3$, $I' = I \cap E$, $\Phi = \{\alpha \in I' | (\alpha, \alpha) \in \{2, 6\}\}$. So $\Phi = \pm\{\epsilon_1 - \epsilon_2, \epsilon_2 - \epsilon_3, \epsilon_1 - \epsilon_3, 2\epsilon_1 - \epsilon_2 - \epsilon_3, 2\epsilon_2 - \epsilon_1 - \epsilon_3, 2\epsilon_3 - \epsilon_1 - \epsilon_2\}$. As a base choose $\epsilon_1 - \epsilon_2, -2\epsilon_1 + \epsilon_2 + \epsilon_3$.

$\mathcal{W}$ is a normal subgroup of $\text{Aut}\Phi$. Let $\Gamma = \{\sigma \in \text{Aut}\Phi | \sigma(\Delta) = \Delta\}$, Δ a fixed base of Φ . Then Γ is a subgroup of $\text{Aut}\Phi$. If $\tau \in \Gamma \cap \mathcal{W}$, $\tau = 1$ by the simple transitivity of $\mathcal{W}$. Moreover, if $\tau \in \text{Aut}\Phi$ is arbitrary, $\tau(\Delta)$ is another base of Φ , so there exists $\sigma \in \mathcal{W}$ s.t. $\sigma\tau(\Delta) = \Delta$ and therefore $\tau \in \Gamma\mathcal{W}$.

$\rightarrow \text{Aut}\Phi$ is the semidirect product of Γ and $\mathcal{W}$.

$\{ \forall \tau \in \text{Aut} \Phi, \forall \alpha, \beta \in \Phi, \langle \alpha, \beta \rangle = \langle \tau(\alpha), \tau(\beta) \rangle \}$. Therefore $\forall \tau \in \Gamma$, it determines an automorphism of the Dynkin diagram of Φ . If τ acts trivially on the diagram, then $\tau = 1$ because Δ spans E . On the other hand, each automorphism of the Dynkin diagram obviously determines an automorphism of Φ by the proposition 3.1. So Γ **may be identified with the group of diagram automorphisms**.

4 Isomorphism and Conjugacy Theorems

4.1 Isomorphism Theorem

Let L be a semisimple Lie algebra over the algebraically closed field F of characteristic 0, H be a maximal toral subalgebra of L , $\Phi \subset H^*$ the set of roots of L relative to H . Then the rational span of Φ in H^* is of dimension l over $\mathbb{Q}$ where $l = \dim_F H^*$. By extending the base field from $\mathbb{Q}$ to $\mathbb{R}$, we obtain an l -dimensional real vector space E spanned by Φ . Additionally, the symmetric bilinear form dual to the Killing form is carried along to E , making E a Euclidean space. Then Φ is a root system in E .

Proposition 4.1. *Let L be a simple Lie algebra, H, Φ as above. Then Φ is an irreducible root system.*

pf. Suppose the contrary. Then Φ decomposes into $\Phi_1 \cup \Phi_2$, where Φ_i are orthogonal. If $\begin{cases} \alpha \in \Phi_1 \\ \beta \in \Phi_2 \end{cases}$, then $\begin{cases} (\alpha + \beta, \alpha) \\ (\alpha + \beta, \beta) \end{cases} \neq 0$, so $\alpha + \beta$ cannot be a root, and $[L_\alpha L_\beta] = 0$. This shows the subalgebra K of L generated by all $L_\alpha, \forall \alpha \in \Phi_1$ is centralised by all $L_\beta, \forall \beta \in \Phi_2$; in particular, K is a proper subalgebra of L , because $Z(L) = 0$. Furthermore, K is normalised by all $L_\alpha, \forall \alpha \in \Phi_1$, hence by all $L_\alpha, \forall \alpha \in \Phi$, hence by L . Therefore, K is a proper ideal of L , different from 0, a contradiction to the simplicity of L .

Corollary 4.1. *Let L be a semisimple Lie algebra, with maximal toral subalgebra H and root system Φ . If $L = L_1 \oplus \cdots \oplus L_t$ is the decomposition of L into simple ideals, then $H_i = H \cap L_i$ is a maximal toral subalgebra of L_i and the corresponding irreducible root system Φ_i may be regarded canonically as a subsystem of Φ in such a way that $\Phi = \Phi_1 \cup \cdots \cup \Phi_t$ is the decomposition of Φ into its irreducible components.*

pf. Let L be an arbitrary semisimple Lie algebra. Then L can be written uniquely as a direct sum $L_1 \oplus \cdots \oplus L_t$ of simple ideals. If H is a maximal toral subalgebra of L , then $H = H_1 \oplus \cdots \oplus H_t$, where $H_i = L_i \cap H$. Evidently, each H_i is a toral subalgebra of L_i , in fact maximal toral (any toral subalgebra of L_i larger

than H_i would be automatically toral in L , centralise all H_j , $j \neq i$, and generate with them a toral subalgebra of L larger than H , a contradiction).

Let Φ_i denote the root system of L_i relative to H_i in the real vector space E_i . If $\alpha \in \Phi_i$, we can view α as a linear function on H , by decreeing that $\alpha(H_j) = 0$, $\forall j \neq i$. Then α is a root of L relative to H with $L_\alpha \subset L_i$. Conversely, if $\alpha \in \Phi$, $\exists i$ s.t. $[H_i L_\alpha] \neq 0$, since otherwise H would centralise L_α . Then $L_\alpha \subset L_i$ so $\alpha|_{H_i}$ is a root of L_i relative to H_i . Thus Φ may be decomposed as $\Phi_1 \cup \dots \cup \Phi_t$, and $E \cong E_1 \oplus \dots \oplus E_t$. $\rightarrow$ by the corollary, we can reduce the problem of characterising semisimple Lie algebras to the problem of characterising simple ones.

Proposition 4.2. *Let L be a semisimple Lie algebra, H be a maximal toral subalgebra of L , Φ be the root system of L relative to H . Fix a base Δ of Φ . Then L is generated as Lie algebra by the root spaces $L_\alpha, L_{-\alpha}$, $\forall \alpha \in \Delta$; equivalently, L is generated by arbitrary nonzero root vectors $\begin{cases} x_\alpha \in L_\alpha \\ y_\alpha \in L_{-\alpha} \end{cases}$.*

pf. Let β be an arbitrary positive root relative to Δ . By the corollary 3.1., β may be written in the form $\beta = \alpha_1 + \dots + \alpha_s$ where $\alpha_i \in \Delta$ and where each partial sum $\alpha_1 + \dots + \alpha_i$ is a positive root. Also $[L_\gamma L_\delta] = L_{\gamma+\delta}$ whenever $\gamma, \delta, \gamma+\delta \in \Phi$. Using induction on s , L_β lies in the subalgebra of L generated by all L_α , $\forall \alpha \in \Delta$. Similarly, if β is negative, then L_β lies in the subalgebra of L generated by all $L_{-\alpha}$, $\forall \alpha \in \Delta$. Then $\begin{cases} L = H + \bigoplus_{\alpha \in \Phi} L_\alpha \\ H = \sum_{\alpha \in \Phi} [L_\alpha L_{-\alpha}] \end{cases}$ implies the proposition.
 $\rightarrow$ if $0 \neq \begin{cases} x_\alpha \in L_\alpha \\ y_\alpha \in L_{-\alpha} \end{cases}$, $\forall \alpha \in \Delta$ with $[x_\alpha y_\alpha] = h_\alpha$, we shall call $\{x_\alpha, y_\alpha\}$, a standard set of generators for L .

If (L, H) , (L', H') are two pairs, each consisting of a simple Lie algebra and a maximal toral subalgebra, an isomorphism $\Phi \rightarrow \Phi'$ is induced by an isomorphism $E \rightarrow E'$ of the ambient Euclidean spaces. The root system axioms are unaffected if we multiply the inner product on E or E' by a positive real number. Therefore, it does no harm to assume that the isomorphism $\Phi \rightarrow \Phi'$ comes from an isometry of the Euclidean spaces. Additionally, the isomorphism $\Phi \rightarrow \Phi'$ extends uniquely to an isomorphism of vector spaces $\psi : H^* \rightarrow H'^*$ since $\begin{Bmatrix} \Phi \\ \Phi' \end{Bmatrix}$ spans $\begin{Bmatrix} H^* \\ H'^* \end{Bmatrix}$. In turn ψ induces an isomorphism $\pi : H \rightarrow H'$ via the Killing form identification of H, H' with their duals. Explicitly, if $\alpha \mapsto \alpha'$ denotes the given map $\Phi \mapsto \Phi'$, then $\pi(t_\alpha) = t'_{\alpha'}$, where $\begin{Bmatrix} t_\alpha \\ t'_{\alpha'} \end{Bmatrix}$ correspond to $\begin{Bmatrix} \alpha \\ \alpha' \end{Bmatrix}$ via the Killing form. Since the given isomorphism of Φ, Φ' comes from an isometry between the respective Euclidean

spaces, $\pi(h_\alpha) = h'_\alpha$ since $h_\alpha = \frac{2t_\alpha}{(\alpha, \alpha)}$. Since H, H' are abelian Lie algebras, π can even be regarded as an isomorphism of Lie algebras.

Theorem 4.1. *Let L, L' be simple Lie algebras over F , with respective maximal toral subalgebras H, H' and corresponding root systems Φ, Φ' . Suppose there is an isomorphism of Φ onto Φ' denoted $\alpha \mapsto \alpha'$, inducing $\pi : H \rightarrow H'$. Fix a base $\Delta \subset \Phi$ so $\Delta' = \{\alpha' | \alpha \in \Delta\}$ is a base of Φ' . $\forall \left\{ \begin{array}{l} \alpha \in \Delta \\ \alpha' \in \Delta' \end{array} \right.$, choose arbitrary nonzero $\left\{ \begin{array}{l} x_\alpha \in L_\alpha \\ x'_\alpha \in L'_\alpha \end{array} \right.$, i.e. choose an arbitrary Lie algebra isomorphism $\pi_\alpha : L_\alpha \rightarrow L'_\alpha$. Then there exists a unique isomorphism $\pi : L \rightarrow L'$ extending $\pi : H \rightarrow H'$ and extending all the π_α , $\alpha \in \Delta$.*

pf.

- Existence

If L, L' are to be essentially the same, then their direct sum $L \oplus L'$ should include a subalgebra D resembling the diagonal subalgebra $\{(x, x) | x \in L\}$ of $L \oplus L$, which is isomorphic to L under the projection of $L \oplus L$ onto each factor. x_α , $\alpha \in \Delta$ determines unique $y_\alpha \in L_{-\alpha}$ for which $[x_\alpha y_\alpha] = h_\alpha$ and similarly in

L' . Let D be generated by the elements $\left\{ \begin{array}{l} \bar{x}_\alpha = (x_\alpha, x'_{\alpha'}) \\ \bar{y}_\alpha = (y_\alpha, y'_{\alpha'}) \\ \bar{h}_\alpha = (h_\alpha, h'_{\alpha'}) \end{array} \right.$, $\forall \left\{ \begin{array}{l} \alpha \in \Delta, \alpha' \in \Delta' \end{array} \right.$..

Because L, L' are simple, Φ, Φ' are irreducible by the proposition 4.1. Therefore Φ, Φ' have unique maximal roots β, β' relative to Δ, Δ' respectively, which correspond under the given isomorphism $\Phi \rightarrow \Phi'$. Choose arbitrary nonzero

$\left\{ \begin{array}{l} x \in L_\beta \\ x' \in L'_{\beta'} \end{array} \right.$ Set $\bar{x} = (x, x') \in L \oplus L'$, and let M be the subspace of $L \oplus L'$ spanned by all $\text{ad} \bar{y}_{\alpha_1} \dots \text{ad} \bar{y}_{\alpha_m}(\bar{x})$ where $\alpha_i \in \Delta$ with repetitions allowed. Then that belongs to $L_{\beta - \sum \alpha_i} \oplus L'_{\beta' - \sum \alpha'_i}$; in particular, $M \cap (L_\beta \oplus L'_{\beta'})$ is only one dimensional, forcing M to be a proper subspace of $L \oplus L'$.

- Subalgebra D stabilises M .

by definition $\text{ad} \bar{y}_\alpha$, $\forall \alpha \in \Delta$ stabilises M and by the fact that $[h y_\alpha]$ is a multiple of y_α , $\text{ad} \bar{h}_\alpha$ does likewise. On the other hand, for simple α , $\text{ad} x_\alpha$ commutes with all $\text{ad} y_\gamma$, γ simple, except $\gamma = \alpha$, since $\alpha - \gamma$ is not a root by the lemma 3.4. If we apply $\text{ad} \bar{x}_\alpha$ to $\text{ad} \bar{y}_{\alpha_1} \dots \text{ad} \bar{y}_{\alpha_m}(\bar{x})$, we can move it past each $\text{ad} \bar{y}_\gamma$ except $\text{ad} \bar{y}_\alpha$, in which case an extra summand involving $\text{ad} \bar{h}_\alpha$ is introduced. Since $\text{ad} \bar{x}_\alpha(\bar{x}) = 0$ whenever $\alpha \in \Delta$, $\alpha + \beta \notin \Phi$ by maximality, we see that $\text{ad} \bar{x}_\alpha$ stabilises M .

Then D is a proper subalgebra; otherwise M would be a proper nonzero ideal of $L \oplus L'$ but L, L' are unique ideals of this type and obviously $M \neq L, L'$.

- The projections of D onto the first and second factors of $L \oplus L'$ are Lie algebra isomorphisms.

These projections are Lie algebra homomorphisms by general principles and they are onto by the previous proposition and definition of D . On the other hand, suppose D has nonzero intersection with L , a kernel of projection onto second factor. Then D contains some $(w, 0)$, $w \neq 0$, so D also contains all $(\text{ad}z_{\alpha_1} \dots \text{ad}z_{\alpha_s}(w), 0)$, $\pm \alpha_i \in \Delta$, $z_{\alpha} \in \{x_{\alpha}, y_{\alpha}\}$. These elements form a nonzero ideal of L by the previous proposition, which must be L itself given L being simple. Thus D includes L . For any $(a, b) \in D$, $(0, b) = (a, b) - (a, 0) \in D$ and the second projection is onto, so $D \supset 0 \oplus L'$, i.e. D must also include L' , hence all of $L \oplus L'$, a contradiction.

The isomorphism $L \rightarrow L'$ just obtained via D sends $\begin{Bmatrix} x_{\alpha} \\ h_{\alpha} \end{Bmatrix}$ to $\begin{Bmatrix} x'_{\alpha'} \\ h'_{\alpha'} \end{Bmatrix}$, $\forall \alpha \in \Delta$, hence coincides with π on H .

- Uniqueness

x_{α} , $\alpha \in \Delta$ determines unique $y_{\alpha} \in L_{-\alpha}$ for which $[x_{\alpha}y_{\alpha}] = h_{\alpha}$ and L is generated by x_{α}, y_{α} by the previous proposition. So the uniqueness follows.

→ it can be extended to the semisimple L by projecting it onto its components.

CLAIM : The map $\alpha \mapsto -\alpha$ is an automorphism of Φ .

pf.

Proposition 4.3. *L as in the previous theorem, but not necessarily simple. Fix a nonzero $x_{\alpha} \in L_{\alpha}$, $\alpha \in \Delta$ and let $y_{\alpha} \in L_{-\alpha}$ satisfy $[x_{\alpha}y_{\alpha}] = h_{\alpha}$. Then there exists an*

automorphism σ of L of order 2 satisfying
$$\begin{cases} \sigma(x_{\alpha}) = -y_{\alpha} \\ \sigma(y_{\alpha}) = -x_{\alpha} \\ \sigma(h) = -h \end{cases}.$$

pf. A map sending each root to its negative, which evidently belongs to $\text{Aut}\Phi$ and the induced map on H sends h to $-h$. Then if $\sigma : H \rightarrow H$ is this isomorphism, $\sigma(h_{\alpha}) = -h_{\alpha}$ which is the same as $h_{-\alpha}$. Let x_{α} be sent to $-y_{\alpha}$, $\alpha \in \Delta$. Then according to the theorem, σ extends to an automorphism of L sending $x_{\alpha} \mapsto -y_{\alpha}$. Then $y_{\alpha} \mapsto -x_{\alpha}$ since $\sigma(y_{\alpha})$ is the unique element of L_{α} with $[\sigma(x_{\alpha})\sigma(y_{\alpha})] = \sigma(h_{\alpha}) = -h_{\alpha}$. Moreover σ has order 2, because σ^2 fixes a set of generators of L .

e.g. $L = \mathfrak{sl}(2, F)$, the automorphism σ .

The Weyl group $\mathcal{W}$ of Φ accounts for most of the automorphisms of Φ . By the previous theorem, the existence of corresponding automorphisms of L , which extend the action of $\mathcal{W}$ on H . If $\sigma \in \mathcal{W}$, the extension of σ to an automorphism of L must map L_β to $L_{\sigma\beta}$.

4.2 Conjugacy Theorem

If $t \in \text{End} V$, where V is a finite dimensional vector space, then V is the direct sum of all $V_a = \ker(t - a \cdot 1)^m$ where m is the multiplicity of a as root of the characteristic polynomial of t . Each V_a is invariant under t and the restriction of t to V_a is the sum of the scalar a and a nilpotent endomorphism. For $x \in L$, write $L = \bigoplus_{a \in F} L_a(\text{adx}) = L_0(\text{adx}) \oplus L_*(\text{adx})$, where $L_*(\text{adx})$ denotes the sum of those $L_a(\text{adx})$ for which $a \neq 0$. More generally, if K is a subalgebra of L stable under adx , we may write $K = K_0(\text{adx}) \oplus K_*(\text{adx})$ even if $x \notin K$.

Lemma 4.1. *If $a, b \in F$, then $[L_a(\text{adx})L_b(\text{adx})] \subset L_{a+b}(\text{adx})$. In particular, $L_0(\text{adx})$ is a subalgebra of L , and when $\text{char} F = 0$, $a \neq 0$, each element of $L_a(\text{adx})$ is ad-nilpotent.*

pf. $(\text{adx} - a - b)^m[yz] = \sum_{i=0}^m \binom{m}{i} [(\text{adx} - a)^i(y)(\text{adx} - b)^{m-i}(z)]$ so for sufficiently large m , all terms on the RHS are 0, when $\begin{cases} y \in L_a(\text{adx}) \\ z \in L_b(\text{adx}) \end{cases}$. adx shifts L_b to L_{a+b} and with $a \neq 0$ in characteristic 0, repeated shifts eventually leave the finitely many nonzero L_b .

$\rightarrow L_0(\text{adx})$ is a subalgebra of L , for $x \in L$, called an **Engel subalgebra**.

Lemma 4.2. *Let K be a subalgebra of L . Choose $z \in K$ s.t. $L_0(\text{adz})$ is minimal in the collection of all $L_0(\text{adx})$, $x \in K$. Suppose $K \subset L_0(\text{adz})$. Then $L_0(\text{adz}) \subset L_0(\text{adx})$, $\forall x \in K$.*

pf. Fix $x \in K$ and consider the family $\{\text{ad}(z + cx) | c \in F\}$ of endomorphisms of L ; since $K_0 = L_0(\text{adz})$ is a subalgebra of L including K , these endomorphisms stabilise K_0 , hence induce endomorphisms of the quotient vector space L/K_0 as well. If T is an indeterminate, we can therefore express the characteristic polynomial of $\text{ad}(z + cx)$ as the product $f(T, c)g(T, c)$ of its characteristic polynomials on $K_0, L/K_0$ respectively. If $\begin{cases} r = \dim K_0 \\ n = \dim L \end{cases}$, we can write $\begin{cases} f(T, c) = T^r + f_1(c)T^{r-1} + \dots + f_r(c) \\ g(T, c) = T^{n-r} + g_1(c)T^{n-r-1} + \dots + g_{n-r}(c) \end{cases}$, where the coefficients $f_i(c), g_i(c)$ are polynomials in c of degree at most i . By definition, the eigenvalue 0 of adz occurs only on the subspace K_0 , which means g_{n-r} is not identically 0 on F . Therefore, we can find as many scalars as we please which are not zeros of g_{n-r} ; say $c_1, \dots, c_{r+1}$ are distinct scalars of this kind.

$g_{n-r}(c) \neq 0$ is equivalent to 0 is not an eigenvalue of $\text{ad}(z + cx)$ on the quotient space; this forces all of $L_0(\text{ad}(z + cx))$ to lie in the subspace K_0 . But K_0 is chosen to be minimal, so $L_0(\text{ad}z) = L_0(\text{ad}(z + c_i x))$, $1 \leq i \leq r + 1$. This means $\text{ad}(z + c_i x)$ has the sole eigenvalue 0 on $L_0(\text{ad}z)$, i.e. $f(T, c_i) = T^r$. So each of the polynomials $f_1, \dots, f_r$ has $r + 1$ distinct zeros $c_1, \dots, c_{r+1}$. This gives the polynomials to be identically 0. So $K_0 \subset L_0(\text{ad}(z + cx))$, $\forall c \in F$. Since x was arbitrary, we may replace it by $x - z$, take $c = 1$ and get $L_0(\text{ad}z) \subset L_0(\text{ad}x)$.

Lemma 4.3. *If K is a subalgebra of L containing an Engel subalgebra, then $N_L(K) = K$. In particular, Engel subalgebras are self-normalising.*

pf. $x \in K$, $L_0(\text{ad}x) \subset K$. Then $\text{ad}x$ acts on $N_L(K)/K$ without eigenvalue 0 since every generalised 0-eigenvector lies in $L_0(\text{ad}x) \subset K$. On the other hand, $x \in K$ implies $[N_L(K)x] \subset K$, so $\text{ad}x$ acts trivially on $N_L(K)/K$. Together, these force $K = N_L(K)$.

Definition 4.1. *A Cartan subalgebra, abbreviated CSA, of a Lie algebra L is a nilpotent subalgebra which equals its normaliser in L .*

Theorem 4.2. *Let H be a subalgebra of the Lie algebra L . Then H is a CSA of L iff H is a minimal Engel subalgebra.*

pf. Suppose $H = L_0(\text{ad}z)$ is an Engel subalgebra of L . Then by the previous lemma, H is self-normalising. If in addition H properly contains no other Engel subalgebra, then the lemma 4.2's hypotheses are satisfied. So $H = L_0(\text{ad}z) \subset L_0(\text{ad}x)$, $\forall x \in H$. In particular $\text{ad}_H x$ is nilpotent for $x \in H$. Therefore by Engel's theorem H is nilpotent.

Let H be a CSA of L . Since H is nilpotent, $H \subset L_0(\text{ad}x)$, $\forall x \in H$. Suppose $\forall x \in H$, $H \subsetneq L_0(\text{ad}x)$. Take $L_0(\text{ad}z)$, $z \in H$ to be as small as possible. Then by the lemma 4.2., $L_0(\text{ad}x) \supset L_0(\text{ad}z)$, $\forall x \in H$. This means in the representation of H induced on the nonzero vector space $L_0(\text{ad}z)/H$, each $x \in H$ acts as a nilpotent endomorphism. Then H annihilates some nonzero $y + H$, i.e. $\exists y \notin H$ for which $[Hy] \subset H$, a contradiction to the self-normalising H . Then $\exists x \in H$, $H = L_0(\text{ad}x)$, so H is an Engel subalgebra. If $L_0(\text{ad}w) \subseteq H$, then since $[ww] = 0$, $w \in L_0(\text{ad}w)$ so $w \in H$ and nilpotence gives $H \subseteq L_0(\text{ad}w)$, so it is minimal.

Corollary 4.2. *Let L be semisimple with $\text{char} F = 0$. Then the CSA's of L are precisely the maximal toral subalgebras of L .*

pf. If L is semisimple with $\text{char} F = 0$, then a maximal toral subalgebra H is abelian hence nilpotent and $N_L(H) = H$ because $L = H + \coprod_{\alpha \in \Phi} L_\alpha$ with $[HL_\alpha] = L_\alpha$, $\forall \alpha \in \Phi$. So any maximal toral subalgebra is a CSA.

Let H be a CSA. $\forall x \in L$, if $x = x_s + x_n$ is the Jordan decomposition of x in L ,

then $L_0(\text{adx}_s) \subset L_0(\text{adx})$; any y killed by a power of adx_s is also killed by a power of adx , since adx_n is nilpotent and commutes with adx_s . For $x \in L$ semisimple, $L_0(\text{adx}) = C_L(x)$, adx being diagonalisable. The CSA H is minimal Engel, of the form $L_0(\text{adx})$ by the theorem. Also with minimality, $H = L_0(\text{adx}_s) = C_L(x_s)$. But $C_L(x_s)$ evidently includes a maximal toral subalgebra of L , which is a CSA and therefore an Engel subalgebra. H is a minimal Engel subalgebra so it cannot properly contain another Engel subalgebra, which forces the two to be equal, i.e. H is a maximal toral subalgebra.

→ a maximal toral subalgebra of a semisimple Lie algebra with $\text{char} F = 0$ has the form $C_L(s)$, for some s semisimple element. This element is called regular semisimple.

Lemma 4.4. *Let $\phi : L \rightarrow L'$ be an epimorphism of Lie algebras. If H is a CSA of L , then $\phi(H)$ is a CSA of L' .*

pf. $\phi(H)$ is nilpotent. Let $A = \ker \phi$ and identify L' with L/A . If $x+A$ normalises $H+A$, then $x \in N_L(H+A)$. But $H+A$ includes a CSA, a minimal Engel subalgebra, so the subalgebra $H+A$ is self-normalising by the previous lemma. Therefore $x \in H+A$, i.e. $\phi(H)$ is self-normalising.

Lemma 4.5. *Let $\phi : L \rightarrow L'$ be an epimorphism of Lie algebras. Let H' be a CSA of L' , $K = \phi^{-1}(H')$. Then any CSA H of K is also a CSA of L .*

pf. H is nilpotent by assumption. By the previous lemma, $\phi(H)$ is a CSA of $\phi(K) = H'$, forcing $\phi(H) = H'$ because CSAs are minimal Engel. If $x \in L$ normalises H , then $\phi(x)$ normalises $\phi(H)$, and therefore $\phi(x) \in \phi(H)$, i.e. $x \in H + \ker \phi$. But $\ker \phi \subset K$ by construction, so $x \in H + \ker \phi \subset K$. Then $x \in N_K(H) = H$ since H is a CSA of K .

Let L be a Lie algebra.

Definition 4.2. $x \in L$ is called strongly ad-nilpotent if there exists $y \in L$ and some nonzero eigenvalue a of $\text{ad} y$ s.t. $x \in L_a(\text{ad} y)$.

→ denote $\begin{cases} \mathcal{N}(L) & \text{the set of all strongly ad-nilpotent elements of } L \\ \mathcal{E}(L) & \text{the subgroup of } \text{Int} L \text{ generated by all } \exp \text{ad} x, x \in \mathcal{N}(L) \end{cases}$.

Then $\mathcal{N}(L)$ is stable under $\text{Aut} L$ and therefore $\mathcal{E}(L)$ is normal in $\text{Aut} L$. If L is semisimple, $\mathcal{E}(L) = \text{Int} L$. Also define the subgroup $\mathcal{E}(L; K) \subset \mathcal{E}(L)$ generated by all $\exp \text{ad}_L x$, $x \in \mathcal{N}(K)$. Then $\mathcal{E}(K)$ is simply obtained by taking the restriction of $\mathcal{E}(L; K)$ to K . If we take arbitrary $x \in K$ for which $\text{ad}_K x$ is nilpotent, we have no control over $\text{ad}_L x$ and therefore no such direct relationship between $\text{Int} K$ and $\text{Int} L$.

If $\phi : L \rightarrow L'$ is an epimorphism and $y \in L$, then $\phi(L_a(\text{ad} y)) = L_a(\text{ad} \phi(y))$. So $\phi(\mathcal{N}(L)) = \mathcal{N}(L')$.

Lemma 4.6. *Let $\phi : L \rightarrow L'$ be an epimorphism. If $\sigma' \in \mathcal{E}(L')$, then there exists*

$$\sigma \in \mathcal{E}(L) \text{ s.t. the following diagram } \begin{array}{ccc} L & \xrightarrow{\phi} & L' \\ \downarrow \sigma & & \downarrow \sigma' \\ L & \xrightarrow{\phi} & L' \end{array} \text{ commutes.}$$

pf. Let $\sigma' = \exp_{L'} x'$, $x' \in \mathcal{N}(L')$. By the previous stmt, $x' = \phi(x)$ for at least one $x \in \mathcal{N}(L)$. For arbitrary $z \in L$, $(\phi \circ \exp_{L'} x')(z) = \phi(z + [xz] + \frac{1}{2}[x[xz]] + \dots) = \phi(z) + [x'\phi(z)] + \frac{1}{2}[x'[x'\phi(z)]] + \dots = (\exp_{L'} x' \circ \phi)(z)$ so the diagram commutes.

Theorem 4.3. *Let L be solvable, $\mathcal{E}(L)$ defined as previously. Then any two CSAs H_1, H_2 of L are conjugate under $\mathcal{E}(L)$.*

pf. When $\dim L = 1$ or L is nilpotent, the theorem follows trivially.

Assume L is not nilpotent and $\dim L > 1$. Since L is solvable, L possesses nonzero abelian ideals such as the last nonzero term of the derived series. Choose A to be one of the smallest possible dimension. Set $L' = L/A$ and denote the canonical map $\begin{array}{ccc} L & \rightarrow & L/A \\ x & \mapsto & x' \end{array}$. By the lemma 4.4., H'_1, H'_2 are CSAs of the solvable algebra

L' . By inductive hypothesis, there exists $\sigma' \in \mathcal{E}(L')$ sending H'_1 onto H'_2 . Then by the previous lemma, $\exists \sigma \in \mathcal{E}(L)$ s.t. the diagram commutes. This means σ maps the full inverse image $K_1 = \phi^{-1}(H'_1)$ onto $K_2 = \phi^{-1}(H'_2)$. Now $H_2, \sigma(H_1)$ are both CSAs of the algebra K_2 . If K_2 is smaller than L , by the inductive hypothesis $\exists \tau' \in \mathcal{E}(K_2)$ s.t. $\tau'\sigma(H_1) = H_2$; but $\mathcal{E}(K_2)$ consists of the restrictions to K_2 of the elements of $\mathcal{E}(L; K_2) \subset \mathcal{E}(L)$, so $\tau\sigma(H_1) = H_2$, $\exists \tau \in \mathcal{E}(L)$ whose restriction to K_2 is τ' and we are done. Otherwise, we must have $L = K_2 = \sigma(K_1)$ so $K_1 = K_2$, $L = H_2 + A = H_1 + A$. The CSA H_2 is of the form $L_0(\text{adx})$ for suitable $x \in L$ by the theorem 4.2. A being adx stable, $A = A_0(\text{adx}) \oplus A_*(\text{adx})$ and each summand is stable under $L = H_2 + A$. By minimality of A , we have either $A = \begin{cases} A_0(\text{adx}) \\ A_*(\text{adx}) \end{cases}$. The first case is a contradiction because it causes $A \subset H_2$, $L = H_2$. So $A = A_*(\text{adx})$ and therefore $A = L_*(\text{adx})$. Since $L = H_1 + A$, we have an expression $x = y + z$, $y \in H_1$, $z \in L_*(\text{adx})$. Then write $z = [xz']$, $z' \in L_*(\text{adx})$ by the fact that adx is invertible on $L_*(\text{adx})$. Since A is abelian, $(\text{adz}')^2 = 0 \rightarrow \exp \text{adz}' = 1_L + \text{adz}'$. Then applying it to x yields $x - z = y$. In particular, $H = L_0(\text{ady})$ must also be a CSA of L . Since $y \in H_1$, $H_1 \subset H$, $H = H_1$, both being minimal Engel. So H_1 is conjugate to H_2 via $\exp \text{adz}'$. Here z' can be written as sum of certain strongly ad-nilpotent elements z_i of $A = L_*(\text{adx})$, but the latter commute, so $\exp \text{adz}' = \prod_i \exp \text{adz}'_i \in \mathcal{E}(L)$.

Definition 4.3. *A Borel subalgebra of a Lie algebra L is a maximal solvable subalgebra of L .*

Lemma 4.7. *If B is a Borel subalgebra of L , then $B = N_L(B)$.*

pf. Let x normalise B . Then $B + Fx$ is a subalgebra of L , solvable because $[B + Fx, B + Fx] \subset B$ and therefore $x \in B$ by the maximality of B .

Lemma 4.8. *If $\text{Rad}L \neq L$, then the Borel subalgebras of L are in natural one to one correspondence with those of the semisimple Lie algebra $L/\text{Rad}L$.*

pf. $\text{Rad}L$ being a solvable ideal of L , $B + \text{Rad}L$ is a solvable subalgebra of L for any Borel subalgebra B of L i.e. $\text{Rad}L \subset B$ by maximality. Then the lemma follows from the correspondence theorem for the projection $\pi : L \rightarrow L/\text{Rad}L$, since a subalgebra $B \supset \text{Rad}L$ is solvable iff $B/\text{Rad}L$ is solvable, $\text{Rad}L$ being solvable.

Let L be semisimple, $\begin{cases} H & \text{a CSA} \\ \Phi & \text{the root system of } L \end{cases}$. Fix a base Δ and with it a set of positive roots. Set $B(\Delta) = H + \bigoplus_{\alpha > 0} L_\alpha$, $N(\Delta) = \bigoplus_{\alpha > 0} L_\alpha$. Then $B(\Delta)$ is a subalgebra of L , with derived algebra $N(\Delta)$.

CLAIM : $N(\Delta)$ is nilpotent.

pf. If $x \in L_\alpha$, $\alpha > 0$, then application of adx to root vectors for roots of positive height increases height by at least one. This shows how to make the descending central series which goes to zero. Then $B(\Delta)$ is solvable. Let K be any subalgebra of L properly including $B(\Delta)$. Then K , being stable under $\text{ad}H$, must include some L_α , $\alpha < 0$. This forces K to include the simple algebra S_α , in particular K cannot be solvable. So $B(\Delta)$ is a Borel subalgebra.

Lemma 4.9. *Let L be semisimple with CSA H and root system Φ . For each base $\Delta \subset \Phi$, $B(\Delta)$ is a Borel subalgebra of L , called standard relative to H . All standard subalgebras of L relative to H are conjugate under $\mathcal{E}(L)$.*

pf. By the previous stmt, the first part of the lemma follows. The reflection σ_α , acting on H , may be extended to an inner automorphism τ_α of L , which is in $\mathcal{E}(L)$ by construction. Then it sends $B(\Delta)$ to $B(\sigma_\alpha \Delta)$. Since Weyl group is generated by reflections and acts transitively on bases, $\mathcal{E}(L)$ acts transitively on the standard Borels relative to H .

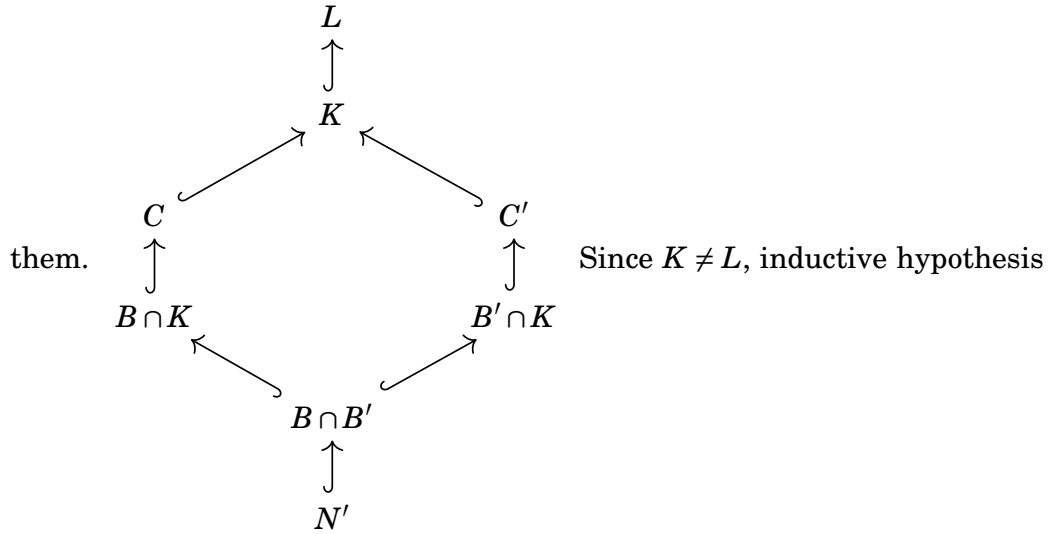
Theorem 4.4. *The Borel subalgebras of an arbitrary Lie algebra L are all conjugate under $\mathcal{E}(L)$.*

pf. When $\dim L = 1$, theorem follows trivially.

Given $\text{Rad}L \neq L$, assume L is semisimple by the lemma 4.6. and 4.8. along with the inductive hypothesis. Fix a standard Borel subalgebra B relative to some CSA. Let B' be another Borel subalgebra. If $B \cap B' = B$, there is nothing to prove since this forces $B = B'$ by the maximality. Do second downward induction on dimension of $B \cap B'$, and consider the case where $\dim B \cap B' < \dim B$. Then by inductive hypothesis, any Borel algebra whose intersection with B has larger dimension is already conjugate to B . $B \cap B' \neq 0$ gives

- the set N' of nilpotent elements of $B \cap B'$ is nonzero

Since B is standard, N' is a subspace and the derived algebra of $B \cap B'$ consists of nilpotent elements. This implies N' is an ideal in $B \cap B'$. N' is not an ideal of L , so its normaliser K is a proper subalgebra of L . Consider action of N' on $B/(B \cap B')$ induced by ad . Each $x \in N'$ acts nilpotently on this vector space, so there must be nonzero $y + (B \cap B')$ killed by all $x \in N'$, i.e. s.t. $\begin{cases} [xy] \in B \cap B' \\ y \notin B \cap B' \end{cases}$. But $[xy] \in [BB]$, so it is nilpotent. This forces $[xy] \in N'$, i.e. $y \in N_B(N') = B \cap K$ while $y \notin B \cap B'$. Similarly, $B \cap B'$ is properly contained in $B' \cap K$. On the other hand, $B \cap K$, $B' \cap K$ are solvable subalgebras of K . Let C, C' be respective Borel subalgebras of K including



yields $\sigma \in \mathcal{E}(L; K) \subset \mathcal{E}(L)$ s.t. $\sigma(C') = C$. Because $B \cap B'$ is a proper nonzero subalgebra of both C, C' , the second inductive hypothesis yields $\tau \in \mathcal{E}(L)$ s.t. $\tau\sigma(C') \subset B$, i.e. τ sends a Borel subalgebra of L including $\sigma(C') = C$ onto B . Here $B \cap \tau\sigma(B') \supset \tau\sigma(C') \cap \tau\sigma(B') \supset \tau\sigma(B' \cap K) \supsetneq \tau\sigma(B \cap B')$, so the former has greater dimension than $B \cap B'$. Then by the second inductive hypothesis, B is conjugate under $\mathcal{E}(L)$ to $\tau\sigma(B')$.

- the set N' of nilpotent elements of $B \cap B'$ is zero.

Any Borel subalgebra of L contains the semisimple and nilpotent parts of its elements by the lemma 4.7 and the Jordan decomposition. This shows at once that $B \cap B' = T$ is a toral subalgebra. Then since B is a standard Borel subalgebra, $B = B(\Delta)$, $N = N(\Delta)$, $B = H + N$. Since $\begin{cases} [BB] = N \\ T \cap N = 0 \end{cases}$, $N_B(T) = C_B(T)$. Let C be any CSA of $C_B(T)$; in particular C is nilpotent and $T \subset$

$N_{C_B(T)}(C) = C$. If $\left\{ \begin{array}{l} n \in N_B(C) \\ t \in T \subset C \end{array} \right.$, then $(\text{ad}t)^k n = 0$, $\exists k \in \mathbb{Z}_{++}$, given C nilpotent. But $\text{ad}t$ is semisimple, so $k = 1$ and $n \in C_B(T)$. Thus $N_B(C) = N_{C_B(T)}(C) = C$. As a nilpotent self-normalising subalgebra of B , C is therefore a CSA of B which includes T . Then by the previous theorem, C is a maximal toral subalgebra of L conjugate under $\mathcal{E}(B)$ hence under $\mathcal{E}(L)$ to H , so wlog we may assume $T \subset H$. Suppose $T = H$. Then $H \subsetneq B'$, so B' must include at least one L_α , $\alpha < 0$ relative to Δ . Applying τ_α to B' yields a Borel subalgebra B'' whose intersection with B includes $H + L_{-\alpha}$; so by the second inductive hypothesis B'' is conjugate to B and we are done. Suppose T is properly included in H . Then either B' centralises T or not. If $B' \subset C_L(T)$, then $\dim C_L(T) < \dim L$. So $H \subset C_L(T)$ gives a Borel subalgebra B'' of $C_L(T)$ including H , and therefore we can use the inductive hypothesis to find $\sigma \in \mathcal{E}(L; C_L(T)) \subset \mathcal{E}(L)$ sending B' onto B'' . In particular, B'' is a Borel subalgebra of L , including H , so it is conjugate to B under $\mathcal{E}(L)$ due to the second inductive hypothesis. If $B' \not\subset C_L(T)$, there exists a common eigenvector $x \in B'$ for $\text{ad}T$ and an element $t \in T$ for which $[tx] = \alpha x$ with α rational and positive. Define $S = H + \bigoplus L_\alpha$, $\alpha \in \Phi$ running over those roots for which $\alpha(t)$ is rational and positive. Then S is a subalgebra of L and $x \in S$. Moreover, S is solvable by the previous lemma. Let B'' be a Borel subalgebra of L which includes S . Then $B'' \cap B' \supset T + Fx \supsetneq T = B' \cap B$ so $\dim B'' \cap B' > \dim B \cap B'$. Similarly, $B'' \cap B \supset H \supsetneq T$ so $\dim B'' \cap B > \dim B' \cap B$. By the second inductive hypothesis B'' is conjugate to B . In particular, B'' is standard relative to a CSA conjugate to H . Transporting the pair (B'', B') back to B by that conjugation, another application of the second inductive hypothesis gives B'' conjugate to B' . Therefore B is conjugate to B' .

- $B \cap B' = 0$.

$\dim L \geq \dim B + \dim B'$, since B is standard we know $\dim B > \frac{1}{2} \dim L$. Take T to be a maximal toral subalgebra of B' . If $T = 0$, B' consists of nilpotent elements; B' is therefore nilpotent by Engel's theorem as well as self-normalising by the lemma 4.7., i.e. B' is a CSA, a contradiction since all CSA of L are toral. Therefore $T \neq 0$. If H_0 is a maximal toral subalgebra of L including T , then B' has nonzero intersection with any standard Borel subalgebra B'' relative to H_0 . Therefore B' is conjugate to B'' by the first part of the proof. Thus $\dim B' = \dim B'' > \frac{1}{2} \dim L$, a contradiction.

Corollary 4.3. *The Cartan subalgebras of an arbitrary Lie algebra L are conjugate under $\mathcal{E}(L)$.*

pf. Let H, H' be two CSAs of L . Being nilpotent, each lies in at least one Borel subalgebra, say B and B' respectively. By the previous theorem, $\exists \sigma \in \mathcal{E}(L)$ s.t.

$\sigma(B) = B'$. Now $\sigma(H)$, H' are both CSAs of the solvable algebra B' , so by the theorem 4.3., $\exists \tau' \in \mathcal{E}(B')$ for which $\tau'\sigma(H) = H'$. But τ' is the restriction to B' of some $\tau \in \mathcal{E}(L; B') \subset \mathcal{E}(L)$, so $\tau\sigma(H) = H'$, $\tau\sigma \in \mathcal{E}(L)$.

Definition 4.4. For arbitrary Lie algebra L over F , the dimension of a CSA of L is called the rank.

→ if L is semisimple, $\text{rank} L$ coincides with the $\text{rank} \Phi$, where Φ is the root system of L relative to any maximal toral subalgebra.

5 Existence Theorem

5.1 Universal Enveloping Algebras

Fix a finite dimensional vector space V over F . Let $T^0V = F$, $T^1V = V$, $T^2V = V \otimes V, \dots, T^mV = V \otimes \dots \otimes V$. Define $\mathcal{S}(V) = \bigoplus_{i=0}^{\infty} T^iV$, and introduce an associative product, defined on homogeneous generators of $\mathcal{S}(V)$ s.t. $(v_1 \otimes \dots \otimes v_k)(w_1 \otimes \dots \otimes w_m) = v_1 \otimes \dots \otimes v_k \otimes w_1 \otimes \dots \otimes w_m \in T^{k+m}V$. This makes $\mathcal{S}(V)$ an associative graded algebra with 1, which is generated by 1 along with any basis of V . This is called the **tensor algebra** on V .

→ it is universal in the sense that given any F -linear map $\phi : V \rightarrow \mathcal{U}$ where $\mathcal{U}$ is an associative algebra with 1 over F , there is a unique homomorphism of F -

algebras $\psi : \mathcal{S}(V) \rightarrow \mathcal{U}$ s.t. $\psi(1) = 1$ and

$$\begin{array}{ccc} V & \xrightarrow{i} & \mathcal{S}(V) \\ & \searrow \phi & \downarrow \psi \\ & & \mathcal{U} \end{array} \quad \text{commutes. Then let } I$$

be the ideal in $\mathcal{S}(V)$ generated by all $x \otimes y - y \otimes x$, $x, y \in V$ and call $\mathcal{G}(V) = \mathcal{S}(V)/I$ **the symmetric algebra** on V . Let $\sigma : \mathcal{S}(V) \rightarrow \mathcal{G}(V)$ denote the canonical map. Here the generators of I lie in T^2V , which implies $I = (I \cap T^2V) \oplus (I \cap T^3V) \oplus \dots$

Therefore, σ is injective on $\begin{cases} T^0V = F \\ T^1V = V \end{cases}$, allowing to identify V with a subspace of

$\mathcal{G}(V)$. Then $\mathcal{G}(V)$ inherits a grading $\mathcal{G}(V) = \bigoplus_{i=0}^{\infty} S^iV$. The effect of factoring out I is to make the elements of V commute, so $\mathcal{G}(V)$ is universal for linear maps of V into commutative associative F -algebras with 1. Moreover, if $(x_1, \dots, x_n)$ is any fixed basis of V , then $\mathcal{G}(V)$ is canonically isomorphic to the polynomial algebra over F in n variables, with basis consisting of 1 and all $x_{i(1)} \dots x_{i(t)}$, $1 \leq i(1) \leq \dots \leq i(t) \leq n$.

Definition 5.1. A universal enveloping algebra of L is a pair $(\mathcal{U}, i)$ where $\mathcal{U}$ is an associative algebra with 1 over F , $i : L \rightarrow \mathcal{U}$ is a linear map satisfying $i([xy]) = i(x)i(y) - i(y)i(x)$, $\forall x, y \in L$ and the following holds; for any associative F -algebra $\mathcal{U}'$ with 1 and any linear map $j : L \rightarrow \mathcal{U}'$ satisfying $j([xy]) = j(x)j(y) - j(y)j(x)$,

there exists a unique homomorphism of algebras $\phi : \mathcal{U} \rightarrow \mathcal{U}'$, sending 1 to 1 s.t. $\phi \circ i = j$.

- existence

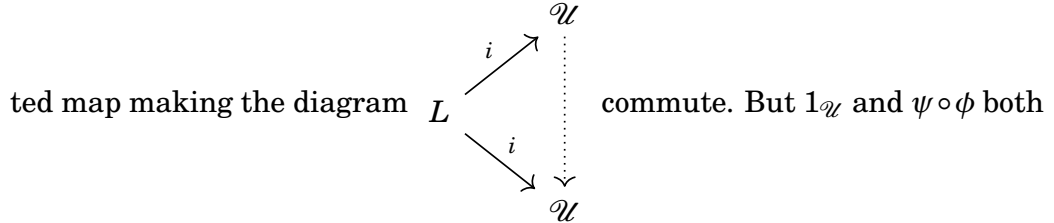
Let $\mathcal{J}(L)$ be the tensor algebra on L and let J be the two-sided ideal in $\mathcal{J}(L)$ generated by all $x \otimes y - y \otimes x - [xy]$, $x, y \in L$. Define $\mathcal{U}(L) = \mathcal{J}(L)/J$ and let $\pi : \mathcal{J}(L) \rightarrow \mathcal{U}(L)$ be the canonical homomorphism. Then $J \subset \bigoplus_{i>0} T^i L$ so π maps $T^0 L = F$ isomorphically into $\mathcal{U}(L)$ and therefore it contains at least the scalars.

- $(\mathcal{U}(L), i)$ is a universal enveloping algebra of L where $i : L \rightarrow \mathcal{U}(L)$ is the restriction of π to L .

Let $j : L \rightarrow \mathcal{U}'$ be as in the definition. The universal property of $\mathcal{J}(L)$ yields an algebra homomorphism $\phi' : \mathcal{J}(L) \rightarrow \mathcal{U}'$ which extends j and sends 1 to 1. Then the property of j forces all $x \otimes y - y \otimes x - [xy]$ to lie in $\ker \phi'$, so ϕ' induces a homomorphism $\phi : \mathcal{U}(L) \rightarrow \mathcal{U}'$ s.t. $\phi \circ i = j$. $\mathcal{U}(L)$ is generated by 1 and $i(L)$ and $\phi(1) = 1$, $\phi \circ i = j$ fix ϕ on these, so ϕ is unique.

- Uniqueness

Assume there is another pair $(\mathcal{B}, i')$ satisfying the same hypotheses. Then we get homomorphisms $\begin{cases} \phi : \mathcal{U} \rightarrow \mathcal{B} \\ \psi : \mathcal{B} \rightarrow \mathcal{U} \end{cases}$. By definition, there is a unique dot-



do the trick so $\psi \circ \phi = 1_{\mathcal{U}}$ and similarly $\phi \circ \psi = 1_{\mathcal{B}}$, so the uniqueness follows.

Write $\mathcal{J} = \mathcal{J}(L)$, $\mathcal{G} = \mathcal{G}(L)$, $\mathcal{U} = \mathcal{U}(L)$. Define a filtration on $\mathcal{J}$ by $T_m = T^0 \oplus T^1 \oplus \dots \oplus T^m$ and let $U_m = \pi(T_m)$, $U_{-1} = 0$. Then $\begin{cases} U_m U_p = U_{m+p} \\ U_m \subset U_{m+1} \end{cases}$. Set $G^m = U_m/U_{m-1}$ and let the multiplication in $\mathcal{U}$ define a bilinear map $G^m \times G^p \rightarrow G^{m+p}$. It is well-defined, given $U_{m-1} U_p \subset U_{m+p-1}$. This extends to a bilinear map $\mathfrak{G} \times \mathfrak{G} \rightarrow \mathfrak{G}$ where $\mathfrak{G} = \bigoplus_{m=0}^{\infty} G^m$, making $\mathfrak{G}$ a graded associative algebra with 1. Since π maps T^m into U_m , the composite linear map $\phi_m : T^m \rightarrow U_m \rightarrow G^m = U_m/U_{m-1}$ makes sense. It is also surjective because $U_m = \pi(T^m) + U_{m-1}$ imply T^m to U_m/U_{m-1}

onto. Therefore the maps ϕ_m combine to yield a linear map $\phi : \mathcal{J} \rightarrow \mathfrak{G}$ which is surjective and sends 1 to 1.

Lemma 5.1. $\phi : \mathcal{J} \rightarrow \mathfrak{G}$ is an algebra homomorphism. Moreover, $\phi(I) = 0$ so ϕ induces a homomorphism ω of $\mathcal{G} = \mathcal{J}/I$ onto $\mathfrak{G}$.

pf. Let $\begin{cases} x \in T^m \\ y \in T^p \end{cases}$ be homogeneous tensors. By the definition of the product in $\mathfrak{G}$, $\phi(xy) = \phi(x)\phi(y)$, so ϕ is multiplicative on $\mathcal{J}$. Let $x \otimes y - y \otimes x$, $x, y \in L$ be a typical generator of I . Then $\pi(x \otimes y - y \otimes x) \in U_2$ by definition. On the other hand, $\pi(x \otimes y - y \otimes x) = \pi([xy]) \in U_1$, and therefore $\phi(x \otimes y - y \otimes x)$ is the class in U_2/U_1 of an element of U_1 , hence 0. So $I \subset \ker \phi$.

Fix an ordered basis $(x_\lambda | \lambda \in \Omega)$ of L . This choice identifies $\mathcal{G}$ with the polynomial algebra in indeterminates z_λ , $\lambda \in \Omega$. For each sequence $\Sigma = (\lambda_1, \dots, \lambda_m)$ of indices, let $\begin{cases} z_\Sigma = z_{\lambda_1} \dots z_{\lambda_m} \in S^m \\ x_\Sigma = x_{\lambda_1} \otimes \dots \otimes x_{\lambda_m} \in T^m \end{cases}$. Call Σ increasing if $\lambda_1 \leq \dots \leq \lambda_m$ in the given ordering of Ω . Set $\emptyset$ is increasing and $z_\emptyset = 1$. So $\{z_\Sigma | \Sigma \text{ increasing}\}$ is a basis of $\mathcal{G}$. Associated with the grading $\mathcal{G} = \bigoplus S^m$ is the filtration $S_m = S^0 \oplus \dots \oplus S^m$. Write $\lambda \leq \Sigma$ if $\lambda \leq \mu$, $\forall \mu \in \Sigma$.

Lemma 5.2. $\forall m \in \mathbb{Z}_+$, there exists a unique linear map $f_m : L \otimes S_m \rightarrow \mathcal{G}$ satisfying

$$\begin{cases} f_m(x_\lambda \otimes z_\Sigma) = z_\lambda z_\Sigma, \quad \forall \lambda \leq \Sigma & \forall z_\Sigma \in S_m \\ f_m(x_\lambda \otimes z_\Sigma) - z_\lambda z_\Sigma \in S_k & \forall k \leq m, z_\Sigma \in S_k \\ f_m(x_\lambda \otimes f_m(x_\mu \otimes z_T)) = f_m(x_\mu \otimes f_m(x_\lambda \otimes z_T)) + f_m([x_\lambda x_\mu] \otimes z_T) & \forall z_T \in S_{m-1} \end{cases}$$

Moreover, the restriction of f_m to $L \otimes S_{m-1}$ agrees with f_{m-1} .

pf. Here terms in the third conditions make sense once the second condition is proved. Also the restriction of f_m to $L \otimes S_{m-1}$ automatically satisfies all three conditions in $m-1$, so this restricted map must coincide with f_{m-1} by the uniqueness.

- When $m = 0$, $z_\Sigma = 1$ and therefore we may let $f_0(x_\lambda \otimes 1) = z_\lambda$ and extend linearly to $L \otimes S_0$. Then the above three conditions are satisfied and by the first condition f_0 is unique.
- Assuming the existence of a unique f_{m-1} satisfying all three conditions. For the case $\lambda \leq \Sigma$, the first condition cannot hold unless we define $f_m(x_\lambda \otimes z_\Sigma) = z_\lambda z_\Sigma$. In case $\lambda \leq \Sigma$ fails, the first index μ in Σ must be strictly less than λ , so $\Sigma = (\mu, T)$ where $\mu \leq T$ and T has length $m-1$. Then by the first condition in $m-1$, $z_\Sigma = z_\mu z_T = f_{m-1}(x_\mu \otimes z_T)$. Since $\mu \leq T$, $f_m(x_\mu \otimes z_T) = z_\mu z_T$ is already defined. So $f_m(x_\lambda \otimes f_m(x_\mu \otimes z_T)) = f_m(x_\lambda \otimes z_\Sigma)$ by the first condition in $m-1$.
defined. So $f_m(x_\lambda \otimes z_T) = f_{m-1}(x_\lambda \otimes z_T) \equiv z_\lambda z_T \pmod{S_{m-1}}$ the second condition in $m-1$.

The RHS of the third condition equals $f_m(x_\mu \otimes f_m(x_\lambda \otimes z_T)) + f_m([x_\lambda x_\mu] \otimes z_T) = z_\mu z_\lambda z_T + f_{m-1}(x_\mu \otimes y) + f_{m-1}([x_\lambda x_\mu] \otimes z_T)$, $y \in S_{m-1}$, which involves only values already defined. The third condition forces $f_m(x_\lambda \otimes z_\Sigma)$ to equal it, so we define it this way. Hence f_m can be defined in only one way and all three conditions hold in case $\mu < \lambda$, $\mu \leq T$. But $[x_\mu x_\lambda] = -[x_\lambda x_\mu]$, the conditions also hold for $\lambda < \mu$, $\lambda \leq T$. When $\lambda = \mu$, all conditions are also satisfied.

When neither $\lambda \leq T$ nor $\mu \leq T$ is true, write $T = (v, \Psi)$ where $v \leq \Psi$, $v < \lambda, v < \mu$. Abbreviate $f_m(x \otimes z)$ by xz , $\forall x \in L$, $z \in S_m$. The inductive hypothesis in-

sures $\begin{cases} x_\mu z_T = x_\mu(x_v z_\Psi) = x_v(x_\mu z_\Psi) + [x_\mu x_v]z_\Psi \\ x_\mu z_\Psi = z_\mu z_\Psi + \omega, \omega \in S_{m-1} \end{cases}$. Since $v \leq \Psi$, $v < \mu$, the third

condition applies to $x_\lambda(x_v(z_\mu z_\Psi))$. By the inductive hypothesis, the third con-

dition also applies to $x_\lambda(x_v \omega)$ and therefore to $x_\lambda(x_v(x_\mu z_\Psi))$. So $x_\lambda(x_\mu z_T) =$

$x_v(x_\lambda(x_\mu z_\Psi)) + [x_\lambda x_v](x_\mu z_\Psi) + [x_\mu x_v](x_\lambda z_\Psi) + [x_\lambda[x_\mu x_v]]z_\Psi$. Since λ, μ are in-

terchangeable, if we interchange them and subtract the two resulting equa-

tions, we get $x_\lambda(x_\mu z_T) - x_\mu(x_\lambda z_T) = x_v(x_\lambda(x_\mu z_\Psi)) - x_v(x_\mu(x_\lambda z_\Psi)) + [x_\lambda[x_\mu x_v]]z_\Psi -$

$[x_\mu[x_\lambda x_v]]z_\Psi = x_v([x_\lambda x_\mu]z_\Psi) + [x_\lambda[x_\mu x_v]]z_\Psi + [x_\mu[x_v x_\lambda]]z_\Psi = [x_\lambda x_\mu](x_v z_\Psi) + ([x_v[x_\lambda x_\mu]] + [x_\lambda[x_\mu x_v]] + [x_\mu[x_v x_\lambda]])z_\Psi = [x_\lambda x_\mu]z_T$ so the conditions are satisfied.

Lemma 5.3. *There exists a representation $\rho : L \rightarrow \mathfrak{gl}(\mathcal{G})$ satisfying*

$$\begin{cases} \rho(x_\lambda)z_\Sigma = z_\lambda z_\Sigma & \forall \lambda \leq \Sigma \\ \rho(x_\lambda)z_\Sigma \equiv z_\lambda z_\Sigma \pmod{S_m} & \Sigma \text{ has length } m \end{cases}$$

pf. By the previous lemma, we can define a linear map $f : L \otimes \mathcal{G} \rightarrow \mathcal{G}$ satisfying all three conditions for all m since f_m restricted to $L \otimes S_{m-1}$ is f_{m-1} by the uniqueness part. In other words, $\mathcal{G}$ becomes an L -module by the third condition, affording a representation ρ which satisfies two conditions thanks to the previous lemma's first and second conditions.

Lemma 5.4. *Let $t \in T_m \cap J$, $K = \ker \pi$, $\pi : \mathcal{J} \rightarrow \mathcal{U}$ canonical. Then the homogeneous component t_m of t of degree m lies in I , the kernel of the canonical map $\mathcal{J} \rightarrow \mathcal{G}$.*

pf. t_m can be written as a linear combination of basis elements $x_{\Sigma(i)}$ each $\Sigma(i)$ of length m . The Lie homomorphism $\rho : L \rightarrow \mathfrak{gl}(\mathcal{G})$ constructed in the previous lemma extends by the universal property of tensor algebra $\mathcal{J}$ to an algebra homomorphism $\rho : \mathcal{J} \rightarrow \text{End } \mathcal{G}$ with $J \subset \ker \rho$. So $\rho(t) = 0$. But $\rho(t)1$ is a polynomial whose term of highest degree is the appropriate combination of the $z_{\Sigma(i)}$ by the previous lemma. Thus this combination of the $z_{\Sigma(i)}$ is 0 in $\mathcal{G}$ and $t_m \in I$.

Theorem 5.1 (Poincare-Birkhoff-Witt). *The homomorphism $\omega : \mathcal{G} \rightarrow \mathfrak{G}$ is an isomorphism of algebras.*

pf. Let $t \in T^m$, $\pi : \mathcal{J} \rightarrow \mathcal{U}$ the canonical map. Let $\pi(t) \in U_{m-1}$. Then $\pi(t) = \pi(t')$ for some $t' \in T_{m-1}$ and therefore $t - t' \in J$. $t - t' \in T_m \cap J$. So the homogeneous

component of degree m being t , $t \in I$ by the previous lemma. Thus $\ker \phi \subset I$ degree by degree, so the induced $\omega : \mathcal{G} \rightarrow \mathfrak{G}$ has zero kernel and it was already onto. That makes it an isomorphism.

References

- [1] James E. Humphreys. *Introduction to Lie Algebras and Representation Theory*. Vol. 9. Graduate Texts in Mathematics. New York: Springer-Verlag, 1972. ISBN: 978-0-387-90053-7. DOI: 10.1007/978-1-4612-6398-2.